

Financial Mathematics

Lecture notes

Jarek Kędra Markus Upmeyer

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Abstract

Lecture notes for the course Financial Mathematics at the University of Aberdeen.

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Introduction

Financial Mathematics is the study of mathematical models motivated by financial markets. The models, as usual, are idealisations of reality and as such have many simplifying assumptions. This makes their applicability a delicate matter. Also, due to the probabilistic nature of the subject, even a super-accurate model can only offer predictions which are correct on average. In simple language, we should always expect that predictions offered by any model can be wrong.

From the mathematical viewpoint all the material presented here consists of applications of Linear Algebra and Probability Theory, and later, Stochastic Calculus. It is presented in a rigorous way with the emphasis on the mathematical content and precision.

The material is split into sections each of which is concluded by a set of exercises. Hints and examples of some of the solutions are typeset in blue.

September 2026: Due to a substantial progress of the Large Language Models (LLM) we added exercises with an intended use of LLMs. The main purpose is that a suitable interaction with an LLMs might improve human understanding. Also, we believe that practicing using LLMs in concrete problems and tasks is a good preparation for the future job market.

1 Warm up: price dynamics

1.1 Intuition

Suppose you have a share of stock worth 100 now. What will its price be tomorrow, in a week, in a month, in a year? Nobody knows. We may think that it is quite likely that tomorrow's price will be between 98 and 102 and very unlikely that it will increase above 1000. We may list possible outcomes and assign the likelihood of obtaining each particular price.

For example, we may assign probabilities as follows:

$$\text{Price next week} = \begin{cases} 98 & \text{with probability } \frac{1}{8}, \\ 99 & \text{with probability } \frac{1}{4}, \\ 100 & \text{with probability } \frac{1}{2}, \\ 101 & \text{with probability } \frac{1}{8}. \end{cases}$$

Of course, our guess can be much more complicated.

This mental exercise shows that the price of a share of stock at a given future time is a **random variable**. Moreover, it shows that we are **guessing** the future prices. This is an important step in applications of financial mathematics: the choice of a good model.

1.2 Examples of models

1.2.1 1-step binomial model with one asset

Let $\Omega = \{u, d\}$ and let the probability be defined by $\mathbf{P}\{u\} = p$ and $\mathbf{P}\{d\} = 1 - p$. Let $S(1): \Omega \rightarrow \mathbf{R}$ be the random variable given by

$$S(1)(u) = S(0)U, \quad S(1)(d) = S(0)D,$$

where $0 < D < U$ are real numbers and $S(0)$ is the current price of the asset.¹ The distribution of $S(1)$ is given by

$$S(1) = \begin{cases} S(0)U & \text{with probability } p, \\ S(0)D & \text{with probability } 1 - p. \end{cases}$$

¹If $D < 1 < U$, then these indeed represent price jumps down and up.

1.2.2 1-step trinomial model with one asset

Let $\Omega = \{u, m, d\}$ and let the probability be defined by $\mathbf{P}\{u\} = p_1$, $\mathbf{P}\{m\} = p_2$ and $\mathbf{P}\{d\} = 1 - p_1 - p_2$. Let $S(1): \Omega \rightarrow \mathbf{R}$ be a random variable given by

$$S(1)(u) = S(0)U, \quad S(1)(m) = S(0)M, \quad S(1)(d) = S(0)D,$$

where $0 < D < M < U$ are real numbers and $S(0)$ is the current price of the asset.

1.2.3 2-step binomial model with one asset

In a model with two time steps let $\Omega = \{u, d\}^2 = \{(u, u), (u, d), (d, u), (d, d)\}$ and define the probability by

$$\mathbf{P}(u, u) = p^2, \quad \mathbf{P}(u, d) = \mathbf{P}(d, u) = p(1 - p), \quad \mathbf{P}(d, d) = (1 - p)^2.$$

Let $0 < D < U$ be real numbers. Define a pair of random variables $S(1), S(2): \Omega \rightarrow \mathbf{R}$ by (see Figure 1.2)

$$\begin{array}{ll} S(1)(u, u) = S(0)U^2 & S(2)(u, u) = S(0)U^2 \\ S(1)(u, d) = S(0)UD & S(2)(u, d) = S(0)UD^2 \\ S(1)(d, u) = S(0)DU & S(2)(d, u) = S(0)DU^2 \\ S(1)(d, d) = S(0)D^2 & S(2)(d, d) = S(0)D^2 \end{array}$$

The meaning of the elements of the sample space is as follows. Suppose that $D < 1 < U$. The event (u, u) means that the price of the asset increased at time $t = 1$ and time $t = 2$; the event (d, u) means that the price of the asset decreased at time $t = 1$ and increased at time $t = 2$; and so on.

Observe, that the probability $\mathbf{P}$ is defined in such a way that the price goes up at each time with probability p and does down with probability $1 - p$.

1.2.4 1-step binomial model with two assets

Let $\Omega = \{u, d\}^2 = \{(u, u), (u, d), (d, u), (d, d)\}$ and let $0 < p_1, p_2 < 1$. The probability is defined by

$$\begin{array}{ll} \mathbf{P}(u, u) = p_1 p_2, & \mathbf{P}(u, d) = p_1(1 - p_2), \\ \mathbf{P}(d, u) = (1 - p_1)p_2, & \mathbf{P}(d, d) = (1 - p_1)(1 - p_2). \end{array}$$

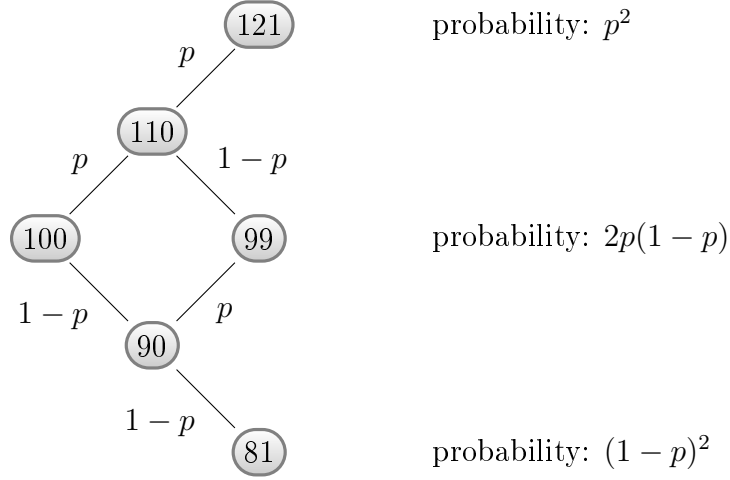


Figure 1.1: The 2-step binomial model with one asset can be visualized as a **graph**. In this example, $D = 0.9$ and $U = 1.1$.

Let $0 < D_1 < U_1$ and $0 < D_2 < U_2$ be real numbers. Let $S_1(1), S_2(1): \Omega \rightarrow \mathbf{R}$ be two random variables given by

$$\begin{aligned}
 S_1(1)(u, u) &= S_1(0)U_1 & S_2(1)(u, u) &= S_2(0)U_2 \\
 S_1(1)(u, d) &= S_1(0)U_1 & S_2(1)(u, d) &= S_2(0)D_2 \\
 S_1(1)(d, u) &= S_1(0)D_1 & S_2(1)(d, u) &= S_2(0)U_2 \\
 S_1(1)(d, d) &= S_1(0)D_1 & S_2(1)(d, d) &= S_2(0)D_2
 \end{aligned}$$

Although the sample space Ω is the same as in the previous example the meanings of events are different. For example, the event (u, d) means that the price of the first asset increased and the price of the second asset decreased.

There is a more efficient notation which can be used in this and other examples. Namely, let $\Omega = \{0, 1\}^2 = \{(1, 1), (1, 0), (0, 1), (0, 0)\}$ and let $S_i(1): \Omega \rightarrow \mathbf{R}$ for $i = 1, 2$ be defined by

$$S_i(1)(u_1, u_2) = S_i(0)U_i^{u_i}D_i^{1-u_i}.$$

The efficiency is clear: we can express both random variables in one formula. This notation is also practical for programming purposes.

1.2.5 2-step model with two assets, each of which follows the binomial model

Let $\Omega = M_{2 \times 2}(\{0, 1\})$ be the set of 2×2 matrices with entries from $\{0, 1\}$. Let the probability be defined by

$$\mathbf{P} \left(\begin{smallmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{smallmatrix} \right) = p_1^{u_{11}+u_{12}} (1-p_1)^{2-u_{11}-u_{12}} p_2^{u_{21}+u_{22}} (1-p_2)^{2-u_{21}-u_{22}},$$

where $0 < p_1, p_2 < 1$. Let $S_i(t): \Omega \rightarrow \mathbf{R}$, where $i = 1, 2$ (assets) and $t = 1, 2$ (time), be random variables defined by

$$\begin{aligned} S_i(1) \left(\begin{smallmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{smallmatrix} \right) &= S_i(0) U_i^{u_{i1}} D_i^{1-u_{i1}}, \\ S_i(2) \left(\begin{smallmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{smallmatrix} \right) &= S_i(0) U_i^{u_{i1}+u_{i2}} D_i^{2-u_{i1}-u_{i2}}, \end{aligned}$$

where $0 < D_i < U_i$ for $i = 1, 2$ are real numbers.

1.2.6 n-step binomial model with one asset

Let $\Omega = \{0, 1\}^n$ be the space of all sequences of 0's and 1's of length $n \in \mathbf{N}$. The probability is defined by

$$\mathbf{P}(u_1 \dots u_n) = p^{u_1+\dots+u_n} (1-p)^{n-(u_1+\dots+u_n)},$$

where $0 < p < 1$. Let $S(t): \Omega \rightarrow \mathbf{R}$, for $t = 1, 2, \dots, n$, be random variables defined by

$$S(t)(u_1 \dots u_n) = S(0) U^{u_1+\dots+u_t} D^{t-(u_1+\dots+u_t)},$$

where $0 < D < U$. For example, if $n = 5$ then

$$\begin{aligned} S(3)(01010) &= S(0) U D^2 \\ S(4)(01010) &= S(0) U^2 D^2 \\ S(5)(01010) &= S(0) U^2 D^3. \end{aligned}$$

1.3 Exercises

1a. Let $S(0) = 100$ and let $D = 0.9$, $M = 1$ and $U = 1.2$. For each example in Section 1.2 determine the distribution of each random variable.

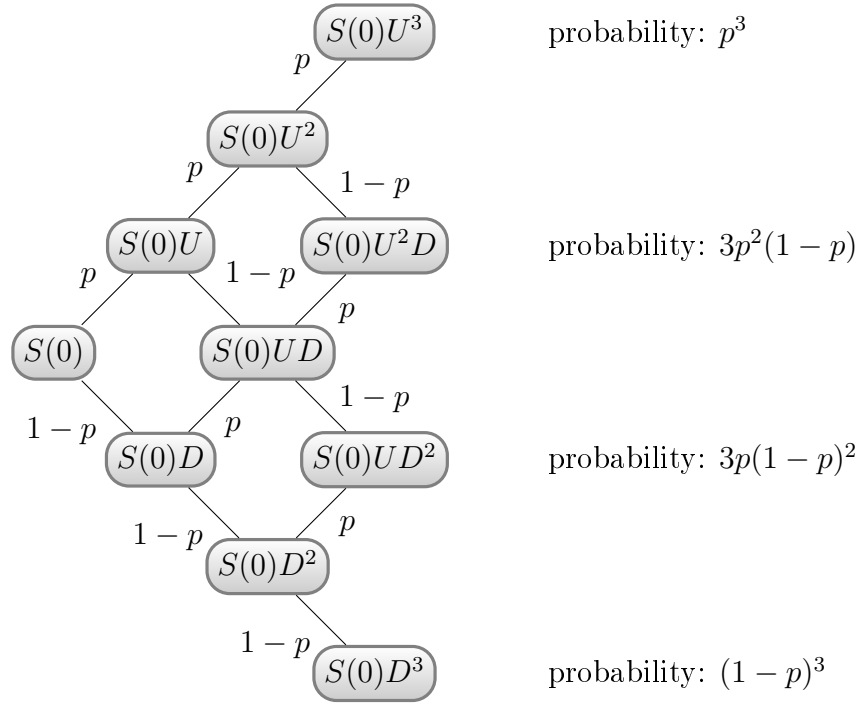


Figure 1.2: The 3-step binomial model with one asset

Solution. In Example 1.2.1 the distribution is given by

$$S(1) = \begin{cases} 90 & \text{with probability } p, \\ 120 & \text{with probability } 1-p. \end{cases}$$

For Example 1.2.2, we have

$$S(1) = \begin{cases} 120 & \text{with probability } p_1, \\ 100 & \text{with probability } p_2, \\ 90 & \text{with probability } 1-p_1-p_2. \end{cases}$$

1b. In Example 1.2.5 express both random variables in a closed formula

$$S_i(t) \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix} = \dots$$

Solution. $S_i(t) = S_i(0)U_i^{\sum_{j=1}^t u_{ij}}D_i^{t-\sum_{j=1}^t u_{ij}}$

1c. For each example discussed in Section 1.2 draw a tree illustrating the price dynamics.

1d. Create several examples of price dynamics.

1e. Consider a 3-step model with 4 assets each following binomial price dynamics. How many elements does the sample space contain? Give an example of a useful sample space and write a formula for the price of the i -th asset at time t :

$$S_i(t) = \dots$$

Solution. For each of the four assets there are two possible outcomes at each step, so

$\Omega = M_{4 \times 3}(\{0, 1\})$ is the set of 4×3 matrices $\begin{pmatrix} u_{11} & u_{12} & u_{13} \\ \vdots & \ddots & \vdots \\ u_{41} & u_{42} & u_{43} \end{pmatrix}$ with entries in $\{0, 1\}$.

The sample space has $2^{12} = 4096$ elements. Generalizing Exercise 1b, for the assets S_i , $i = 1, 2, 3, 4$, the price at time t is $S_i(t) = S_i(0)U_i^{\sum_{j=1}^t u_{ij}}D_i^{t-\sum_{j=1}^t u_{ij}}$ for any choice of price jumps $0 < D_i < U_i$.

1f. Consider an n -step model with m assets each following binomial price dynamics. How many elements does the sample space contain? Describe a suitable sample space and write a formula for the price of the i -th asset at time t .

Solution. For each of the m assets there are two outcomes at each of the n steps, so $\Omega = M_{m \times n}(\{0, 1\})$ is the set of $m \times n$ matrices with entries in $\{0, 1\}$, which has 2^{mn} elements. The formula for the price of the assets is still $S_i(t) = S_i(0)U_i^{\sum_{j=1}^t u_{ij}}D_i^{t-\sum_{j=1}^t u_{ij}}$ for time $t = 1, \dots, n$ and asset $i = 1, \dots, m$, generalizing Exercise 1b.

1g. Exercises with an LLM. Remember that our main objective in this course is to increase *human understanding* of Financial Maths. Keep this in mind when interacting with an LLM.

1. Ask the LLM for a short summary of Section 1 in these Lecture Notes. If you don't understand some phrases or concepts ask the LLM for clarifications. Ask to what extent the binomial model is realistic in practice.

Comment: For your convenience I made the Lecture Notes publicly available at

<https://jarek-kedra.github.io/HTML/fm-lecture-notes.pdf>

You can simply ask your LLM to take a look there instead of uploading the notes.

2. Let $S(0) = 100$, $U = 1.1$, $D = 0.9$ and $p = 0.5$. Ask the LLM to consider the multistep binomial model with this data. Ask them to do the following tasks (or equivalent):

- (a) Determine the distribution of the price at times $t = 1, 2, 3, 4$.
Comment: Observe how quickly the maximal prices grow. This suggests that choosing the parameters U and D as above is not very realistic.
 - (b) Change parameters: $U = 1.01$ and $D = 0.99$. For the new parameters determine the price distribution for time $t = 1, 2, 3, \dots, 10$. Present these distributions graphically.
 - (c) Keep the parameters U and D and change $p = 0.6$. Perform tasks similar to the ones above.
- 3. Try to verify some of the outputs provided by the LLM. Notice that due to the complexity of the LLMs it is hard to verify whether they provide correct answers.
 - 4. Ask the LLM to create a simple computer program to perform computations in the binomial model.
 - 5. Read what Richard Stallman thought about ChatGPT two years ago:
<https://www.stallman.org/chatgpt.html>

2 Basic concepts, notions and questions

2.1 Assets

Examples of assets² are: shares of stock³, foreign currency, commodities, bank deposits, bonds, houses and many more. An asset is called *risk-free* if its future price is known and it is called *risky* if its future price is unknown at present. The future price of a risky asset will be modelled by a random variable. Assets are also called *securities*.

2.2 Time

The time $\mathbf{T}$ is modelled either by a finite set $\mathbf{T} = \{0, 1, 2, \dots, T\}$ or by the closed interval $\mathbf{T} = [0, T]$ of non-negative real numbers. In the first part of the course we mostly consider the first case (discrete time). When $\mathbf{T} = \{0, 1\}$ we say that we consider *single period* model or 1-step model. If $T = n > 1$, then the model is called *multi-period* or n -step. Finally, if $\mathbf{T} = [0, T]$ then we speak about a *continuous time model*.

2.3 Market model

Tentatively, by a market model we mean a probability space $(\Omega, \mathcal{F}, \mathbf{P})$ together with a collection of random variables defined on Ω .

For example, in the previous section we considered simple models in which we specified the probability spaces and random variables modelling the prices of a share of stock.

2.4 The return

Let $S(t)$ be the price of either a risky or risk-free asset at time t . The *return* of the asset at time t is a random variable $K_S(t): \Omega \rightarrow \mathbf{R}$ defined by

$$K_S(t) = \frac{S(t) - S(0)}{S(0)}. \quad (2.1)$$

²The term *asset* can be defined as a resource owned by an individual or legal entity, regarded as having economic value, with the ability to generate cash flow or meet debts.

³A *corporation* is a special type of company that issues *stocks* to raise capital from investors. This divides a portion of the company's equity into *shares*, which entitle the owner to a fraction of the company's equity and profits. In general, *equity* means the ownership of assets that may have debts and liabilities attached to them.

Equivalently,

$$S(t) = (1 + K_S(t))S(0). \quad (2.2)$$

The **expected return** at time t is the expectation of the return: $\mathbf{E}(K_S(t))$.

For example, if the initial price of an asset is $S(0) = 100$ and the future price is

$$S(1) = \begin{cases} 90 & \text{with probability } p, \\ 110 & \text{with probability } 1 - p, \end{cases}$$

then the return is

$$K_S(1) = \begin{cases} -10\% & \text{with probability } p, \\ 10\% & \text{with probability } 1 - p. \end{cases}$$

Hence the expected return is $\mathbf{E}(K_S(1)) = -0.1p + 0.1(1 - p) = -0.2p + 0.1$.

2.5 Money market and the risk free rate

The **risk-free rate** $r \in \mathbf{R}$ is the return of a money deposited in a bank. That is, one unit of money invested at time $t = 0$ yields $R = (1 + r)$ units of money after one time step, as specified by the model. In practice, under normal circumstances $r > 0$ (invested money increases in nominal value), although it can happen that interest rates are negative or zero.⁴

We will mostly assume for simplicity that the risk-free rate is constant over all the periods of the model. A more accurate model should use a **bond process** instead of the constant rate money account as the risk-free asset.

2.6 Portfolio

A **portfolio** is a collection of both risky and risk-free assets. The risky assets may be shares of stocks, forward contracts, options (see below for definitions) and others. In our model, a portfolio is a vector $(x_1, \dots, x_m, y) \in \mathbf{R}^{m+1}$. It represents a collection of assets in which we have x_1 of the first asset $S_1(t)$, x_2 of the second asset $S_2(t)$, etc. and y of the risk-free asset $A(t)$. We also

⁴ Recently, the European Central Bank changed the interest rates as follows: $r = -0.1\%$ in 2014, $r = -0.3\%$ in 2015, $r = -0.4\%$ in 2016, $r = -0.5\%$ in 2019, $r = 0\%$ in early 2022 and $r = 0.75\%$ in late 2022. This era of low interest rates started due to the economic crises in 2008. What was the ECB's rationale for this?

use the notation $x_0 = y$ and $S_0(t) = A(t)$. Since portfolios are vectors, they can be added and multiplied by scalars.

The **value of the portfolio** $(x_1, \dots, x_m, y)$ at time $t \in \mathbf{R}_+$ is the random variable

$$V_{(x,y)}(t) = yA(t) + \sum_{i=1}^m x_i S_i(t) = \sum_{i=0}^m x_i S_i(t), \quad (2.3)$$

where $A(t)$ is the value of a risk-free asset at time t and $S_i(t)$ is the value of a risky asset at time t .

Lemma 2.7. *The value of a portfolio at a fixed time t is linear with respect to the portfolio variables. That is,*

$$V_{(x,y)+(x',y')}(t) = V_{(x,y)}(t) + V_{(x',y')}(t).$$

Proof. Exercise. □

We wish to express the return $K_{(x,y)}(t)$ of the portfolio (x, y) in terms of the returns of each of the individual assets. We compute

$$\begin{aligned} K_{(x,y)}(t) &= \frac{V_{(x,y)}(t) - V_{(x,y)}(0)}{V_{(x,y)}(0)} && \text{by (2.1)} \\ &= \sum_{i=0}^m x_i \frac{S_i(t) - S_i(0)}{V_{(x,y)}(0)} && \text{by (2.3)} \\ &= \sum_{i=0}^m x_i \frac{[1 + K_i(t)]S_i(0) - S_i(0)}{V_{(x,y)}(0)} && \text{by (2.2)} \\ &= \sum_{i=0}^m \frac{x_i S_i(0)}{V_{(x,y)}(0)} K_i(t) \end{aligned}$$

The above computation motivates the following definition. The coefficients

$$w_0 = \frac{x_0 S_0(0)}{V_{(x,y)}(0)} = \frac{yA(0)}{V_{(x,y)}(0)}, \quad w_i = \frac{x_i S_i(0)}{V_{(x,y)}(0)}$$

are called the **weights** of the portfolio (x, y) . It follows that the return of the portfolio is the weighted sum of returns of the contributing assets:

$$K_{(x,y)}(t) = \sum_{i=0}^m w_i K_i(t). \quad (2.4)$$

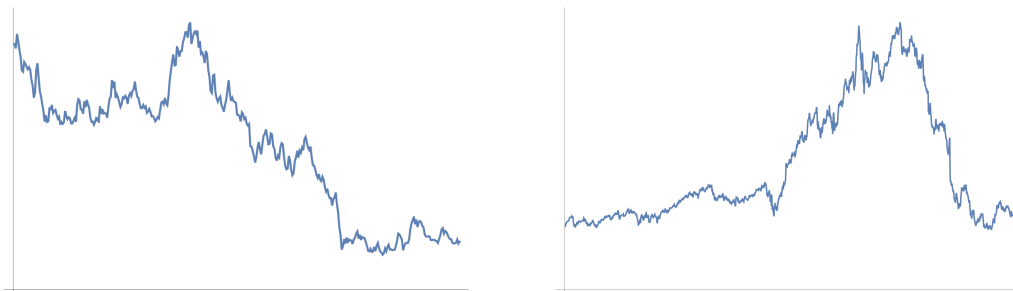


Figure 2.1: Historic share prices of Coinbase Global Inc. (left) and PayPal Holdings Inc. (right). The risks are $\sigma_{\text{Coinbase}} = 0.27$ and $\sigma_{\text{PayPal}} = 0.93$. In which of these would you rather invest?

Consequently the expected return is given by

$$\mathbf{E}(K_{(x,y)}(t)) = \sum_{i=0}^m w_i \mathbf{E}(K_i(t)).$$

Notice that

$$\sum_{i=0}^m w_i = \frac{1}{V_{(x,y)}(0)} \sum_{i=0}^m x_i S_i(0) = \frac{V_{(x,y)}(0)}{V_{(x,y)}(0)} = 1.$$

If the values of the return of a portfolio differ much depending on a scenario, then the investor can sometimes earn much more than the expected return as well as lose money. In this sense, the **variance** of the portfolio, or more precisely, the square root of the variance is interpreted as the measurement of the **risk**:

$$\sigma_{(x,y)}(t) = \sqrt{\text{var}(K_{(x,y)}(t))}.$$

Problem 2.8. *Among the portfolios with given expected return find those which minimise the risk.*

Experiment with your own portfolio in `expectation-risk-hist-shares` on MyAberdeen using Mathematica!

2.9 The axiom of no arbitrage

An *arbitrage* is a way of making profit with no investment. Here are two simple examples.

Example 2.10. Alice offers to buy 1€ for £1 and a Bob offers to sell 1€ for £0.9 (in unlimited quantities). This is an (obvious) arbitrage opportunity: borrow £90,000; buy €100,000 from Bob; sell them for £100,000 to Alice; return £90,000 plus the cost of the loan to the bank; the rest is the arbitrage profit. $\diamond$

Example 2.11. Alice offers to buy 1€ for £0.985 in one years time. Bob sells 1€ for £1 now. Pounds can be borrowed at an interest rate of 4% and euros can be invested at an interest rate of 6%.

Now:

- Borrow £10,000;
- Buy €10,000;
- Put €10,000 in a bank.

After a year:

- Collect invested euro: €10,600;
- Sell collected euro for £10441;
- Pay the loan -£10,400;
- Arbitrage profit: £41. $\diamond$

Axiom of no arbitrage. *There is no portfolio with initial value $V(0) = 0$, all future values $V(t) \geq 0$ non-negative, and such that for some $t > 0$ we have $V(t) > 0$ with positive probability.*

We shall always assume that the axiom of no arbitrage holds for our models. In practice, however, arbitrage opportunities may occur.

2.12 Short selling

The portfolio vector $(x_1, \dots, x_m, y)$ can have negative coordinates. If, say $x_1 < 0$, then we say that the first asset, for example shares of stock, is sold *short*. In practice this means that the investor has to borrow shares of stock and sell them. Normally, short selling is not available to individual investors and in the cases where it is possible it comes with additional costs. In most of our simplified models we will allow short selling. Usually we will mention that a short selling occurs.

The negative coefficient y in the money account means that an investor borrows y unit of money or a risk-free asset.

2.13 Forward contract

An agreement to buy or sell a risky asset at a specified future time T (*delivery date*) for the price F agreed upon at present (*forward price*) is called a *forward contract*. The buyer of the asset enters a *long forward contract* and the seller enters a *short forward contract*.⁵

Example 2.14. The contract states that Alice will sell Bob 10t of manganese ore in six months time for the forward price of £20,000. Therefore, Alice holds the short forward position and Bob has the long forward position in this contract. The current price of manganese ore is £1.98/kg. Is the forward price reasonable? $\diamond$

Problem 2.15. Determine the forward price F so that there is no arbitrage?

We will see in the next section that the axiom of no arbitrage implies that there is a unique forward price.

2.16 Options

A contract giving its holder the *right* (but no obligation) to buy or sell a risky asset for the price X (called the *strike price*) at some future time is called an *option*. We have the following four types of options (these are examples of contingent claims):⁶

European Call	Right to buy an asset for the strike price X only at the <i>exercise</i> time T
European Put	Right to sell an asset for the strike price X only at the exercise time T
American Call	Right to buy an asset for the strike price X at any time $t \in [0, T]$; T is called the <i>expiry</i> time.
American Put	Right to sell an asset for the strike price X at any time $t \in [0, T]$; T is called the expiry time.

⁵The word ‘short’ is used because the seller is often not yet in possession of the goods and is therefore ‘short of the goods’ in question; ‘Long’ is simply the opposite of short.

⁶The terms ‘call’ and ‘put’ are to be understood as follows: If you have a call option, then you have the option to call upon the seller to deliver the asset; If you have a put option, then you have the option to put the asset to the seller.

Example 2.17. Alice offers for sale *the right to buy* her flat in one years time for the price of £100,000. The current value of the flat is £98,000. What price should she charge? Suppose Bob buys the option for £1,000. Is this a good price? After six months the house market is doing very well but Bob gets into financial trouble and he wants to sell the option to Carol for £1,500. Is this a good price? $\diamond$

Since an option gives an investor the right to buy an asset it has some positive value which depends on the actual value of the asset and the prediction for its price in the future.

Problem 2.18. Determine the value $C(t)$ of an option at time $t \in [0, T]$ such that the arbitrage is impossible? In particular, what is the initial price $C(0)$ of an option?

Let's look closer at a European call option. The holder of an option has the right to buy an asset for the price X at time T . If the market price $S(T)$ of the asset is smaller than X , the holder will not exercise the option and can buy the asset cheaper on the market. However, if the market price is higher than the strike price X , the holder will exercise the option and buy it for X . He can immediately sell the asset for $S(T)$ and earn $S(T) - X$ in total.

The random variable

$$\max\{S(T) - X, 0\}$$

is called the **pay-off** of the European call option. Thus at the exercise time the holder of a call option gets either $S(T) - X$ if $S(T) > X$ (paid by the seller of the option) or nothing if $S(T) \leq X$.

There is another fundamental problem to solve. Namely, the seller of the option gets $C(0)$ at time $t = 0$ and is exposed to the risk of paying $S(T) - X$ at the exercise time. So they can invest $C(0)$ at time $t = 0$ with the hope that at time T the investment will allow them to cover the pay-off. This procedure is an example of **hedging**.

Problem 2.19. Find a hedging strategy to cover the pay-off of an option.

2.20 Options abstractly

Recall that the pay-off of the European call option is given by

$$\max\{S(T) - X, 0\}.$$

One may think of the option as a bet between the seller and the buyer. If the final price $S(T)$ of the asset is bigger than X , then the seller pays the buyer the difference $S(T) - X$. This idea evolved further and there are countless types of options which make markets more like a gambling industry. These more general financial instruments are called **contingent claims**. They can be described as follows.

Let $S_1, \dots, S_m$ be the price processes of m risky assets and let $f: \mathbf{R}^m \rightarrow \mathbf{R}$ be a function. A contingent claim with the pay-off function f , the underlying assets $S_1, \dots, S_m$ and exercise time T is a contract such that its holder gets the payment

$$f(S_1(T), \dots, S_m(T))$$

at time T . For example, the function $f: \mathbf{R} \rightarrow \mathbf{R}$ defined by

$$f(x) = \max\{x - X, 0\}$$

is the pay-off function of the European call option. Another example is

$$f(x) = \max\left\{\frac{S(t_1) + S(t_2) + \dots + S(t_n)}{n} - X, 0\right\},$$

which is the pay-off function of the **Asian call option**. Here $0 < t_1 < t_2 < \dots < t_n = T$ are the time instances.

Problem 2.21. *Determine the value of a contingent claim with a given pay-off function.*

2.22 Money market account

The most common risk-free securities we will consider are bank deposits and bonds. In the multiperiod models the value of a bank deposit is given by

$$A(t) = A(t-1)(1 + r_t),$$

where r_t is the interest rate over the period $(t - 1, t]$ for $t = 1, \dots, T$. Hence

$$A(t) = A(0) \prod_{i=1}^t (1 + r_i).$$

In particular, if the interest rate is constant over the whole time, we get

$$A(t) = A(0)(1 + r)^t.$$

This is called **periodic compounding**. For example, interest on mortgages usually compounds monthly and each time step represents a month.

In the case of continuous time and constant interest rate the value of a bank deposit is given by

$$A(t) = A(0) e^{rt},$$

which is called the **continuous compounding**. This is mathematically more convenient, but less important in practice.

A **bond** is a financial security which gives the holder a sequence of guaranteed future payments. The payments are known at present and that is why bonds are considered as risk-free assets.

The **present value** of a bond which promises the payment F , called the **face value**, at time $t = 1$ is given by $B(0) = (1 + r)^{-1}F$, where r is the interest rate over the period of time. Bonds, however, are freely traded and in practice they determine the interest rates by

$$r = \frac{F}{B(0)} - 1.$$

More generally, if $B(t)$ is the price of a bond at time $t \in \{0, 1, 2, \dots, T\}$, then

$$r_t = \frac{B(t) - B(t - 1)}{B(t - 1)}$$

is the interest rate for the period $(t - 1, t]$.

2.23 Summary of problems

1. Find portfolios which minimise the risk among those with the same expected return.
2. Determine the rational forward price.
3. Determine the value of a contingent claim with a given pay-off function.
4. Determine the hedging strategy for the option seller.

2.24 Exercises

2a. Suppose you had brought 1000€ to a bank using the European Central Bank's rates at the beginning of 2014. How much would the bank have returned to you at the end of 2020? Recall here the interest rates from Page 12.

Solution. Working with the standard two decimals of precision, we compute:

Year	r	$S(t)$ at beginning of year t
2014	-0.1%	1000.00€
2015	-0.3%	999.00€
2016	-0.4%	996.00€
2017	-0.4%	992.02€
2018	-0.4%	988.05€
2019	-0.5%	984.10€
2020	-0.5%	979.18€
		974.28€

2b. Give as many examples of both risky and risk-free assets as you can come up with.

2c. Give an example where, after T time steps, periodic compounding and continuous compounding with the same rate r yield different results. Show that the only instances where they give the same result are the trivial cases $T = 0$ or $r = 0$.

Solution. Suppose that $e^{rT} = (1 + r)^T$ for some time T and interest rate r . Setting $x = rT$, we can rewrite this as $e^x = (1 + \frac{x}{T})^T$. Taking the derivative of both sides gives $e^x = T(1 + \frac{x}{T})^{T-1} \frac{1}{T} = (1 + \frac{x}{T})^{T-1}$ by the chain rule. Repeating this T times gives $e^x = (1 + \frac{x}{T})^0 = 1$, which implies $0 = x = rT$.

2d. Let the return of a portfolio be given by $K(u) = 3\%$, $K(d) = -1\%$ and the probabilities by $\mathbf{P}(u) = \mathbf{P}(d) = 1/2$. Compute the expected return, the variance and the standard deviation for K and for $2K$.

Solution.

$$\begin{aligned}\mathbf{E}(K) &= (3\%, -1\%) \cdot (1/2, 1/2) = 1\% \\ \text{var}(K) &= \mathbf{E}((K - \mathbf{E}(K))^2) = (2\%)^2 \\ \sigma_K &= 2\%\end{aligned}$$

If we consider the double return $2K(u) = 6\%$ and $2K(d) = -2\%$ then

$$\begin{aligned}\mathbf{E}(2K) &= (6\%, -2\%) \cdot (1/2, 1/2) = 2\% \\ \text{var}(2K) &= \mathbf{E}((2K - \mathbf{E}(2K))^2) = 4(2\%)^2 \\ \sigma_{2K} &= 4\%\end{aligned}$$

2e. Consider the following situation:

scenario	probability	return K_1	return K_2
ω_1	$1/2$	10%	-5%
ω_2	$1/2$	-5%	10%

Compute the expected return, variance and risk for K_1 , K_2 and for their sum $K_1 + K_2$. Then provide a general formulas for the combination $aK_1 + bK_2$.

Solution.

$$\begin{aligned}\mathbf{E}(K_1) &= 2.5\% \\ \text{var}(K_1) &= (7.5\%)^2 \\ \mathbf{E}(K_2) &= 2.5\% \\ \text{var}(K_2) &= (7.5\%)^2 \\ \mathbf{E}(K_1 + K_2) &= 5\% \\ \text{var}(K_1 + K_2) &= 0 \\ \mathbf{E}(aK_1 + bK_2) &= (a + b) 2.5\% \\ \text{var}(aK_1 + bK_2) &= (a - b)^2 (7.5\%)^2\end{aligned}$$

2f. Prove Lemma 2.7.

Solution.

$$\begin{aligned}V_{(x,y)}(t) + V_{(x',y')}(t) &= yA(t) + \sum_{i=1}^m x_i S_i(t) + y'A(t) + \sum_{i=1}^m x'_i S_i(t) \\ &= (y + y')A(t) + \sum_{i=1}^m (x_i + x'_i) S_i(t) \\ &= V_{(x+x', y+y')}(t) = V_{(x,y)+(x',y')}(t).\end{aligned}$$

2g. Let $A(0) = 90$, $A(T) = 100$, $S(0) = 25$ and let

$$S(T) = \begin{cases} 30 & \text{with probability } p \\ 20 & \text{with probability } 1 - p. \end{cases}$$

1. Calculate the values $V(0)$, $V(T)$ and K_V for the portfolio $x = 10$ shares and $y = 15$ bonds. Compute the weights of this portfolio.
2. Find a portfolio whose value at time $t = T$ is

$$V(T) = \begin{cases} 1,160 & \text{if stock goes up} \\ 1,040 & \text{if stock goes down} . \end{cases}$$

What is the value of this portfolio at time $t = 0$?

Solution. For 1., we compute

$$V(0) = yA(0) + xS(0) = 15 \cdot 90 + 10 \cdot 25 = 1600$$

$$V(T) = yA(T) + xS(T) = \begin{cases} 15 \cdot 90 + 10 \cdot 30 = 1650 & \text{if stock goes up} \\ 15 \cdot 90 + 10 \cdot 20 = 1550 & \text{if stock goes down} \end{cases}$$

$$K_V(T) = \frac{V(T) - V(0)}{V(0)} = \begin{cases} \frac{1650-1600}{1600} = 3.125\% & \text{if stock goes up} \\ \frac{1550-1600}{1600} = -3.125\% & \text{if stock goes down} \end{cases}$$

The weights at $t = 0$ are (the formula above for w_0 was only for $A(0) = 1$)

$$w_0 = \frac{yA(0)}{V(0)} = \frac{15 \cdot 90}{1600} = 27/32 = 84.375\%$$

$$w_1 = \frac{xS(0)}{V(0)} = \frac{10 \cdot 25}{1600} = 5/32 = 15.625\%$$

For 2., we need to find x, y with

$$V(T) = yA(T) + xS(T) = \begin{cases} y \cdot 100 + x \cdot 30 & \text{if stock goes up} \\ y \cdot 100 + x \cdot 20 & \text{if stock goes down} \end{cases}$$

as given. This leads to a pair of linear equations

$$y \cdot 100 + x \cdot 30 = 1160$$

$$y \cdot 100 + x \cdot 20 = 1040$$

Subtracting the second equation from the first gives $x \cdot 10 = 120$, so $x = 12$. Putting this x into the second equation gives $y \cdot 100 + 240 = 1040$, so $y = 9$. We can use this to compute the value of the portfolio at $t = 0$ as

$$V(0) = xA(0) + yS(0) = 12 \cdot 90 + 9 \cdot 25 = 1305.$$

2h. Spot a chance for a profit without initial investment:

Dealer A	buy	sell	Dealer B	buy	sell
€ 1.000	\$ 1.0202	\$ 1.0284	€ 1.000	£ 0.6324	£ 0.6401
£ 1.000	\$ 1.5718	\$ 1.5844	\$ 1.000	£ 0.6299	£ 0.6375

Solution. Borrow \$ 10.000. Sell these to Dealer A and receive 6311.54 £. Then sell your pounds to Dealer B and receive 9860.24 €. Go back to Dealer A and trade your euros for \$ 10059.40. Very nice!

2i. Exercises with an LLM.

1. Request a review of this section and ask for examples and explanations of the notions which have been introduced. Arbitrage is especially important.
2. In our models we assume that arbitrage cannot occur. Ask the LLM what is the situation in real markets. Ask for historical examples of arbitrage.
3. Ask the LLM for an overview about options and for examples of *exotic* ones. Notice that what we are learning in this course is the tip of an iceberg.
4. Let $D_1 = 0.98$, $D_2 = 0.99$, $D_3 = 0.97$, $U_1 = 1.02$, $U_2 = 1.03$, $U_3 = 1.01$ and $p = 0.5$. Consider three risky assets S_1 , S_2 and S_3 each following the binomial model with the corresponding U_i , D_i and p as defined above. Let $S_1(0) = 100$, $S_2(0) = 50$, $S_3(0) = 120$ and let A denoted a risk-free asset with $A(0) = 1$ and $R = 1.02$. Let (x_1, x_2, x_3, y) denote a portfolio. Choose your values of x_i 's and y , for example, start with $(1, 1, 1, -270)$. Ask the LLM to compute the values, returns and expected returns of the portfolio at a few instances of time. Try to verify whether the numbers provided by the LLM are correct.
5. In the above market with one time-step, ask the LLM to create a portfolio which minimises the risk. What is the expected return of that portfolio? Then ask for a portfolio with minimal risk and with expected return 2%. This is a preparation for the next two sections.
6. Create a few examples of options based on the market from the previous item. Explain them to the LLM and ask to compute their pay-offs.

3 Portfolio management I

In this section we consider a single period market model with unspecified probability space and two risky securities. The price processes of these risky securities are denoted by S_1 and S_2 respectively. Their returns are denoted by K_1 and K_2 respectively and the return of the portfolio consisting of these two assets by K_V . The purpose is to find portfolios which minimise risk.

3.1 The risk of a portfolio

Consider a portfolio (x_1, x_2) of two risky assets whose prices at time $t = 0$ are $S_1(0)$ and $S_2(0)$. The initial value of the portfolio is $V(0) = x_1 S_1(0) + x_2 S_2(0)$. Recall that the weights at $t = 0$ are defined as

$$w_1 = \frac{x_1 S_1(0)}{V(0)}, \quad w_2 = \frac{x_2 S_2(0)}{V(0)}, \quad w_1 + w_2 = 1.$$

By (2.4), the return of this portfolio is $K_V = w_1 K_1 + w_2 K_2$.

Example 3.2 (Weights change in time). Let $S_1(0) = 30$, $S_2(0) = 40$ and $(x_1, x_2) = (20, 10)$. Then $V(0) = 1000$ and $w_1 = 60\%$ and $w_2 = 40\%$. Suppose that at time $t = T$ we have $S_1(T) = 35$ and $S_2(T) = 39$. Then $V(T) = 1090$ and $w_1 \approx 64.22\%$ and $w_2 \approx 35.78\%$. $\diamond$

Notation:

$$\begin{array}{lll} \mu_V = \mathbf{E}(K_V) & \sigma_V = \sqrt{\text{var}(K_V)} & c_{12} = \text{cov}(K_1, K_2) \\ \mu_1 = \mathbf{E}(K_1) & \sigma_1 = \sqrt{\text{var}(K_1)} & \rho_{12} = \text{cor}(K_1, K_2) \\ \mu_2 = \mathbf{E}(K_2) & \sigma_2 = \sqrt{\text{var}(K_2)} & \end{array}$$

Using this notation we have

$$\mu_V = w_1 \mu_1 + w_2 \mu_2 \tag{3.1}$$

$$\sigma_V^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 c_{12} \tag{3.2}$$

$$= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12} \tag{3.3}$$

Proposition 3.3. *If $w_1, w_2 \geq 0$ (no short selling is allowed), then*

$$\sigma_V \leq \max\{\sigma_1, \sigma_2\}.$$

Proof. Without loss of generality, we may assume $\sigma_1 \leq \sigma_2$, for otherwise we exchange the labels ‘1’ and ‘2’. Then $w_1\sigma_1 + w_2\sigma_2 \leq (w_1 + w_2)\sigma_2 = \sigma_2$ and

$$\begin{aligned}\sigma_V^2 &= w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2 \\ &\leq (w_1\sigma_1 + w_2\sigma_2)^2 \leq \sigma_2^2\end{aligned}\quad \text{since } \rho_{12} \leq 1. \quad \square$$

Example 3.4. Consider the following situation:

scenario	probability	return K_1	return K_2
ω_1	0.2	−10%	5%
ω_2	0.4	0%	30%
ω_3	0.4	20%	−5%

1. If $w_1 = 40\%$ and $w_2 = 60\%$ then $\sigma_1^2 = 0.0144$, $\sigma_2^2 = 0.0254$,
 $\rho_{12} = -0.6065$ and $\sigma_V^2 = 0.00588$; $\implies \sigma_V^2 < \sigma_1^2 < \sigma_2^2$
2. If $w_1 = 10\%$ and $w_2 = 90\%$ then $\sigma_V^2 = 0.01863$; $\implies \sigma_1^2 < \sigma_V^2 < \sigma_2^2$
3. If $w_1 = -50\%$ and $w_2 = 150\%$ then $\sigma_V^2 = 0.27958$; $\implies$
 $\sigma_1^2 < \sigma_2^2 < \sigma_V^2$

◇

3.5 Minimising the risk

Consider a portfolio (x_1, x_2) with two risky securities. We are interested in the relation between the expected return μ_V and the risk σ_V . More precisely, we want to understand the following subset of the plane

$$\{(\sigma_V, \mu_V) \in \mathbf{R}^2 \mid V \text{ is the value of } (x_1, x_2)\}.$$

In the notation introduced above we have

$$\begin{aligned}\mu_V &= w_1\mu_1 + w_2\mu_2 \\ \sigma_V^2 &= w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2c_{12} \\ 1 &= w_1 + w_2.\end{aligned}$$

Letting $s = w_1$, we have $w_2 = 1 - s$ and therefore

$$\mu_V(s) = s\mu_1 + (1 - s)\mu_2 = s(\mu_1 - \mu_2) + \mu_2, \quad (3.4)$$

$$\begin{aligned} \sigma_V^2(s) &= s^2\sigma_1^2 + (1 - s)^2\sigma_2^2 + 2s(1 - s)c_{12} \\ &= s^2(\sigma_1^2 + \sigma_2^2 - 2c_{12}) + 2s(c_{12} - \sigma_2^2) + \sigma_2^2. \end{aligned} \quad (3.5)$$

The equation for σ_V is that of an hyperbola and the following results can also be deduced from the geometry of hyperbolas, see Section 3.11.

Proposition 3.6. *If $\rho_{12} < 1$ or $\sigma_1 \neq \sigma_2$ then the function σ_V^2 attains its minimum at $s_0 = \frac{\sigma_2^2 - c_{12}}{\sigma_1^2 + \sigma_2^2 - 2c_{12}}$. Moreover,*

$$\begin{aligned} \mu_0 = \mu_V(s_0) &= \frac{\mu_1\sigma_2^2 + \mu_2\sigma_1^2 - (\mu_1 + \mu_2)c_{12}}{\sigma_1^2 + \sigma_2^2 - 2c_{12}} \\ \sigma_0^2 = \sigma_V(s_0)^2 &= \frac{\sigma_1^2\sigma_2^2 - c_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2c_{12}} \end{aligned}$$

Proof. Compute the first derivative of $\sigma_V^2(s)$ and find s_0 for which this derivative is equal to zero. Compute the second derivative and observe that it is positive, hence σ_V attains its minimum at s_0 . $\square$

Proposition 3.7. *If $\rho_{12}^2 < 1$ and $\mu_1 \neq \mu_2$ then $x = \sigma_V$ and $y = \mu_V$ satisfy the equation*

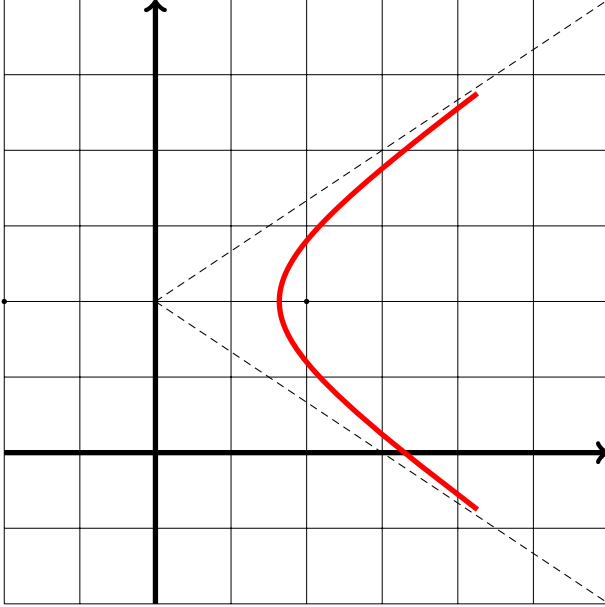
$$x^2 - \alpha(y - \mu_0)^2 = \sigma_0^2,$$

where μ_0 and $\sigma_0^2 > 0$ are as above and $\alpha = \frac{\sigma_1^2 + \sigma_2^2 - 2c_{12}}{(\mu_1 - \mu_2)^2} > 0$. This implies that the set $\{(\sigma_V, \mu_V)\}$ is the hyperbola with asymptotes given by $y = \mu_0 \pm \frac{1}{\sqrt{\alpha}}x$.

Proof. The proof start with the substitutions:

$$\begin{aligned} \sigma_V &\mapsto x \\ \mu_V &\mapsto y \\ \frac{\sigma_1^2 + \sigma_2^2 - 2c_{12}}{(\mu_1 - \mu_2)^2} &\mapsto \alpha \\ \frac{\mu_1\sigma_2^2 + \mu_2\sigma_1^2 - (\mu_1 + \mu_2)c_{12}}{\sigma_1^2 + \sigma_2^2 - 2c_{12}} &\mapsto \mu_0 \\ \frac{\sigma_1^2\sigma_2^2 - c_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2c_{12}} &\mapsto \sigma_V(s_0)^2 \end{aligned}$$

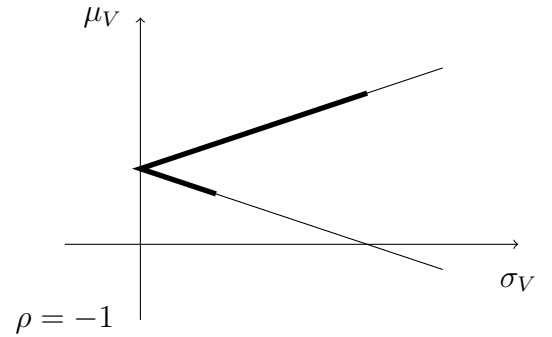
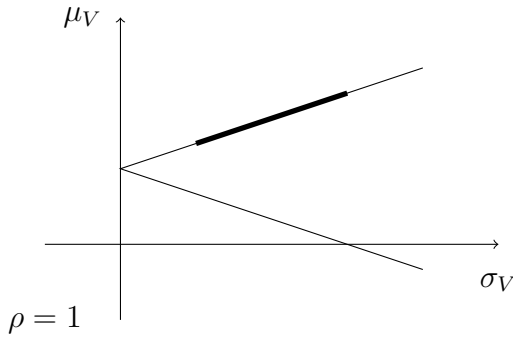
into the equality $x^2 - \alpha(y - \mu_0)^2 = \sigma_0^2$ and verifying that it holds. It is left as an exercise. $\square$



Proposition 3.8. *If $\rho_{12} = 1$ and $\sigma_1 \neq \sigma_2$ then $\sigma_V = 0$ if and only if $w_1 = \frac{-\sigma_2}{\sigma_1 - \sigma_2}$ and $w_2 = \frac{\sigma_1}{\sigma_1 - \sigma_2}$.*

Proposition 3.9. *If $\rho_{12} = -1$ then $\sigma_V = 0$ if and only if $w_1 = \frac{\sigma_2}{\sigma_1 + \sigma_2}$ and $w_2 = \frac{\sigma_1}{\sigma_1 + \sigma_2}$.*

The proofs of both propositions are left as an exercise. The figures below present the curve $(\sigma_V(s), \mu_V(s))$ with thick parts corresponding to $s \in [0, 1]$, that is, portfolios with no short selling.



Corollary 3.10. *This is a summary of the situation. Suppose that $\sigma_1 \leq \sigma_2$. Then*

$$\begin{aligned}
\rho_{12} = 1 &\implies \exists \sigma_V = 0 && \text{with shortselling} \\
\frac{\sigma_1}{\sigma_2} < \rho_{12} < 1 &\implies \exists \sigma_V < \sigma_1 && \text{with shortselling} \\
\rho_{12} = \frac{\sigma_1}{\sigma_2} &\implies \forall \sigma_V \geq \sigma_1 \\
-1 < \rho_{12} < \frac{\sigma_1}{\sigma_2} &\implies \exists \sigma_V < \sigma_1 \\
\rho_{12} = -1 &\implies \exists \sigma_V = 0
\end{aligned}$$

□

3.11 Geometric interpretation

Definition 3.12. The *hyperbola*⁷ with center (x_0, y_0) , semiminor axis $a > 0$, and semimajor axis $b > 0$ is the set of solutions $(x, y) \in \mathbb{R}^2$ to the equation

$$\left(\frac{y - y_0}{b}\right)^2 - \left(\frac{x - x_0}{a}\right)^2 = 1. \quad (3.6)$$

(See Figure 3.1.)

Suppose for simplicity that $y_0 = 0$. Then we can rearrange (3.6) as

$$y^2 = Ax^2 - 2Bx + C, \quad (3.7)$$

where $A = \frac{b^2}{a^2}$, $B = \frac{b^2 x_0}{a^2}$, and $C = \frac{b^2 x_0^2}{a^2} + b^2$. Conversely, starting with (3.7) and $A > 0$, $AC - B^2 > 0$ we can recover the parameters of the hyperbola as

$$a = \frac{\sqrt{AC - B^2}}{A}, \quad b = \frac{\sqrt{AC - B^2}}{\sqrt{A}}, \quad x_0 = \frac{B}{A}. \quad (3.8)$$

(if $A = 0$, then (3.7) is a parabola; if $AC - B^2 = 0$, the hyperbola degenerates into a pair of asymptotes) The asymptotes have slopes $\pm b/a = \pm \sqrt{A}$.

Returning to the portfolio, set $x = s$ and $y = \sigma_V$ and notice that (3.5) is an equation of the form (3.7) with $A = \sigma_1^2 + \sigma_2^2 - 2c_{12}$, $B = \sigma_2^2 - c_{12}$, and $C = \sigma_2^2$. Putting this into (3.8) yields x_0 , a , b and gives the following result.

⁷Hyperbola is derived from the Greek ‘ $\upsilon\pi\epsilon\rho\beta\omicron\lambda\eta$ ’, meaning ‘open at the top’.

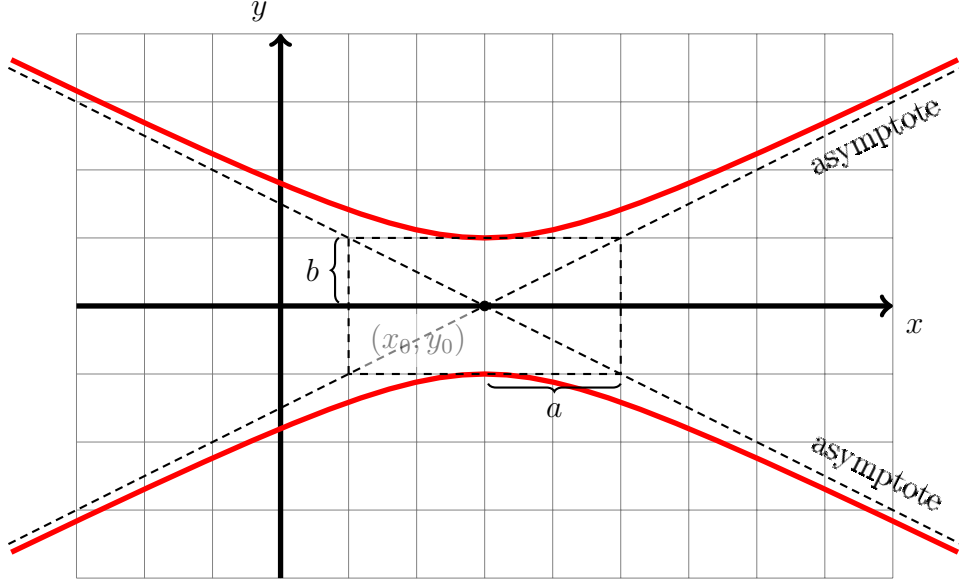


Figure 3.1: A hyperbola with the x -axis as the minor axis and y -axis as the major axis. The **upper branch** (where $y \geq y_0$) and the **lower branch** (where $y \leq y_0$) of the hyperbola are functions of x .

Proposition 3.13. *If $-1 < \rho_{12} < 1$, then the risk $\sigma_V(x)$ of the portfolio with weights $(w_1, w_2) = (x, 1-x)$ as in (3.5) describes a hyperbola with center $(x_0, 0)$, semiminor axis a , and semimajor axis b given by*

$$x_0 = \frac{\sigma_2^2 - c_{12}}{\sigma_1^2 + \sigma_2^2 - 2c_{12}}, \quad a = \frac{\sqrt{\sigma_1^2 \sigma_2^2 - c_{12}^2}}{\sigma_1^2 + \sigma_2^2 - 2c_{12}}, \quad b = \frac{\sqrt{\sigma_1^2 \sigma_2^2 - c_{12}^2}}{\sqrt{\sigma_1^2 + \sigma_2^2 - 2c_{12}}}.$$

In particular, the variance of every portfolio made only of the risky assets S_1 and S_2 satisfies $\sigma_V^2(x) \geq b^2 = \frac{\sigma_1^2 \sigma_2^2 - c_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2c_{12}}$ (see Figure 3.1). The portfolio with the smallest possible variance b^2 has weights

$$w_1 = x_0 = \frac{\sigma_2^2 - c_{12}}{\sigma_1^2 + \sigma_2^2 - 2c_{12}}, \quad w_2 = 1 - w_1 = \frac{\sigma_1^2 - c_{12}}{\sigma_1^2 + \sigma_2^2 - 2c_{12}}$$

and an expected return of

$$\mu_0 = x_0 \mu_1 + (1 - x_0) \mu_2 = \frac{\mu_1 \sigma_2^2 + \mu_2 \sigma_1^2 - (\mu_1 + \mu_2) c_{12}}{\sigma_1^2 + \sigma_2^2 - 2c_{12}}. \quad (3.9)$$

We can use (3.4) to describe a portfolio in terms of its return and substitute $x = \frac{\mu_V - \mu_2}{\mu_1 - \mu_2}$ into (3.5) and get

$$\sigma_V^2 = \underbrace{\frac{\sigma_1^2 + \sigma_2^2 - 2c_{12}}{(\mu_1 - \mu_2)^2}}_{A=} \mu_V^2 - 2 \underbrace{\frac{\mu_1 \sigma_2^2 + \mu_2 \sigma_1^2 - c_{12}(\mu_1 + \mu_2)}{(\mu_1 - \mu_2)^2}}_{B=} \mu_V + \underbrace{\frac{\mu_1^2 \sigma_2^2 + \mu_2^2 \sigma_1^2 - 2c_{12}\mu_1\mu_2}{(\mu_1 - \mu_2)^2}}_{C=}$$

This is again a hyperbola with A, B, C as indicated, see Figure 3.2. The expressions for $a = \frac{\sqrt{AC-B^2}}{A}$ and $b = \frac{\sqrt{AC-B^2}}{\sqrt{A}}$ are a little complicated; Of course, the hyperbola is now centered at $(\mu_0, 0)$, where μ_0 is given by (3.9). The slopes of the asymptotes are

$$b/a = \pm \sqrt{A} = \frac{\pm \sqrt{\sigma_1^2 + \sigma_2^2 - 2c_{12}}}{\mu_1 - \mu_2}.$$

This reproves Proposition 3.7 (with the roles of σ_V and μ_V exchanged).

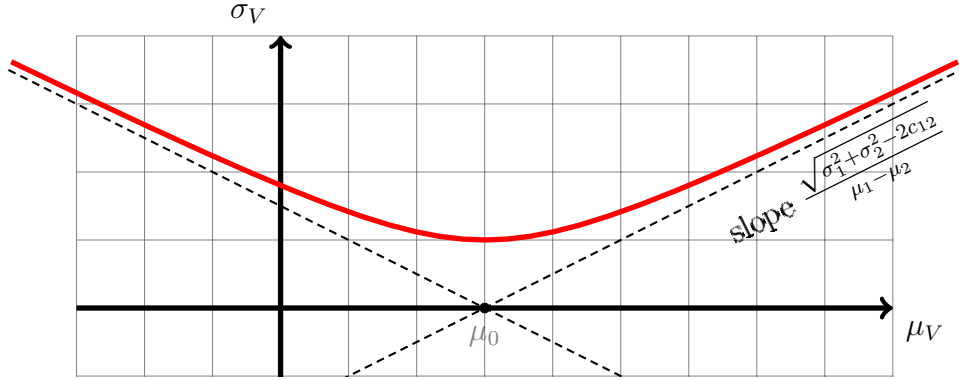


Figure 3.2: The variance σ_V of the return of a portfolio as a function of the expected return μ_V (lower branch omitted, as $\sigma_V \geq 0$). No risk, no fun!

Remark 3.14. If $\rho_{12} = \mp 1$, the hyperbola degenerates into a pair of asymptotes centered at $(\mu_0, 0) = (\frac{\mu_1 \sigma_2 - \rho_{12} \mu_2 \sigma_1}{\sigma_1 - \rho_{12} \sigma_2}, 0)$ of slopes $\pm \frac{\sigma_1 - \rho_{12} \sigma_2}{\mu_1 - \mu_2}$. This is the only case where the equation has a solution on the $\sigma_V = 0$ axis and there exists a zero variance portfolio; it corresponds to the nonsensical combination of an asset and its opposite, for example by short selling. In practice, two assets are never entirely correlated, hence $\rho_{12} \neq \pm 1$.

3.15 Exercises

3a. Verify computations and fill the details in all examples in this section.

3b. Complete the details of Proposition 3.6 and of Proposition 3.7.

3c. Prove Propositions 3.8 and 3.9.

3d. Prove and contemplate Corollary 3.10. In particular, observe that if the two returns are *negatively* correlated then there is a portfolio minimising the risk with no short selling (the fourth case).

3e. Suppose that there are two scenarios (that is, the probability space has two elements) and consider two risky securities with returns K_1 and K_2 . Show that $K_1 = aK_2 + b$ for some numbers a, b and deduce that $\rho_{12} = \pm 1$.

3f. Consider the following situation:

Scenario	Probability	Return K_1	Return K_2
ω_1	0.2	-10%	5%
ω_2	0.4	0%	30%
ω_3	0.4	20%	-5%

(a) What is the sample space for the above situation?

(b) Compute the expected returns $\mathbf{E}(K_1)$ and $\mathbf{E}(K_2)$.

(c) Compute the covariance matrix $[\text{cov}(K_i, K_j)]$ and its determinant.

(d) Find the return of the portfolio with weights $w_1 = 60\%$, $w_2 = 40\%$. Then compute its expected return and the variance.

(e) Find a portfolio with minimal variance.

3g. Let K_1, K_2 be random variables. What is the relationship between the determinant of the covariance matrix and the correlation between K_1 and K_2 ?

3h. Exercises with an LLM.

1. Review the content of this section with the LLM.

2. In this section we define the *risk* as the square root of the variance of the return K_V of a portfolio. It is also called the standard deviation. Ask the LLM for more intuition about the variance or standard deviation of a random variable.
3. Try to solve the above exercises. If you get stuck ask the LLM for an answer. Then try again, knowing what the answer is. If you get stuck ask the LLM for a solution and for further explanations if necessary.
4. Ask the LLM for possible exam questions relevant to this section and try to solve them. First without then with a help from the LLM.

4 Portfolio management II

Let $S_1, \dots, S_n$ denote the price of n risky securities. Consider the set of portfolios with nonzero initial value $\{(x_1, \dots, x_n) \in \mathbf{R}^n \mid V(0) = \sum x_i S_i(0) \neq 0\}$. Let $w_i = \frac{x_i S_i(0)}{V(0)}$ denote the weights of a portfolio $x = (x_1, \dots, x_n)$ in the above set and let

$$\mathbf{A}(S_1, \dots, S_n) = \{(w_1, \dots, w_n) \in \mathbf{R}^n \mid \sum w_i = 1 = \mathbf{w}\mathbf{u}^T\}$$

be the *admissible set*, where

$$\mathbf{w} = \begin{pmatrix} w_1 & w_2 & \cdots & w_n \end{pmatrix}$$

$$\mathbf{u} = \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix}$$

are thought of as one-row matrices.

Let $K_1, \dots, K_n$ be the returns and $\mu_i = \mathbf{E}(K_i)$ their expected values. We also arrange them into a one-row matrix $\mathbf{m} = (\mu_1 \ \mu_2 \ \cdots \ \mu_n)$. Finally, let $c_{ij} = \text{cov}(K_i, K_j)$ be the covariances. They form a *covariance matrix*:

$$\mathbf{C} = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{pmatrix}$$

which is symmetric and nonnegative definite. In what follows, we assume that the covariance matrix is invertible.

Assumption: $\det \mathbf{C} \neq 0$.

Proposition 4.1. *The expected return $\mu_V = \mathbf{E}(K_V)$ and the variance $\text{var}(K_V) = \sigma_V^2$ of the return $K_V = \sum w_i K_i$ of the portfolio with weights $w_1, \dots, w_n$ is given by:*

$$\mu_V = \mathbf{w}\mathbf{m}^T$$

$$\sigma_V^2 = \mathbf{w}\mathbf{C}\mathbf{w}^T.$$

Proof. It is an application of the linearity of expectation and bilinearity of the covariance:

$$\mu_V = \mathbf{E}(K_V) = \mathbf{E}\left(\sum w_i K_i\right) = \sum w_i \mathbf{E}(K_i) = \sum w_i \mu_i = \mathbf{w} \mathbf{m}^T.$$

$$\begin{aligned} \sigma_V^2 &= \text{var}(K_V) = \text{var}\left(\sum \mu_i K_i\right) \\ &= \text{cov}\left(\sum \mu_i K_i, \sum \mu_j K_j\right) \\ &= \sum_{i,j} w_i w_j c_{ij} = \mathbf{w} \mathbf{C} \mathbf{w}^T \end{aligned}$$

□

Aim: To find a portfolio with weights $\mathbf{w}$ with minimal variance. Its weights will be denoted by $\mathbf{w}_{\text{MVP}}$.

Proposition 4.2. *If $\det \mathbf{C} \neq 0$ then $\mathbf{w}_{\text{MVP}} = \frac{\mathbf{u} \mathbf{C}^{-1}}{\mathbf{u} \mathbf{C}^{-1} \mathbf{u}^T}$.*

We present two proofs of this proposition.

A linear algebra proof. **Existence.** Let $W \cong \mathbf{R}^n$ be the space of weights. Since $\mathbf{C}$ is a symmetric and positive definite, it defines an inner product on W by $\langle x, y \rangle_{\mathbf{C}} = x \mathbf{C} y^T$. Let $V = \{x \in W \mid x \mathbf{u}^T = 0\}$. Then the space of admissible weights $\mathbf{A} = \{x \in \mathbf{R}^n \mid x \mathbf{u}^T = 1\}$ is an affine subspace of the form $V + \mathbf{a}$, for some (any) $\mathbf{a} \in \mathbf{A}$. It follows from Proposition 4.18 in the Appendix at the end of this section that there exists a unique $\mathbf{w}_{\text{MVP}} \in \mathbf{A}$ with minimal variance $\mathbf{w}_{\text{MVP}} \mathbf{C} \mathbf{w}_{\text{MVP}}^T$.

The formula. It follows from the proof of Proposition 4.18 that $\mathbf{w}_{\text{MVP}}$ has the form $\mathbf{w}_{\mathbf{a}}^\perp \in V^\perp$. Observe that $\dim V^\perp = 1$ and it follows from the definition of V that

$$\begin{aligned} x \mathbf{u}^T &= 0 \\ x \mathbf{C} (\mathbf{C}^{-1} \mathbf{u}^T) &= 0 \end{aligned}$$

for every $x \in V$. This means that $\mathbf{u} \mathbf{C}^{-1} \in V^\perp$. Thus $\mathbf{w}_{\text{MVP}} = \lambda \mathbf{u} \mathbf{C}^{-1}$ for some $\lambda \in \mathbf{R}$. Since $\mathbf{w}_{\text{MVP}} \in \mathbf{A}$ we get that $\lambda \mathbf{u} \mathbf{C}^{-1} \mathbf{u}^T = 1$, which implies that

$$\mathbf{w}_{\text{MVP}} = \frac{\mathbf{u} \mathbf{C}^{-1}}{\mathbf{u} \mathbf{C}^{-1} \mathbf{u}^T}$$

as claimed. □

Lagrange multipliers proof. In this proof the existence is proven as before using inner products.

Let $f, g: \mathbf{R}^n \rightarrow \mathbf{R}$ be defined by $f(\mathbf{w}) = \mathbf{wCw}^T$ and $g(\mathbf{w}) = \mathbf{wu}^T$ respectively. The problem is to find a minimum of f in the set $g^{-1}(\{1\}) = \{\mathbf{w} \in \mathbf{R}^n \mid g(\mathbf{w}) = \mathbf{wu}^T = 1\}$. It follows from Lagrange's theorem that if x is such a minimum then there exist a number $\lambda \in \mathbf{R}$ (the Lagrange multiplier) such that $\nabla f(x) = \lambda \nabla g(x)$. This is equivalent to the following n equations:

$$\frac{\partial}{\partial x_i} f(x) = \frac{\partial}{\partial x_i} (\lambda g(x)),$$

for $i = 1, \dots, n$. In our case this gives $2\mathbf{wC} = \lambda \mathbf{u}$. That is, if $\mathbf{w}$ is a minimum of the function f then $\mathbf{w} = \frac{\lambda}{2} \mathbf{uC}^{-1}$. We require this minimum to satisfy $\mathbf{wu}^T = 1$, so by multiplying the previous equality by $\mathbf{u}^T$ we get that

$$1 = \frac{\lambda}{2} \mathbf{uC}^{-1} \mathbf{u}^T$$

$$\lambda = \frac{2}{\mathbf{uC}^{-1} \mathbf{u}^T}.$$

Substituting this value of λ we obtain $\mathbf{w} = \frac{\mathbf{uC}^{-1}}{\mathbf{uC}^{-1} \mathbf{u}^T}$. We conclude that if there is a minimum admissible portfolio then it is given by the last formula. $\square$

Definition 4.3. We say that a portfolio with (σ_1, μ_1) *dominates* a portfolio with (σ_2, μ_2) if $\mu_1 \geq \mu_2$ and $\sigma_1 \leq \sigma_2$. This defines a partial order on the set of portfolios.

A portfolio is called *efficient* if it is not dominated by another portfolio. That is, it is a maximal element with respect to the above partial order. The subset of efficient portfolios in the admissible set is called the *efficient frontier*.

Aim: To find the efficient portfolio among those with fixed return μ .

That is, we are looking for $\mathbf{w}$ satisfying $\mathbf{wu}^T = 1$ and $\mathbf{wm}^T = \mu$ with the minimal value $\mathbf{wCw}^T$. As before, we can give a metric argument that such a minimum $\mathbf{w}_0$ exists and we apply the Lagrange theorem to find its value. Let $f, g_1, g_2: \mathbf{R}^n \rightarrow \mathbf{R}$ be given by $f(x) = x\mathbf{C}x^T$, $g_1(x) = x\mathbf{m}^T$ and $g_2(x) = xu^T$. The Lagrange theorem implies that there exist numbers $\lambda_1, \lambda_2 \in \mathbf{R}$ such that

$$\nabla f(\mathbf{w}_0) = \lambda_1 \nabla g_1(\mathbf{w}_0) + \lambda_2 \nabla g_2(\mathbf{w}_0)$$

$$2\mathbf{wC} = \lambda_1 \mathbf{m} + \lambda_2 \mathbf{u}.$$

We thus have to solve the following system of linear equations:

$$\begin{aligned} 2\mathbf{w} - \lambda_1 \mathbf{m} \mathbf{C}^{-1} - \lambda_2 \mathbf{u} \mathbf{C}^{-1} &= 0 \\ \mathbf{w} \mathbf{m}^T - \mu &= 0 \\ \mathbf{w} \mathbf{u}^T - 1 &= 0 \end{aligned}$$

Multiplying the first equation by $\mathbf{m}^T$ and $\mathbf{u}^T$ we get

$$\begin{aligned} \lambda_1 \mathbf{m} \mathbf{C}^{-1} \mathbf{m}^T + \lambda_2 \mathbf{u} \mathbf{C}^{-1} \mathbf{m}^T &= 2\mu \\ \lambda_1 \mathbf{m} \mathbf{C}^{-1} \mathbf{u}^T + \lambda_2 \mathbf{u} \mathbf{C}^{-1} \mathbf{u}^T &= 2 \end{aligned}$$

If the vectors $\mathbf{u}$ and $\mathbf{m}$ are linearly independent (that is, not all μ_i 's are equal) the matrix $M = \begin{pmatrix} \mathbf{m} \mathbf{C}^{-1} \mathbf{m}^T & \mathbf{u} \mathbf{C}^{-1} \mathbf{m}^T \\ \mathbf{m} \mathbf{C}^{-1} \mathbf{u}^T & \mathbf{u} \mathbf{C}^{-1} \mathbf{u}^T \end{pmatrix}$ is invertible and hence the solutions λ_1, λ_2 of the last system of equations are given by

$$\begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = 2M^{-1} \begin{pmatrix} \mu \\ 1 \end{pmatrix}.$$

Finally, we get that $\mathbf{w} = \mu \mathbf{a} + \mathbf{b}$ for some vectors $\mathbf{a}, \mathbf{b} \in \mathbf{R}^n$ which depend only on $\mathbf{m}, \mathbf{u}$ and $\mathbf{C}$. This proves the following result.

Theorem 4.4. *Assume that $\det \mathbf{C} \neq 0$ and that $\mathbf{m}$ and $\mathbf{u}$ are linearly independent. Then there exist vectors $\mathbf{a}, \mathbf{b} \in \mathbf{R}^n$ such that $\mathbf{w}$ represents a portfolio with minimum variance with expected return μ if and only if $\mathbf{w} = \mu \mathbf{a} + \mathbf{b}$.*

Definition 4.5. The line $\{\mathbf{w} \in \mathbf{R}^n \mid \mathbf{w} = \mu \mathbf{a} + \mathbf{b}\}$ is called the **minimum variance line**.

Corollary 4.6. *Assume that $\det \mathbf{C} \neq 0$ and that $\mathbf{m}$ and $\mathbf{u}$ are linearly independent. Let $\mathbf{w}_1, \mathbf{w}_2$ be the weights of two portfolios on the minimum variance line with different expected returns. Then each portfolio on the minimum variance line is of the form $\mathbf{w} = s\mathbf{w}_1 + (1-s)\mathbf{w}_2$ for some $s \in \mathbf{R}$. $\square$*

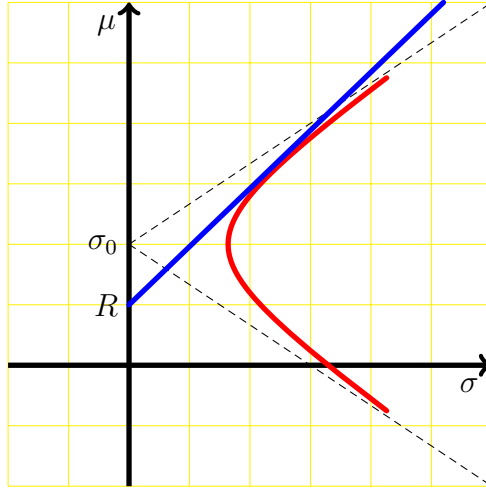
Corollary 4.7. *Consider the map $(\sigma, \mu): \mathbf{R}^n \rightarrow \mathbf{R}^2$ given by the risk and the return of a portfolio. The image of the minimum variance line is a hyperbola on the plane.*

Proof. Consider the weights $\mathbf{w}_1$ and $\mathbf{w}_2$ on the minimum variance line as two risky securities. Then the statement follows from Proposition 3.7. $\square$

The image of the minimum variance line under the risk-return map will also be called the minimum variance line.

Consider a portfolio formed from n risky securities $S_1, \dots, S_n$ and one risk-free security A with return R . Look at the figure below. The minimal variance line of all risky assets is the red hyperbola. There is a portfolio V_0 on the minimal variance line which minimises the risk (the point (σ_0, μ_0) with minimal σ ; that is, $\sigma_0 = 2$ on the picture). If our risk-free asset has return $R < \sigma_0$ then the line passing through $(0, R)$ and tangent to the upper part of the hyperbola is the efficient frontier for the union $S_1, \dots, S_n, A$ (the blue line). It is called the **capital market line** and the portfolio corresponding to the tangency point is called the **market portfolio**. Notice that if $R \geq \sigma_0$ then no line through $(0, R)$ is tangent to the upper half of the hyperbola.

Aim: To find the weights of the market portfolio.



Let (σ_M, μ_M) be the variance and the return of the capital market portfolio. The slope of the capital market line is equal to $\frac{\mu_M - R}{\sigma_M}$ and hence we have

$$\frac{\mathbf{w}\mathbf{m}^T - R}{\sqrt{\mathbf{w}\mathbf{C}\mathbf{w}^T}} = \frac{\mu_M - R}{\sigma_M}$$

Since the tangency point (σ_M, μ_M) is the one for which the slope is maximal we need to find $\max \frac{\mathbf{w}\mathbf{m}^T - R}{\sqrt{\mathbf{w}\mathbf{C}\mathbf{w}^T}}$ over all $\mathbf{w} \in \mathbf{R}^n$ subject to the constraint $\mathbf{w}\mathbf{u}^T = 1$.

This is again an application of the Lagrange theorem. We write the Lagrange function

$$F(\mathbf{w}, \lambda) = \frac{\mathbf{w}\mathbf{m}^T - R}{\sqrt{\mathbf{w}\mathbf{C}\mathbf{w}^T}} - \lambda(\mathbf{w}\mathbf{u}^T - 1)$$

and solve the equation for vanishing of the gradient:

$$\nabla_{\mathbf{w}} F(\mathbf{w}, \lambda) = \frac{\mathbf{m}\sqrt{\mathbf{w}\mathbf{C}\mathbf{w}^T} - \frac{\mathbf{w}\mathbf{m}^T - R}{\sqrt{\mathbf{w}\mathbf{C}\mathbf{w}^T}} \mathbf{w}\mathbf{C}}{\mathbf{w}\mathbf{C}\mathbf{w}^T} - \lambda \mathbf{u} = 0.$$

This is equivalent to the first equation in the following sequence:

$$\begin{aligned} \frac{\mathbf{w}\mathbf{m}^T - R}{\sqrt{\mathbf{w}\mathbf{C}\mathbf{w}^T}} \mathbf{w}\mathbf{C} &= \mathbf{m}\sqrt{\mathbf{w}\mathbf{C}\mathbf{w}^T} - \lambda \mathbf{u}\mathbf{w}\mathbf{C}\mathbf{w}^T \\ \frac{\mu_M - R}{\sigma_M} \mathbf{w}\mathbf{C} &= \mathbf{m}\sigma_M - \lambda\sigma_M^2 \mathbf{u} && \text{divide by } \sigma_M \\ \frac{\mu_M - R}{\sigma_M^2} \mathbf{w}\mathbf{C} &= \mathbf{m} - \lambda\sigma_M \mathbf{u} && \text{multiply by } \mathbf{w}^T \\ \frac{\mu_M - R}{\sigma_M^2} \mathbf{w}\mathbf{C}\mathbf{w}^T &= \mathbf{m}\mathbf{w}^T - \lambda\sigma_M \mathbf{u}\mathbf{w}^T \\ \mu_M - R &= \mu_M - \lambda\sigma_M \\ \lambda &= \frac{R}{\sigma_M} && \text{substitute to the third equality} \\ \frac{\mu_M - R}{\sigma_M^2} \mathbf{w}\mathbf{C} &= \mathbf{m} - R\mathbf{u} && \text{multiply by } \mathbf{C}^{-1} \\ \frac{\mu_M - R}{\sigma_M^2} \mathbf{w} &= (\mathbf{m} - R\mathbf{u})\mathbf{C}^{-1} \\ \mathbf{w} &= \frac{(\mathbf{m} - R\mathbf{u})\mathbf{C}^{-1}}{(\mathbf{m} - R\mathbf{u})\mathbf{C}^{-1}\mathbf{u}^T}, \end{aligned}$$

provided $(\mathbf{m} - R\mathbf{u})\mathbf{C}^{-1}\mathbf{u}^T \neq 0$. To see that $\frac{\mu_M - R}{\sigma_M^2} = (\mathbf{m} - R\mathbf{u})\mathbf{C}^{-1}\mathbf{u}^T$ multiply the penultimate equality by $\mathbf{u}^T$.

Theorem 4.8. *Assume that $\det \mathbf{C} \neq 0$ and that $R < \mu_{\text{MVP}}$ (the return of the risk-free security is smaller than the return on the minimum variance portfolio). Then the market portfolio exists and its weights are given by*

$$\mathbf{w}_M = \frac{(\mathbf{m} - R\mathbf{u})\mathbf{C}^{-1}}{(\mathbf{m} - R\mathbf{u})\mathbf{C}^{-1}\mathbf{u}^T}.$$

Proof. We need to check that $(\mathbf{m} - R\mathbf{u})\mathbf{C}^{-1}\mathbf{u}^T \neq 0$.

$$\begin{aligned} R &< \mu_{\text{MVP}} = \mathbf{w}_{\text{MVP}}\mathbf{m}^T = \frac{\mathbf{u}\mathbf{C}^{-1}\mathbf{m}^T}{\mathbf{u}\mathbf{C}^{-1}\mathbf{u}^T} \\ R\mathbf{u}\mathbf{C}^{-1}\mathbf{u}^T &< \mathbf{u}\mathbf{C}^{-1}\mathbf{m}^T \\ 0 &< \mathbf{u}\mathbf{C}^{-1}\mathbf{m}^T - R\mathbf{u}\mathbf{C}^{-1}\mathbf{u}^T \\ 0 &< (\mathbf{m} - R\mathbf{u})\mathbf{C}^{-1}\mathbf{u}^T \end{aligned}$$

as required. The last equality is a consequence of the symmetry of $\mathbf{C}^{-1}$ because $\mathbf{u}\mathbf{C}^{-1}\mathbf{m}^T = (\mathbf{u}\mathbf{C}^{-1}\mathbf{m}^T)^T = \mathbf{m}\mathbf{C}^{-1}\mathbf{u}^T$. $\square$

Consider the market portfolio with weights $\mathbf{w}_M$ and another portfolio with weights $\mathbf{w}_V$. The image of the line $s\mathbf{w}_M + (1-s)\mathbf{w}_V$, $s \in \mathbf{R}$ with respect to the risk-return map is a hyperbola tangent to the capital market line (why?).

Theorem 4.9 (Capital Asset Pricing Model). *Suppose that $\det \mathbf{C} \neq 0$ and that the return R of the risk-free security is smaller than the return μ_{MVP} of the minimum variance portfolio. Then the expected return of μ_V of any admissible portfolio is given by $\mu_V = R + \beta_V(\mu_M - R)$, where $\beta_V = \frac{\text{cov}(K_V, K_M)}{\sigma_M^2}$.*

Proof. Let $P = s\mathbf{w}_V + (1-s)\mathbf{w}_M$ for some $s \in \mathbf{R}$. We know that

$$\begin{aligned} \sigma_P &= (s^2\sigma_V^2 + (1-s)^2\sigma_M^2 + 2s(1-s)\text{cov}(K_V, K_M))^{1/2} \\ \mu_P &= s\mu_V + (1-s)\mu_M. \end{aligned}$$

by computing the derivatives at $s = 0$ we get

$$\begin{aligned} \frac{\partial \sigma_P}{\partial s} \Big|_{s=0} &= \frac{\text{cov}(K_V, K_M) - \sigma_M^2}{\sigma_M} \\ \frac{\partial \mu_P}{\partial s} \Big|_{s=0} &= \mu_V - \mu_M. \end{aligned}$$

Since the hyperbola given by $(\sigma_P(s), \mu_P(s))$ is tangent to the capital market line for $s = 0$, the ratio of the above partial derivatives is equal to the slope of the capital market line:

$$\frac{\mu_M - R}{\sigma_M} = \frac{\mu_V - \mu_M}{\frac{\text{cov}(K_V, K_M) - \sigma_M^2}{\sigma_M}}.$$

From this equation we get that

$$\mu_V = R + \frac{\text{cov}(K_V, K_M)}{\sigma_M^2}(\mu_M - R)$$

as claimed. $\square$

The term $\beta_V(\mu_M - R)$ is called the **risk premium**. The CAPM Theorem describes the state of equilibrium of the market in the sense that if, for example, $\mu_V > R + \beta_V(\mu_M - R)$ then the investors will be investing in V increasing the price of this security and consequently decreasing μ_V . Notice, however, that μ_V is only an expected return and its value is based on predictions which may be inaccurate.

Estimating the return

The CAPM formula relates the expected returns of an admissible and the market portfolio. In this section we will relate the actual returns. Let $K_V: \Omega \rightarrow \mathbf{R}$ be the return of an admissible portfolio and let K_M be the return of the market portfolio. We would like to find numbers $\alpha, \beta \in \mathbf{R}$ such that the random variable $\beta K_M + \alpha$ well approximates the random variable K_V in the sense that $\sum_{i=1}^N (K_V(\omega_i) - (\beta K_M + \alpha)(\omega_i))^2 p_i$ is minimal. It means that we want to minimise the expectation of the square of the random variable $\epsilon = \epsilon(\alpha, \beta) = K_V - (\beta K_M + \alpha)$.

$$\begin{aligned} \mathbf{E}(\epsilon^2) &= \mathbf{E}((K_V - (\beta K_M + \alpha))^2) \\ &= \mathbf{E}(K_V^2 - 2K_V(\beta K_M + \alpha) + (\beta K_M + \alpha)^2) \\ &= \mathbf{E}(K_V^2) - 2\beta \mathbf{E}(K_V K_M) - 2\alpha \mathbf{E}(K_V) + \beta^2 \mathbf{E}(K_M^2) + 2\alpha \beta \mathbf{E}(K_M) + \alpha^2 \\ &= \alpha^2 + 2\alpha \beta \mathbf{E}(K_M) + \beta^2 \mathbf{E}(K_M^2) - 2\alpha \mathbf{E}(K_V) - 2\beta \mathbf{E}(K_V K_M) + \mathbf{E}(K_V^2). \end{aligned}$$

Since the coefficients of α^2 and β^2 are positive the function ϵ has a minimum attained at its critical point.

$$\begin{aligned} \frac{\partial \epsilon(\alpha, \beta)}{\partial \alpha} &= 2\alpha + 2\beta \mathbf{E}(K_M) - 2\mathbf{E}(K_V) \\ \frac{\partial \epsilon(\alpha, \beta)}{\partial \beta} &= 2\beta \mathbf{E}(K_M^2) + 2\alpha \mathbf{E}(K_M) - 2\mathbf{E}(K_V K_M) \end{aligned}$$

Thus the critical point of ϵ is the solution of the following system of linear equations

$$\begin{aligned}\alpha + \beta \mathbf{E}(K_M) &= \mathbf{E}(K_V) \\ \alpha \mathbf{E}(K_M) + \beta \mathbf{E}(K_M^2) &= \mathbf{E}(K_V K_M).\end{aligned}$$

which gives $\alpha = \mu_V - \beta_V \mu_M$ and $\beta = \frac{\text{cov}(K_V, K_M)}{\text{var}(K_M)} \mu_M = \beta_V \mu_M$. It follows that

$$\begin{aligned}\epsilon_V &= K_V - (\beta K_M + \alpha) \\ &= K_V - \beta_V K_M - \mu_V + \beta_V \mu_M \\ &= K_V - \mu_V - \beta_V (K_M - \mu_M).\end{aligned}$$

We can now compute the variance of the portfolio V :

$$\begin{aligned}\sigma_V^2 &= \text{var}(K_V) = \text{var}(K_V - \mu_V) \\ &= \text{var}(\epsilon_V + \beta_V (K_M - \mu_M)) \\ &= \text{var}(\epsilon_V) + \beta_V^2 \text{var}(K_M) + 2\beta_V \text{cov}(\epsilon_V, K_M) \\ &= \text{var}(\epsilon_V) + \beta_V^2 \text{var}(K_M) \\ &= \text{var}(\epsilon_V) + \text{cov}(K_V, K_M)\end{aligned}$$

since $\text{cov}(\epsilon_V, K_M) = 0$ (exercise).

Example 4.10. Let $w_i = \frac{1}{n}$ and $c_{ij} = c$ for $i \neq j$. We have that

$$\frac{1}{n} \min \sigma_i^2 + \frac{n(n-1)}{n^2} c \leq \sigma_V^2 \leq \frac{1}{n} \max \sigma_i^2 + \frac{n(n-1)}{n^2} c$$

Passing to the limit as $n \rightarrow \infty$ and assuming that the variances of assets are uniformly bounded we get $\sigma_V^2 = c$. This means that for large number of assets the risk of the portfolio comes from the correlations rather than from the risks of individual assets. $\diamond$

Recall that the weights of the market portfolio satisfy $\gamma \mathbf{w}_M \mathbf{C} = (\mathbf{m} - R\mathbf{u}) \mathbf{C}^{-1}$ for some $\gamma \in \mathbf{R}$. It follows that

$$\beta_V = \frac{\text{cov}(K_V, K_M)}{\sigma_M^2} = \frac{\gamma \mathbf{w}_V \mathbf{C} \mathbf{w}_M^T}{\gamma \mathbf{w}_M \mathbf{C} \mathbf{w}_M^T} = \frac{\mathbf{w}_V (\mathbf{m} - R\mathbf{u})^T}{\mathbf{w}_M (\mathbf{m} - R\mathbf{u})^T} = \frac{\mu_V - R}{\mu_M - R}.$$

Computing μ_V from this equation gives another proof of the CAPM formula:

$$\mu_V = \beta_V (\mu_M - R) + R.$$

This is an affine function of β . Its graph on the β, μ plane is called the *security market line*.

Uncorrelated portfolios

The observation in this section is motivated by the following question asked by a student:

Question 4.11. *What is the financial meaning of an eigenvalue of a covariance matrix?*

Theorem 4.12. *Let $K_1, K_2, \dots, K_m$ be the returns of m risky assets. Let $C = (\text{cov}(K_i, K_j))$ be the covariance matrix and let $\lambda_1, \dots, \lambda_m$ be its eigenvalues. Then there exists portfolios $W_1, \dots, W_m$ with returns $\hat{K}_1, \dots, \hat{K}_m$ such that*

$$\text{cov}(\hat{K}_i, \hat{K}_j) = \begin{cases} 0 & \text{if } i \neq j \\ \lambda_i & \text{if } i = j. \end{cases}$$

If $\det C \neq 0$ then $\lambda_i > 0$ for all $i = 1, \dots, m$.

Proof. Since the covariance matrix C is real and symmetric, it is diagonalisable by an orthogonal matrix. That is, we have

$$D = \text{diag}(\lambda_1, \dots, \lambda_m) = UCU^{-1},$$

where D is diagonal and $U = (u_{ij})$ is orthogonal, which means $U^{-1} = U^T$. Let $\hat{K}_i = K_i U$. Then

$$\begin{aligned} \text{cov}(\hat{K}_i, \hat{K}_j) &= K_i U \cdot C \cdot (K_j U)^T \\ &= K_i \cdot UCU^{-1} \cdot K_j^T \\ &= K_i D K_j^T = \begin{cases} 0 & \text{if } i \neq j \\ \lambda_i & \text{if } i = j. \end{cases} \end{aligned}$$

□

Observe that if $U = (u_{ij})$ is the diagonalising matrix as in the proof above then the weights of the portfolio $W_i = (u_{i1}, u_{i2}, \dots, u_{im})$.

Assume that $\det C \neq 0$ and consider the market model in which the portfolios $W_1, \dots, W_m$ are the basic securities. In this case the covariance matrix is diagonal, $D = \text{diag}(\lambda_1, \dots, \lambda_m)$, and we can apply Proposition 4.2 to obtain the weights of the minimum variance portfolio

$$\hat{\mathbf{w}}_{\text{MVP}} = \left(\frac{\mathbf{u} D^{-1}}{\mathbf{u} D^{-1} \mathbf{u}^T} \right) = \frac{1}{\lambda_1^{-1} + \dots + \lambda_m^{-1}} (\lambda_1^{-1}, \dots, \lambda_m^{-1}).$$

Notice that $\lambda_1^{-1} + \dots + \lambda_m^{-1} = \text{tr} D^{-1} = \text{tr} C^{-1}$.

4.13 Geometric interpretation

We can also derive the formula for the minimal variance portfolio using elementary geometry.

Definition 4.14. An *affine plane* P is the set of solutions $(x_1, \dots, x_n) \in \mathbb{R}^n$ to an equation of the form

$$a_1x_1 + \dots + a_nx_n = \beta, \quad (4.1)$$

where $a_1, \dots, a_n, \beta$ are real numbers, see Figure 4.1. In vector notation, $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{x} = (x_1, \dots, x_n)$, we can abbreviate (4.1) as $\mathbf{a}^T \mathbf{x} = \beta$.

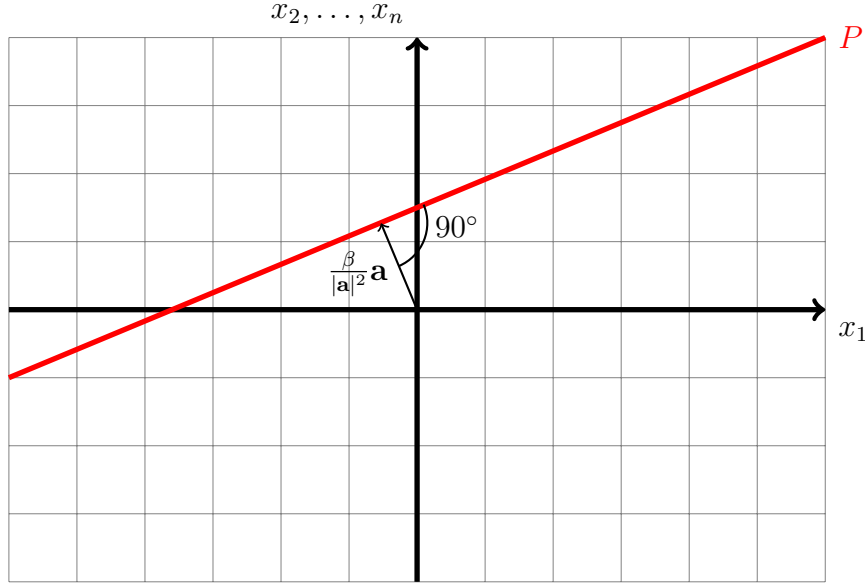


Figure 4.1: The point on the affine plane P closest to the origin is $\mathbf{x}_{\text{MIN}} = \frac{\beta}{|\mathbf{a}|^2} \mathbf{a}$ (since it is perpendicular to P); It has distance $|\beta|/|\mathbf{a}|$.

Definition 4.15. An *ellipse* E centered at the origin is the set of solutions $(x_1, \dots, x_n) \in \mathbb{R}^n$ to an equation of the form

$$\mathbf{x}^T \mathbf{C} \mathbf{x} = \sigma^2, \quad (4.2)$$

where $\sigma > 0$ and C is a positive-definite symmetric matrix⁸.

⁸This means $\mathbf{C} = \mathbf{C}^T$ and that $\mathbf{x}^T \mathbf{C} \mathbf{x} > 0$ for all $\mathbf{x} \neq 0$.

For example, if $\lambda_1, \dots, \lambda_n > 0$ are positive, then the diagonal matrix

$$\mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \quad (4.3)$$

is a positive-definite symmetric matrix which yields the familiar equation

$$\lambda_1 x_1^2 + \dots + \lambda_n x_n^2 = \sigma^2 \quad (4.4)$$

for an ellipse with k^{th} axis $\mathbf{e}_k/\lambda_k$.

According to the following mathematical theorem, there is always a co-ordinate transform that puts an ellipse (4.2) into the standard form (4.4).

Theorem 4.16 (Spectral theorem). *There is an orthogonal coordinate transform $\mathbf{S} \in \text{O}(n)$ ⁹ such that $\mathbf{\Lambda} = \mathbf{S}\mathbf{C}\mathbf{S}^T$ is a diagonal matrix (4.3). In particular, the equation (4.2) in the coordinates $\mathbf{y} = \mathbf{S}\mathbf{x}$ becomes*

$$\lambda_1 y_1^2 + \dots + \lambda_n y_n^2 = \sigma^2. \quad (4.5)$$

Hence in the **distorted** coordinates $\mathbf{z} = \mathbf{\Lambda}^{1/2}\mathbf{y}$ (equivalently, $z_k = \sqrt{\lambda_k}y_k$), the equation (4.2) becomes that of a circle

$$z_1^2 + \dots + z_n^2 = \sigma^2. \quad (4.6)$$

Dropping the admissibility condition for a portfolio, by Proposition 4.1 the set of all portfolios with a given variance σ^2 is an ellipse, where we set $\mathbf{x} = \mathbf{w}$. The set of admissible portfolios is an affine plane. The **minimum variance portfolio** is the admissible portfolio with the minimal variance, the smallest σ for which the ellipse (4.2) intersects the plane $\mathbf{w}\mathbf{u}^T = 1$, see Figure 4.2. In terms of the coordinate $\mathbf{z} = \mathbf{\Lambda}^{1/2}\mathbf{S}\mathbf{x}$, the admissible plane is

$$1 = \mathbf{u}^T \mathbf{x} = \mathbf{u}^T (\mathbf{S}^T \mathbf{\Lambda}^{-1/2} \mathbf{z}) = \underbrace{(\mathbf{\Lambda}^{-1/2} \mathbf{S} \mathbf{u})^T}_{\mathbf{a}^T} \mathbf{z}$$

As in Figure 4.1, the smallest radius of the circle meeting the plane is

$$\sigma_V = \frac{1}{|\mathbf{a}|} = \frac{1}{\sqrt{\mathbf{a}^T \mathbf{a}}} = \frac{1}{\sqrt{\mathbf{u}^T \mathbf{S}^T \mathbf{\Lambda}^{-1/2} \mathbf{\Lambda}^{-1/2} \mathbf{S} \mathbf{u}}} = \frac{1}{\sqrt{\mathbf{u}^T \mathbf{C}^{-1} \mathbf{u}}}.$$

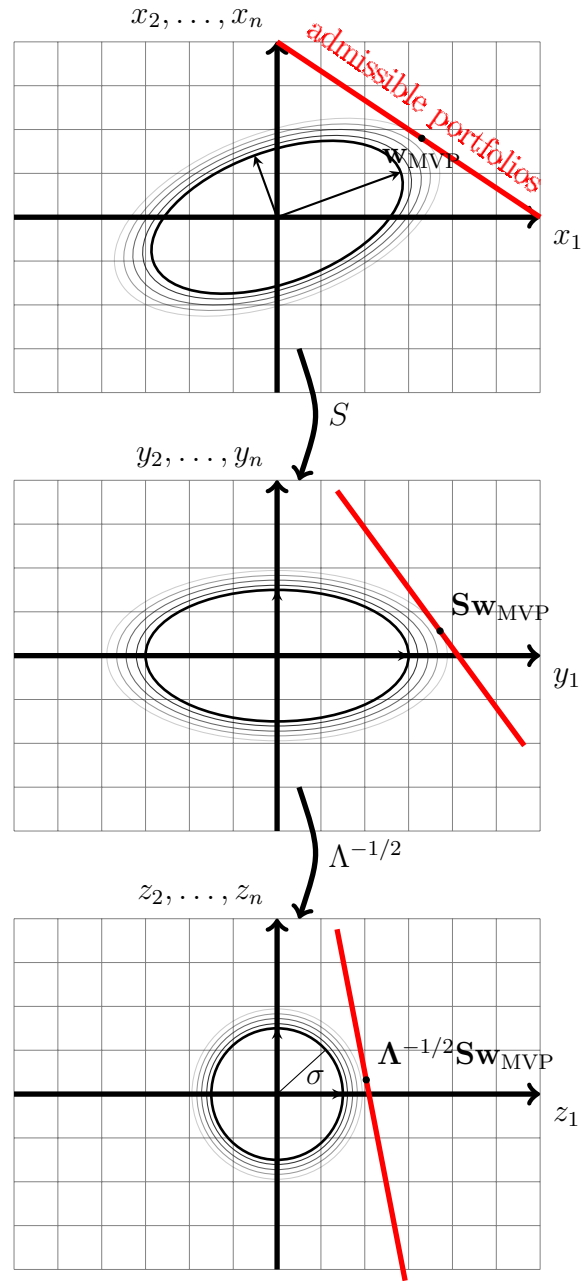


Figure 4.2: The minimum variance portfolio is given by the smallest radius σ such that the circle meets the plane.

The intersection point is, expressed in the old coordinates $\mathbf{x} = \mathbf{S}^T \Lambda^{-1/2} \mathbf{z}$,

$$\mathbf{w}_{\text{MVP}} = \mathbf{S}^T \Lambda^{-1/2} \left(\frac{\mathbf{a}}{|\mathbf{a}|^2} \right) = \frac{\mathbf{S}^T \Lambda^{-1/2} \Lambda^{-1/2} \mathbf{S} \mathbf{u}}{\mathbf{u}^T \mathbf{C}^{-1} \mathbf{u}} = \frac{\mathbf{C}^{-1} \mathbf{u}}{\mathbf{u}^T \mathbf{C}^{-1} \mathbf{u}}.$$

This is a geometric proof of Proposition 4.2. There is a similar geometric proof of the formula for the efficient portfolio with given return.

⁹This means that the matrices $\mathbf{S}^T \mathbf{S}$ and $\mathbf{S} \mathbf{S}^T$ are both the identity matrix.

Appendix: Inner products

Let V be a real n -dimensional vector space. An inner product is a map $\langle \cdot, \cdot \rangle: V \rightarrow \mathbf{R}$ such that the following conditions hold for all $x, y, z \in V$ and $a \in \mathbf{R}$.

- $\langle x, y \rangle = \langle y, x \rangle$,
- $\langle ax, y \rangle = a\langle x, y \rangle$,
- $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$.
- $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ if and only if $x = 0$.

Example 4.17. If V has a basis and if $\mathbf{C}$ is a symmetric positive definite $n \times n$ matrix then it defines an inner product by $\langle x, y \rangle_{\mathbf{C}} = x\mathbf{C}y^T$. $\diamond$

Proposition 4.18. *Let V be a real vector space equipped with an inner product $\langle \cdot, \cdot \rangle$. Let $W \subset V$ be a vector subspace and let $a \in V$ be a vector so that $W + a \subset V$ is an affine subspace.. Let $f: V \rightarrow \mathbf{R}$ be a function defined by $f(x) = \langle x, x \rangle$. Then f has a unique minimum on the affine subspace $W + a$.*

Proof. Let $W^\perp = \{v \in V \mid \langle v, w \rangle = 0 \text{ for every } w \in W\}$ be the orthogonal complement of W . Observe that $V = W \oplus W^\perp$. Express $a = w_a + w_a^\perp$. We claim that

$$\langle w_a^\perp, w_a^\perp \rangle \leq \langle w + a, w + a \rangle$$

for every $w \in W$. That is, w_a is the required minimum. Indeed,

$$\begin{aligned} \langle w + a, w + a \rangle &= \langle w + w_a + w_a^\perp, w + w_a + w_a^\perp \rangle \\ &= \langle w + w_a, w + w_a \rangle + 2\langle w + w_a, w_a^\perp \rangle + \langle w_a^\perp, w_a^\perp \rangle \\ &= \langle w + w_a, w + w_a \rangle + \langle w_a^\perp, w_a^\perp \rangle \\ &\geq \langle w_a^\perp, w_a^\perp \rangle \end{aligned}$$

Observe that the equality holds if and only if $w + w_a = 0$ which implies that $w + a = w_a^\perp$. That is, $w_a^\perp$ is the unique minimum. $\square$

Appendix: Lagrange's theorem

Let $f, g_1, \dots, g_m: \mathbf{R}^n \rightarrow \mathbf{R}$ be differentiable functions, where $m < n$. Let $M = \{x \in \mathbf{R}^n \mid g_i(x) = c_i\}$, where $c_i \in \mathbf{R}$. Suppose that the gradients $\nabla g_i(x)$ are linearly independent at each $x \in M$. If f has a minimum at

$x_0 \in M$ then there exist numbers $\lambda_1, \dots, \lambda_m \in \mathbf{R}$ such that $\nabla f(x_0) = \lambda_1 \nabla g_1(x_0) + \dots + \lambda_m \nabla g_m(x_0)$. Moreover, x_0 is a solution of the following system of equations:

$$\begin{aligned}\nabla f(x) &= \lambda_1 \nabla g_1(x) + \dots + \lambda_m \nabla g_m(x) \\ g_1(x) &= c_1 \\ &\vdots \\ g_m(x) &= c_m\end{aligned}$$

Let $L: \mathbf{R}^{n+m} \rightarrow \mathbf{R}$ be defined by $L(x, \lambda_1, \dots, \lambda_m) = f(x) + \lambda_1(g_1(x) - c_1) + \dots + \lambda_m(g_m(x) - c_m)$. Solving the last system of equations is equivalent to solving the following:

$$\begin{aligned}\nabla_x L(x, \lambda_1, \dots, \lambda_m) &= 0 \\ \frac{\partial L(x, \lambda_1, \dots, \lambda_m)}{\partial \lambda_1} &= 0 \\ &\vdots \\ \frac{\partial L(x, \lambda_1, \dots, \lambda_m)}{\partial \lambda_m} &= 0\end{aligned}$$

The function L is called the **Lagrange function** and the numbers λ_i the **Lagrange multipliers**.

4.19 Exercises

4a. Prove that the covariance is a symmetric bilinear form. That is, prove that

- $\text{cov}(K, L) = \text{cov}(L, M)$ and
- $\text{cov}(aK + bL, M) = a \text{cov}(K, M) + b \text{cov}(L, M)$.

for all $a, b \in \mathbf{R}$ and all random variables $K, L, M: \Omega \rightarrow \mathbf{R}$.

4b. Let $K_1, K_2: \Omega \rightarrow \mathbf{R}$ be random variables. Let $\mathbf{C} = [\text{cov}(K_i, K_j)]$ be the covariance matrix.

- (1) Prove that the covariance matrix is symmetric: $\mathbf{C}^T = \mathbf{C}$.
- (2) Prove that the covariance matrix is nonnegative definite: $x\mathbf{C}x^T \geq 0$ for every $x \in \mathbf{R}^2$.
- (3) Assume that $\det \mathbf{C} \neq 0$. Show that both $\mathbf{C}$ and its inverse $\mathbf{C}^{-1}$ are symmetric and positive definite.

4c. Let S_1, S_2, S_3 be three risky securities with returns K_1, K_2, K_3 . Suppose that the expected returns and the covariances are as follows:

$$\begin{array}{lll} \text{Expected returns:} & \mu_1 = 0.20 & \mu_2 = 0.13 \quad \mu_3 = 0.17 \\ \text{Standard deviations:} & \sigma_1 = 0.25 & \sigma_2 = 0.28 \quad \sigma_3 = 0.20 \\ \text{Correlations:} & \rho_{12} = 0.30 & \rho_{23} = 0.00 \quad \rho_{31} = 0.15 \end{array}$$

- (1) Find the weights of the minimum variance portfolio.
- (2) Compute the expected return and the variance of this portfolio.
- (3) Find the portfolio with minimum variance among those portfolios with expected return $\mu = 20\%$. Compute the weights and the standard deviation of this portfolio.
- (4) Compute the minimum variance line.
- (5) Draw the minimum variance line on the w_2, w_3 plane.
- (6) Draw the image of the minimum variance line on the σ, μ plane.
- (7) Choose $0 < R < \mu_{\text{MVP}}$ and compute the capital market line for the risk-free security with the return R and find the corresponding market portfolio.
- (8) Let V be an admissible portfolio. Compute its beta factor β_V , expected return and the risk premium.

4d. Create one more example like the previous one and solve all the items for it. Convince yourself that given a problem of this type you can solve it comfortably.

4e. Show that the matrix M from the proof of Theorem 4.4 is indeed invertible.

5 Pricing results in single period models

In this section we mostly consider the 1-step binomial market model with one risky and one risk-free asset. That is, the probability space of the model is

$$(\Omega = \{u, d\}, 2^\Omega, \mathbf{P}),$$

where the probability measure is defined by $\mathbf{P}(u) = p$ and $\mathbf{P}(d) = 1 - p$, with $0 < p < 1$. We consider two time instances $t = 0$ and $t = 1$. The prices of the assets are denoted by $A(0), A(1)$ – the risk-free assets and $S(0), S(1)$ – the risky asset, where $S(1): \Omega \rightarrow \mathbf{R}$ is a random variable with distribution

$$S(1) = \begin{cases} S(1)(u) & \text{with probability } p \\ S(1)(d) & \text{with probability } 1 - p. \end{cases}$$

We also assume that $S(1)(u) > S(1)(d)$.

5.1 Inequalities on stock prices

Proposition 5.2. *The axiom of no arbitrage implies*

$$\frac{S(1)(d)}{S(0)} < \frac{A(1)}{A(0)} < \frac{S(1)(u)}{S(0)}.$$

Proof. We prove the first inequality. It is a proof by contradiction. Suppose the first inequality does not hold. That is, $\frac{S(1)(d)}{S(0)} \geq \frac{A(1)}{A(0)}$, which is equivalent to

$$S(1)(d) - \frac{S(0)}{A(0)}A(1) \geq 0. \quad (5.1)$$

Consider the portfolio $(x, y) = \left(1, -\frac{S(0)}{A(0)}\right)$ and compute its values at time $t = 0$ and $t = 1$:

$$\begin{aligned} V_{(x,y)}(0) &= xS(0) + yA(0) = S(0) - \frac{S(0)}{A(0)}A(0) = 0. \\ V_{(x,y)}(1) &= \begin{cases} xS(1)(d) + yA(1) &= S(1)(d) - \frac{S(0)}{A(0)}A(1) \geq 0. \\ xS(1)(u) + yA(1) &= S(1)(u) - \frac{S(0)}{A(0)}A(1) > 0. \end{cases} \end{aligned}$$

This portfolio has initial value equal to zero and the final value non-negative and positive with non-zero probability and hence it is an arbitrage. Since,

the axiom of no arbitrage, prevents existence of such portfolio it implies that the inequality (5.1) cannot be true. This shows that the first inequality in the statement is true.

The proof of the second inequality is similar and it is left as an exercise. $\square$

Remark 5.3. We guessed the arbitrage portfolio in the above proof by looking at the inequality (5.1). Its left hand side looks like the value of a portfolio at time $t = 1$.

Remark 5.4. The portfolio $\left(1, -\frac{S(0)}{A(0)}\right)$ means that at time $t = 0$ we borrow $\frac{S(0)}{A(0)}$ of the risk-free asset, sell it, which gives us $\frac{S(0)}{A(0)}A(0) = S(0)$ money and for this money we buy 1 share of stock. At time $t = 1$ we sell the stock for $S(1)$ (whatever it is) and pay $\frac{S(0)}{A(0)}A(1)$ to repay the risk-free loan.

5.5 The price of the forward contract

Proposition 5.6. *The axiom of no arbitrage implies that the forward price is given by*

$$F = S(0)\frac{A(1)}{A(0)} = S(0)(1 + K_A(1))$$

Proof. Suppose that $F > S(0)(1 + K_A(1))$. This is equivalent to

$$F - \frac{S(0)}{A(0)}A(1) > 0.$$

Create a portfolio (x, y, z) with $V(0) = 0$ and $V(1) = F - \frac{S(0)}{A(0)}A(1) > 0$. By comparing the last expression with the general formula for $V(1)$ we find that $(1, -\frac{S(0)}{A(0)}, -1)$ is a required portfolio which violates the axiom of no arbitrage. This implies that $F \leq S(0)(1 + K_A(1))$.

The remaining part of the proof, that is, to show that $F \geq S(0)(1 + K_A(1))$ is left as an exercise. $\square$

Again, let us discuss what does the portfolio $\left(1, -\frac{S(0)}{A(0)}, -1\right)$ mean in terms of financial actions. At time $t = 0$:

- Borrow $S(0)$;
- Buy a share for $S(0)$;

- Enter into short forward contract with the price F .
- We have $V(0) = S(0) - \frac{S(0)}{A(0)}A(0) = 0$.

At time $t = T$:

- Close short forward contract (sell a share for F);
- Close the risk-free part by paying $\frac{S(0)}{A(0)}A(1) = S(0)(1 + K_A(1))$.
- We have $V(1) = S(1) - \frac{S(0)}{A(0)}A(1) - (S(1) - F) = F - S(0)(1 + K_A(1)) > 0$ as required.

5.7 The price of an option

Let $C(1): \Omega \rightarrow \mathbf{R}$ be the pay-off of a European option. We say than an option can be **replicated** if there exists a portfolio (x, y) such that its value at time $t = 1$ is equal to the pay-off $C(1)$:

$$V_{(x,y)}(1) = xS(1) + yA(1) = C(1).$$

Proposition 5.8. *The axiom of no arbitrage implies that if an option can be replicated by a portfolio (x, y) then its price at time $t = 0$ is given by*

$$C(0) = xS(0) + yA(0).$$

Proof. Suppose that $C(0) > xS(0) + yA(0)$. Then at time $t = 0$:

- issue and sell a call option for $C(0)$;
- take a long position in the portfolio (x, y) ; for example, buy share and borrow money; the balance is $C(0) - xS(0) - yA(0) > 0$;
- invest the balance risk free; the result is $V(0) = 0$.

At time $t = 1$:

- If the prices have gone up then settle the option by paying $S(1)(u) - X$; if the prices have gone down pay nothing; the cost is $C(1)$;
- close the portfolio (x, y) ; the profit is $xS(1) + yA(1)$; and the balance is $-C(1) + xS(1) + yA(1) = 0$;
- close the risk-free investment with positive profit

$$C(0) - xS(0) - yA(0) + \text{interest} > 0.$$

This violates the axiom of no arbitrage and proves that $C(0) \leq xS(0) + yA(0)$. The proof of the opposite inequality is left as an exercise. $\square$

Remark 5.9. The corresponding portfolio for the above proof is

$$(x, y, -1, \epsilon),$$

where x is the position in shares, y in bonds, -1 means that a call option has been issued and sold, $\epsilon > 0$ is the positive balance invested in the bank. Notice that it is not specified whether x or y are positive or negative here. In many examples they are of opposite signs. The initial value of this portfolio is

$$xS(0) + yA(0) - C(0) + \epsilon = 0.$$

The final value is $(1 + r)\epsilon > 0$ due to replication. It has been implicitly assumed that the return of the money deposit is positive.

5.10 Pricing done abstractly

Consider a contingent claim with the initial price $C(0)$ and the pay-off $C(1)$. For example, in the case of the forward contract $C(0) = 0$ and $C(1) = S(1) - F$, where F was unknown. For European call option, however, $C(0)$ was unknown and $C(1) = \max\{S(1) - X, 0\}$. In both cases, in order to find missing ingredients we constructed a suitable portfolio (x, y) with the underlying risky asset and money such that $V(0) = C(0)$ and $V(1) = C(1)$. Then we read off the values we were looking for from the portfolio.

Example 5.11. Consider a long forward contract with the forward price F . Its initial value is zero (entering the contract costs nothing) and the pay-off is $S(1) - F$. Portfolios with initial value zero have the form $\left(x, -\frac{xS(0)}{A(0)}\right)$. The final value implies that $x = 1$, otherwise F would depend on $S(1)$ which would make it a random variable. Thus the portfolio we are looking for is $\left(1, -\frac{S(0)}{A(0)}\right)$; compare with the argument in Proposition 5.6. $\diamond$

Example 5.12. Consider a contingent claim with the pay-off $C(1): \Omega \rightarrow \mathbf{R}$. In Proposition 5.8 we argued that the initial value of the replicating portfolio is the initial price of the option. Observe that if the final price $S(1)$ of the risky asset is a non-constant random variable, that is, $S(1)(d) \neq S(1)(u)$

then there replicating portfolio exists. Indeed, we need to solve the following system of linear equations:

$$\begin{cases} xS(1)(d) + yA(1) &= C(1)(d) \\ xS(1)(u) + yA(1) &= C(1)(u). \end{cases}$$

The solutions are

$$x = \frac{C(1)(u) - C(1)(d)}{S(0)(U - D)} \quad y = \frac{C(1)(d)U - C(1)(u)D}{(U - D)A(0)R},$$

provided $U \neq D$, where $S(1)(d) = S(0)D$ and $S(1)(u) = S(0)U$ and $R = (1 + r)$. $\diamond$

Example 5.13. Consider a trinomial model with $\Omega = \{u, m, d\}$ and a European call option. The system of linear equations to solve in order to find the replication portfolio is now

$$\begin{cases} xS(1)(d) + yA(1) &= 0 \\ xS(1)(m) + yA(1) &= \max\{S(1)(m) - X, 0\} \\ xS(1)(u) + yA(1) &= S(1)(u) - X. \end{cases}$$

It is easy to find values of $S(1)$ such that the system will have no solutions. $\diamond$

Example 5.14. Consider again single period binomial model and a European call option based on two assets with the pay-off $\max\left\{\frac{S_1(1)+S_2(1)}{2} - X, 0\right\}$. The replicating portfolio (x_1, x_2, y) is a solution of the following system of equations:

$$\begin{cases} x_1S_1(1)(d) + x_2S_2(1)(d) + yA(1) &= 0 \\ x_1S_1(1)(u) + x_2S_2(1)(u) + yA(1) &= \frac{S_1(1)(u)+S_2(1)(u)}{2} - X \end{cases}$$

It is easy to find suitable values for $S_1(1)$ and $S_2(1)$ so that the system will have *infinitely many* solutions. It follows that there will be infinitely many pairwise distinct initial prices of the option none of which will violate the axiom of no arbitrage. $\diamond$

5.15 Risk-neutral probability measure

Consider a contingent claim with the initial price $C(0)$ and the pay-off $C(1)$. In Example 5.12 we found the replicating portfolio

$$x = \frac{C(1)(u) - C(1)(d)}{S(0)(U - D)} \quad y = \frac{C(1)(d)U - C(1)(u)D}{(U - D)A(0)R},$$

under the assumption that $S(1)(d) \neq S(1)(u)$, which is equivalent to $U \neq D$. It follows from Proposition 5.8 that the initial value of the contingent claim is equal to the initial value of the replicating portfolio. Let us compute this value.

$$\begin{aligned} C(0) &= xS(0) + yA(0) \\ &= \frac{C(1)(u) - C(1)(d)}{U - D} + \frac{C(1)(d)U - C(1)(u)D}{(U - D)R} \\ &= \frac{C(1)(u)R - C(1)(d)R + C(1)(d)U - C(1)(u)D}{(U - D)R} \\ &= \frac{C(1)(u)(R - D) + C(1)(d)(U - R)}{(U - D)R} \\ &= \frac{R - D}{U - D} \cdot R^{-1}C(1)(u) + \frac{U - R}{U - D} \cdot R^{-1}C(1)(d) \end{aligned}$$

Observe that if $D < R < U$ then

$$\frac{R - D}{U - D} + \frac{U - R}{U - D} = 1$$

and both summands are positive which suggests that

$$\mathbf{P}_*(u) = p_* = \frac{R - D}{U - D}, \quad \mathbf{P}_*(d) = (1 - p_*) = \frac{U - R}{U - D}$$

is some kind of probability. The initial price of the contingent claim is the expectation of $R^{-1}C(1)$, the discounted pay-off of the contingent claim:

$$C(0) = R^{-1}\mathbf{E}_{\mathbf{P}_*}(C(1)).$$

Let us look at the behaviour of the price of the risky asset with respect to this probability. The proof of the following observation is a straightforward computation which is left as an exercise.

Proposition 5.16. *If $D < R < U$ then*

$$\begin{aligned}\mathbf{E}_{\mathbf{P}_*}(S(1)) &= RS(0). \\ \mathbf{E}_{\mathbf{P}_*}(K_S(1)) &= r.\end{aligned}$$

Proof.

$$\begin{aligned}\mathbf{E}_{\mathbf{P}_*}(S(1)) &= \frac{R-D}{U-D}S(0)U + \frac{U-R}{U-D}S(0)D \\ &= \frac{RU - DU + UD - RD}{U-D}S(0) \\ &= \frac{R(U-D)}{U-D}S(0) = RS(0)\end{aligned}$$

□

The above results show that, with respect to the probability $\mathbf{P}_*$, the price of a risky asset behaves like a risk-free security: the discounted expected price is the initial price. Also, the return of the risky asset is equal to r , the risk-free rate. That is why the probability measure $\mathbf{P}_*$ is called ***risk-neutral***.

5.17 Exercises

5a. Complete the proofs of Propositions 5.2, 5.6 and 5.8.

5b. Let $A(0) = 100$, $A(1) = 112$ and $S(0) = 34$. Is it possible to find an arbitrage opportunity if the forward price of stock is $F = 38.60$ with delivery date $t = 1$?

5c. Let (x, y, z) denote a portfolio consisting of x shares, y money (so that $A(0) = 1$) and z forward contracts. Assume that $S(0) = 100$ and $S(1) \in \{90, 110\}$ and that the forward price is $F = 105$, and that the interest rate $r = 10\%$. In the following examples, list the financial operations corresponding to the given portfolio and compute its initial and the final value:

- | | |
|--------------------|--------------------|
| 1. $(2, -30, 5);$ | 3. $(-2, 30, 5);$ |
| 2. $(2, -30, -5);$ | 4. $(-2, 30, -5);$ |

5d. Let $A(0) = 100$, $A(1) = 110$, $S(0) = 100$ and let

$$S(1) = \begin{cases} 120 & \text{with probability } p \\ 80 & \text{with probability } 1 - p. \end{cases}$$

1. Compute the price $C(0)$ of a call option with exercise time 1 and a strike price 90.
 2. Compute the price $C(0)$ of a call option with exercise time 1 and a strike price 110.
- 5e.** Let $A(1) = 105$ (the rest of the data are as above). Compute the price $C(0)$ with strike price 100 and exercise time 1.
- 5f.** Let $A(1) = 115$ (the rest of the data are as above). Compute the price $C(0)$ with strike price 100 and exercise time 1.
- 5g.** In Example 5.13 find the suitable value of $S(1)$.
- 5h.** In Example 5.14 find the suitable values of $S_1(1)$ and $S_2(1)$.
- 5i.** Prove the second claim in Proposition 5.16.
- 5j.** For the setup in Exercises 5d, 5e and 5f compute the risk-neutral probability.

6 Risk-neutral probability measures in single period models

In this section we consider a single period market model with finite probability space and m risky assets. In the previous section we discovered a certain useful probability which we will now define in a more general context and further investigate.

6.1 Definition of a risk-neutral probability measure

Definition 6.2. Consider a single period market model with finite probability space $(\Omega = \{\omega_1, \dots, \omega_N\}, 2^\Omega, \mathbf{P})$ and risky assets $S_1, \dots, S_m$. Let r be the risk-free rate and let $R = (1 + r)$. A probability measure $\mathbf{P}_*$ is called *risk-neutral* if:

1. $\mathbf{P}_*(\omega) > 0$ for every $\omega \in \Omega$, and
2. $\mathbf{E}_{\mathbf{P}_*}(S_i(1)) = R S_i(0)$ for each $i = 1, \dots, m$.

Remark 6.3. The first condition means that the probability $\mathbf{P}_*$ is equivalent to $\mathbf{P}$. The second condition is equivalent to $\mathbf{E}_{\mathbf{P}_*}(K_i(1)) = r$.

Definition 6.4. If we only require that $\mathbf{P}_*$ is a probability, not necessarily, equivalent to $\mathbf{P}$ then such $\mathbf{P}_*$ is called a *martingale probability measure*.

6.5 Risk-neutral pricing

The following result shows the main usage of a risk-neutral probability. One can price an option without even specifying that is the physical probability.

Theorem 6.6. Consider a single period model with finite probability space and m risky assets. Let $C(1): \Omega \rightarrow \mathbf{R}$ be the pay-off of a contingent claim based on m risky assets. Then for every risk-neutral probability $\mathbf{P}_*$ the initial price given by

$$C(0) = \frac{1}{R} \mathbf{E}_{\mathbf{P}_*}(C(1))$$

does not violate the axiom of no arbitrage.

Proof. Consider a portfolio $(x_1, \dots, x_m, y, z)$ consisting of x_i amount of the i -th risky asset, y amount of risk-free asset and z amount of the contingent claim such that:

$$V_{(x,y,z)}(0) = 0 \qquad V_{(x,y,z)}(1) \geq 0.$$

We shall show that the final value of this portfolio has to be identically equal to zero which shows that arbitrage is not possible. We start with computing the expectation of the final value of the portfolio with respect to the risk-neutral measure.

$$\begin{aligned} \mathbf{E}_{\mathbf{P}_*}(V_{(x,y,z)}(1)) &= \mathbf{E}_{\mathbf{P}_*} \left(\sum_{i=1}^m x_i S_i(1) + yA(1) + zC(1) \right) \\ &= \sum_{i=1}^m x_i \mathbf{E}_{\mathbf{P}_*}(S_i(1)) + yRA(0) + z\mathbf{E}_{\mathbf{P}_*}(C(1)) \\ &= \sum_{i=1}^m x_i RS_i(0) + yRA(0) + zRC(0) \\ &= RV_{(x,y,z)}(0) = 0. \end{aligned}$$

Since the risk-neutral probability measure satisfies $\mathbf{P}_*(\omega) > 0$ for every $\omega \in \Omega$ the above computation implies that $V_{(x,y,z)}(1)$ has to be identically zero. Consequently, there is no arbitrage which proves the theorem. $\square$

6.7 Existence of risk-neutral probability measures

Now the question is whether a risk-neutral probability exists. We saw in Example 5.13 that it may not exist at all and in Example 5.14 that there may be infinitely many risk-neutral probabilities.

Intuition. Definition 6.2 and Remark 6.3 say that a risk-neutral probability is a solution of the following system of m linear equations:

$$\mathbf{E}_{\mathbf{P}_*}(K_i(1)) = r$$

which expand to

$$\begin{cases} p_{1*} + p_{2*} + \dots + p_{N*} & = 1 \\ K_1(1)(\omega_1)p_{1*} + \dots + K_1(1)(\omega_N)p_{N*} & = r \\ \dots & \dots \\ K_m(1)(\omega_1)p_{1*} + \dots + K_m(1)(\omega_N)p_{N*} & = r \end{cases}$$

(the first equation says that $\mathbf{P}_*$ is a probability). It also has to satisfy the N inequalities: $0 < p_{j*}$ for $j = 1, \dots, N$. So it is the system of $m + 1$ equations with N unknowns and generically it should have infinitely many solutions if $N > m + 1$, a unique solution if $N = m + 1$ and no solutions if $N < m + 1$. Since N is the number of possible scenarios, we want it to be relatively big to cover many possible outcomes. So our intuition should be that in sensible models $N > m + 1$ and there are infinitely many risk-neutral probability measures. ♠

The next result shows that in our models there are always risk-neutral probability measures.

Theorem 6.8. *Consider a single period market model with finite probability space. Then there exists a risk-neutral probability if and only if there is no arbitrage.*

Proof. Assume that there exists a risk-neutral probability $\mathbf{P}_*$. Consider a portfolio with $V(0) = 0$ and $V(1) \geq 0$. It follows, by the computation analogous to the one in the proof of Theorem 6.6 that $\mathbf{E}_{\mathbf{P}_*}(V(1)) = 0$ which implies that $V(1) = 0$ and consequently there is no arbitrage.

For the converse implication suppose that there is no arbitrage. Let $\Omega = \{\omega_1, \dots, \omega_N\}$. Let $g_j: \mathbf{R}^m \rightarrow \mathbf{R}$ for $j = 1, \dots, N$ be defined by

$$g_j(x_1, \dots, x_m) = \sum_{i=1}^m x_i(\omega_j) \left(\frac{S_i(1)(\omega_j)}{1 + R} - S_i(0) \right).$$

Let $G: \mathbf{R}^m \rightarrow \mathbf{R}^N$ be given by $G(x_1, \dots, x_m) = (g_1(x), \dots, g_N(x))$, where $x = (x_1, \dots, x_m)$. The map G is linear and hence its image W is a vector subspace of $\mathbf{R}^N$. We **claim** that the only element of W with non-negative coordinates is the origin $(0, \dots, 0)$. Indeed, let $x = (x_1, \dots, x_m)$ be such that $g_j(x) \geq 0$ for all $j = 1, \dots, N$ and let y be such that the portfolio (x, y) has

initial value $V(0) = 0$. Then

$$\tilde{V}(1)(\omega_j) = \frac{1}{R} \left(\sum_{i=1}^m x_i(\omega_j) S_i(1)(\omega_j) + yA(1) \right) \quad (6.1)$$

$$= \frac{1}{R} \left(\sum_{i=1}^m x_i(\omega_j) S_i(1)(\omega_j) \right) + yA(0) - V(0) \quad (6.2)$$

$$= \sum_{i=1}^m x_i(\omega_j) \left(\frac{S_i(1)(\omega_j)}{R} - S_i(0) \right) \quad (6.3)$$

$$= g_j(x) \geq 0. \quad (6.4)$$

The axiom of no arbitrage implies that $V(1) = 0$ and hence also $\tilde{V}(1) = 0$ which means that $g_j(x) = 0$ for all $j = 1, \dots, N$. This proves the **claim**.

Let $\Delta = \{(q_1, \dots, q_N) \in \mathbf{R}^N \mid \sum q_j = 1 \text{ and } q_j \geq 0\}$ denote the set of all probabilities on $\Omega = \{\omega_1, \dots, \omega_N\}$. The above claim proves that the subspace W is disjoint from Δ . It follows from Lemma ?? that there exists a vector $\mathbf{P}_* = (p_1^*, \dots, p_N^*) \in W^\perp \cap \Delta$ with strictly positive coordinates. The condition that $\mathbf{P}_* \in W^\perp$ is equivalent to

$$0 = \langle \mathbf{P}_*, G(x) \rangle = \sum_{j=1}^N p_{j*} g_j(x) = \mathbf{E}_{\mathbf{P}_*}(G(x))$$

for every $x \in \mathbf{R}^m$. In particular, if $x = (0, \dots, 1, \dots, 0)$ (all coordinates zero except the i -th one which equal to 1) we get that $G(x) = \frac{S_i(1)}{R} - S_i(0)$ and hence

$$0 = \mathbf{E}_{\mathbf{P}_*}(G(x)) = \mathbf{E}_{\mathbf{P}_*} \left(\frac{S_i(1)}{R} - S_i(0) \right) = \mathbf{E}_{\mathbf{P}_*} \left(\frac{S_i(1)}{R} \right) - S_i(0)$$

which implies that $E_*(S_i(1)) = R S_i(0)$. That is, $\mathbf{P}_*$ is a risk-neutral probability. $\square$

Remark 6.9. Lemma ?? provides a vector $v \in V^\perp$ such that $\langle v, q \rangle > 0$ for every $q \in A$. Taking $q = (0, \dots, 1, \dots, 0)$ we get that coordinates of v are all positive. Thus the required $\mathbf{P}_*$ is given by $p_{j*} = \frac{v_j}{\sum_j v_j}$.

6.10 Pricing options in incomplete markets

Consider a European contingent claim with exercise time $t = 1$ and the pay-off $C(1): \Omega \rightarrow \mathbf{R}$. Suppose that an investor sells the above option for $C(0) > 0$. If the option can be replicated then the price $C(0)$ is equal to the initial value of the replicating portfolio. The investor then invest $C(0)$ in the replicating portfolio and at time $t = 1$ they have money to settle the option. This situation is not interesting.

Definition 6.11. A market model is called *complete* if for every European contingent claim there exists a replication portfolio. Otherwise, the market is called *incomplete*.

Theorem 6.12. *The market is complete and satisfies the axiom of no arbitrage if and only if it admits a unique risk neutral measure.*

Proof. Suppose that the market is complete (and there is no arbitrage, which we always assume). We shall show that there is a unique risk-neutral probability.

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be the probability space of the model. Let $B \in \mathcal{F}$ be any subset. Consider a contingent claim with the pay-off $C(1) = I_B$, the indicator function of the set B . Since the market is complete, there is a replicating portfolio (x, y) . Let $\mathbf{P}_*$ be any risk-neutral probability measure (which exists according to Theorem 6.8). We compute the $\mathbf{P}_*$ -measure of B .

$$\begin{aligned} \mathbf{P}_*(B) &= \mathbf{E}_{\mathbf{P}_*}(I_B) = \mathbf{E}_{\mathbf{P}_*}(C(1)) \\ &= \mathbf{E}_{\mathbf{P}_*} \left(\sum_{i=1}^m x_i S_i(1) + yA(1) \right) \\ &= \sum_{i=1}^m x_i \mathbf{E}_{\mathbf{P}_*}(S_i(1)) + yRA(0) \\ &= \sum_{i=1}^m x_i RS_i(0) + yRA(0) = R V_{(x,y)}(0). \end{aligned}$$

This shows the uniqueness of risk-neutral probability.

Let $\mathbf{P}_* = (p_1^*, \dots, p_N^*)$ be a unique risk-neutral probability. We know that the model satisfies the No Arbitrage principle and we need to show its

completeness. Suppose it is not complete and consider an option with pay-off $H(1)$ which cannot be replicated. We shall construct another risk-neutral probability $\mathbf{P}'_*$.

Let $V = \{(V_{(x,y)}(1)(\omega_1), \dots, V_{(x,y)}(1)(\omega_N)) \in \mathbf{R}^N \mid \omega_j \in \Omega, (x, y) \in \mathbf{R}^{n+1}\}$ be the set of all values at time one of all portfolios (x, y) with respect to all scenarios ω_j . It is a vector subspace of $\mathbf{R}^N$ and $(H(1)(\omega_1), \dots, H(1)(\omega_N)) \notin V$ because $H(1)$ cannot be replicated.

Let $\langle x, y \rangle_* = \sum_{j=1}^N x_j y_j p_j^* = \mathbf{E}_{\mathbf{P}_*}(xy)$ be an inner product on $\mathbf{R}^N$. Consider an element $w \in \mathbf{R}^N$ which does not belong to V and apply the lemma to the space V and $A = \{w\}$. It follows that there exists $v \in V^{\perp_*}$ such that $\langle v, w \rangle > 0$ (this implies that $v \neq 0$). Since $(1, \dots, 1) \in V$ we get that $0 = \langle v, (1, \dots, 1) \rangle = \mathbf{E}_{\mathbf{P}_*}(v)$. Let

$$q_j^* = \left(1 + \frac{v_j}{2a}\right) p_j^*,$$

where $a = \max |v_j|$. The numbers q_j^* define a probability because

$$\sum q_j^* = \sum p_j^* + \sum \frac{v_j}{2a} p_j^* = 1 + \frac{1}{2a} E_*(v) = 1.$$

Moreover, since $1 + \frac{v_j}{2a}$ is positive and different from 1 for at least one j , we get that the probability $\mathbf{Q}$ defined by q_j^* 's is different from $\mathbf{P}_*$. It remains to check that it is risk-neutral. Consider a portfolio $(0, \dots, 1, \dots, 0, 0)$ (all positions zero except the i -th one equal to 1). Its value at time 1 is given by $V(1) = S_i(1)$.

$$\begin{aligned} \mathbf{E}_{\mathbf{Q}_*}(S_i(1)) &= \sum_j S_i(1)(\omega_j) q_j^* \\ &= \mathbf{E}_{\mathbf{P}_*} \left(\left(1 + \frac{v}{2a}\right) S_i(1) \right) \\ &= \mathbf{E}_{\mathbf{P}_*}(S_i(1)) + \frac{1}{2a} E_*(v S_i(1)) \\ &= \mathbf{E}_{\mathbf{P}_*}(S_i(1)) = S_i(0)(1 + R), \end{aligned}$$

which proves that the new probability is risk-neutral. $\square$

As we mentioned in the beginning of this section, incomplete markets are more interesting and the above theorem tells us that in such markets there are many risk-neutral probability measures. Thus the following problem naturally arises.

Question 6.13. Which risk-neutral probability should be used to price a contingent claim?

Recall that the no-arbitrage principle implies that the price of an option is the risk-neutral expectation of the pay-off:

$$\begin{aligned} C &= \mathbf{E}_{\mathbf{P}_*}(f(S_1(1), \dots, S_m(1))) \\ &= \sum_{i=1}^N p_{i*} f(S_1(\omega_i), \dots, S_m(\omega_N)). \end{aligned}$$

Let $F: \Omega \rightarrow \mathbf{R}$ be the simplified notation for the pay-off. That is,

$$F(\omega) = f(S_1(\omega), \dots, S_m(\omega))$$

and we consider F as a vector in $\mathbf{R}^N$ with coordinates $(F(\omega_1), \dots, F(\omega_N))$. Then we can consider the price of the option as a function of a risk-neutral probability

$$\mathbf{P}_* \mapsto C(\mathbf{P}_*) = \mathbf{E}_{\mathbf{P}_*}(F) = \langle \mathbf{P}_*, F \rangle$$

and ask the question of extremising it.

Let us summarize these observations. Let $\Delta_0 \subset \mathbf{R}^N$ be the set of martingale probability measures. It is obtained as the intersection of the set

$$\Delta = \left\{ (p_1, \dots, p_N) \in \mathbf{R}^N \mid \sum p_i = 1, p_i \geq 0 \right\}$$

of all probability measures with the set of vectors satisfying the system of m linear equations

$$\mathbf{E}_{\mathbf{P}_*}(S_i(1)) = RS_i(0), \quad i = 1, 2, \dots, m.$$

Thus the set Δ_0 is a bounded convex polytope and the set of risk-neutral measures is its interior. The no-arbitrage price of the option is a function $C: \text{int}\Delta_0 \rightarrow \mathbf{R}$ given by

$$C(\mathbf{P}_*) = \langle \mathbf{P}_*, F \rangle.$$

Thus it is a restriction of a linear functional given by the inner product with the pay-off vector to the interior of the polytope Δ_0 . When we consider the restriction of this functional to the whole polytope Δ_0 (not only to its interior) then its maxima and minima are attained on vertices of Δ_0 and finding them is a standard problem of linear programming.

6.14 Example: Two assets, each following the one-step binomial model

The initial set up

Consider the model with two risky assets with final prices $S_1, S_2: \Omega \rightarrow \mathbf{R}$, where $\Omega = \{u, d\}^2$. Thus the event $\{ud\}$ means that the price of the first asset went up and the price of the second asset decreased. Moreover,

$$\begin{aligned} S_1(u*) &= s_1(0)U_1 & S_1(d*) &= s_1(0)D_1 \\ S_2(*u) &= s_2(0)U_2 & S_2(*d) &= s_2(0)D_2, \end{aligned}$$

where $D_1, D_2 < R < U_1, U_2$. This means that each of the assets follow its own binomial model. Let, $(p_1, 1 - p_1)$ and $(p_2, 1 - p_2)$ be the risk-neutral measures for each asset within their respective one-step binomial models. They are given by

$$p_i = \frac{R - D_i}{U_i - D_i}$$

for $i = 1, 2$. Finally, let

$$q_1 = P(uu), \quad q_2 = P(ud), \quad q_3 = P(du), \quad q_4 = P(dd).$$

The set of risk-neutral measures

The risk-neutral measures $(q_1, q_2, q_3, q_4) \in \mathbf{R}^4$ are solutions of the following system of linear equations:

$$\begin{cases} q_1 + q_2 + q_3 + q_4 &= 1 \\ S_1(uu)q_1 + S_1(ud)q_2 + S_1(du)q_3 + S_1(dd)q_4 &= s_1(0)R \\ S_2(uu)q_1 + S_2(ud)q_2 + S_2(du)q_3 + S_2(dd)q_4 &= s_2(0)R \end{cases} \quad (6.5)$$

and they satisfy the inequalities $0 < q_i < 1$.

There is a one dimensional set of solutions of the above system of equations:

$$\begin{cases} q_1 &= t \\ q_2 &= p_1 - t \\ q_3 &= p_2 - t \\ q_4 &= 1 - p_1 - p_2 + t, \end{cases} \quad (6.6)$$

where $\max\{p_1 + p_2 - 1, 0\} < t < \min\{p_1, p_2\}$.

Thus the set Δ_0 of martingale measures is the intersection of the set of all measures with a line which is an interval. Observe that since $p_1, p_2 > 0$, this interval is nontrivial.

Extremal prices of a contingent claim

Observe that since the set of risk-neutral probabilities is the interior of the closed polytope Δ_0 , the extremal prices are never achieved. In other words, the set of the no-arbitrage prices is an open interval $(H_{\min}, H_{\max})$, where $H_{\min}$ and $H_{\max}$ are the minimum and the maximum of the function $\mathbf{Q} \mapsto \mathbf{E}_{\mathbf{Q}}(F)$ for $\mathbf{Q} \in \Delta_0$. Our aim is to compute these values.

Let $F = (F_1, F_2, F_3, F_4) \in \mathbf{R}^4$ be the pay-off of a contingent claim which depends on the final prices of both assets. That is, $F_i = f(S_1(\omega_i), S_2(\omega_i))$ for a suitable function $f \in \mathbf{R}_+^2 \rightarrow \mathbf{R}$. The no-arbitrage price of such a contingent claim is given by the expectation of the pay-off with respect to a risk-neutral measure. We get that

$$\begin{aligned} H(t) = H(Q_t) &= F_1 t + F_2(p_1 - t) + F_3(p_2 - t) + F_4(1 - p_1 - p_2 + t) \\ &= F_2 p_1 + F_3 p_2 + F_4(1 - p_1 - p_2) + t(F_1 - F_2 - F_3 + F_4). \end{aligned}$$

Determining the extremal prices is broken into several cases depending on whether $H(t)$ is increasing or decreasing.

Case 1: $F_1 - F_2 - F_3 + F_4 = 0$. In this case the price is constant (it means that the contingent claim has a replication portfolio).

Case 2: $F_1 - F_2 - F_3 + F_4 > 0$. In this case the price function $H(t)$ is increasing and:

$$\begin{aligned} \min H(t) &= \begin{cases} H(p_1 + p_2 - 1), & \text{if } p_1 + p_2 > 1, \\ H(0), & \text{otherwise.} \end{cases} \\ &= \begin{cases} F_2(1 - p_1) + F_3(1 - p_2) + F_4(p_1 + p_2 - 1), & \text{if } p_1 + p_2 > 1, \\ F_1(1 - p_1 - p_2) + F_2 p_2 + F_3 p_1, & \text{otherwise.} \end{cases} \end{aligned}$$

$$\begin{aligned}\max H(t) &= \begin{cases} H(p_1), & \text{if } p_1 \leq p_2, \\ H(p_2), & \text{if } p_1 \geq p_2. \end{cases} \\ &= \begin{cases} F_1(1 - p_2) + F_2(p_2 - p_1) + F_4 p_1, & \text{if } p_1 \leq p_2, \\ F_1(1 - p_1) + F_3(p_1 - p_2) + F_4 p_2, & \text{if } p_1 \geq p_2. \end{cases}\end{aligned}$$

Case 3: $F_1 - F_2 - F_3 + F_4 < 0$. In this case the price is decreasing and the extrema are given by the reversed case above. That is, the minimum is given by the above maximum and vice versa.

Drawing

The following figure shows several examples of martingale measures drawn in the 3-dimensional simplex of all probability measures on the sample space with 4 elements. The typical case is drawn in red: the interval ends in the interior of two faces of the simplex. A less typical cases are drawn in blue, where $t = 0$ and $p_1 = p_2 = \frac{1}{2}$ and both ends are in the edges.

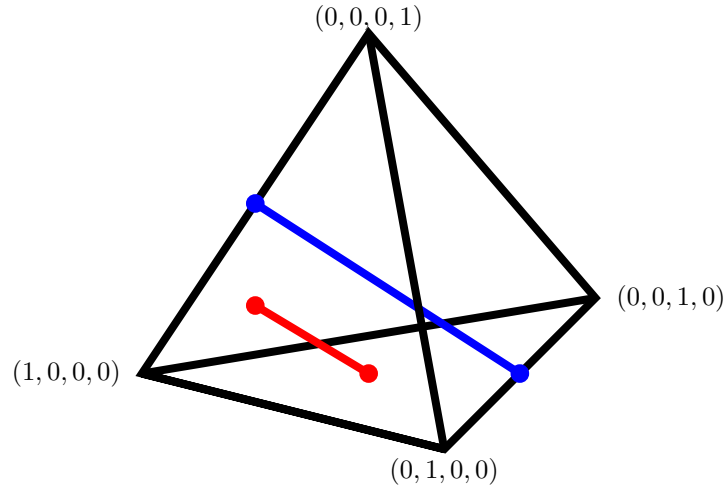


Figure 6.1: Risk-neutral measures

The efficient portfolio

Recall that we are looking for a portfolio $(\beta, \alpha_1, \alpha_2)$ with minimal initial value and the final value higher than the pay-off of the contingent claim. The latter means that

$$V(\omega) \geq F(S_1(\omega), S_2(\omega)),$$

where $\omega \in \{uu, ud, du, dd\}$. These four inequalities define a subset $\Pi \subset \mathbf{R}^3$ which is the intersection of four half-spaces. We are then looking for a minimum of the functional

$$(\beta, \alpha_1, \alpha_2) \mapsto \beta + \alpha_1 s_1(0) + \alpha_2 s_2(0)$$

over the subset Π . Since this subset is a convex, possibly unbounded, polyhedron, the minimum is attained at its boundary. Since, according to the Strong Duality, we know that the value of this minimum is equal to the $\max H(t)$ computed above, it is enough to check for the solution of the systems

$$\begin{cases} \beta + s_1(0)\alpha_1 + s_2(0)\alpha_2 & = \max H(t) \\ \beta(1+R) + S_1(\omega_i)\alpha_1 + S_2(\omega_i)\alpha_2 & = F(\omega_i) \\ \beta(1+R) + S_1(\omega_j)\alpha_1 + S_2(\omega_j)\alpha_2 & = F(\omega_j) \end{cases}$$

for $\omega_i \neq \omega_j$ from the set Ω .

We will apply Cramer's rule to solve the above system. Let A_{ω_i, ω_j} denote the coefficient matrix of the above system and let A_{ω_i, ω_j}^k , where $k = 1, 2, 3$. Denote the coefficient matrix with the k -th row replaced by $(\max H(t), F(\omega_i), F(\omega_j))^T$. Then, if $\det A_{\omega_i, \omega_j} \neq 0$ then, according to Cramer's rule, we get

$$\beta = \frac{\det A_{\omega_i, \omega_j}^1}{\det A_{\omega_i, \omega_j}^1}, \quad \alpha_1 = \frac{\det A_{\omega_i, \omega_j}^2}{\det A_{\omega_i, \omega_j}^2}, \quad \alpha_2 = \frac{\det A_{\omega_i, \omega_j}^3}{\det A_{\omega_i, \omega_j}^3}.$$

Consider the case of $\omega_1 = uu$ and $\omega_2 = ud$. Then

$$\det A_{uu, ud} = s_1(0)s_2(0)(U_1 - R)(D_2 - U_2) \neq 0,$$

because $D_i < R < U_i$. Thus this system has a unique solution given by

$$\beta = \frac{\max H(t)U_1(D_2 - U_2) + F(uu)(U_1 - D_2) + F(ud)(U_2 - U_1)}{s_1(0)s_2(0)(U_1 - R)(D_2 - U_2)}$$

$$\alpha_1 = \frac{\max H(t)R(U_2 - D_2) + F(uu)(D_2 - R) + F(ud)(R - U_2)}{s_1(0)s_2(0)(U_1 - R)(D_2 - U_2)}$$

$$\alpha_2 = \frac{(F(uu) - F(ud))(R - U_1)}{s_1(0)s_2(0)(U_1 - R)(D_2 - U_2)}$$

6.15 Summary

1. An investor wants to sell a call option with pay-off based on final prices of n risky assets and they want to know:
 - The price they can charge.
 - How to reduce the risk.
2. We create the model with finite number of scenarios $\Omega = \{\omega_1, \dots, \omega_N\}$. We want the model to be reasonably accurate and that is why we need a large number of scenarios: $N > n$. This implies that the model is incomplete.
3. We determine the set of risk-neutral probabilities. Since this set comprises of solutions of linear equations intersected with the standard simplex Δ of all probabilities, we conclude that it is the interior of a convex bounded polytope Δ_0 of martingale measures.
4. Any no-arbitrage price of the option is equal to the risk-neutral expectation with respect to a risk-neutral probability $Q \in \text{int}\Delta_0$. Thus we consider the price of the option as a linear function $H: \Delta_0 \rightarrow \mathbf{R}$. We conclude that such a linear function attains its maximum and minimum at a vertex of Δ_0 . Computing the extrema of such function is the standard task for a Linear Programming. There are efficient algorithms. We tell the investor this maximum value and the price they can charge for the option within is any value smaller than this maximum.
5. To answer the second question of the investor we consider the set of portfolios defined by the linear inequalities

$$V(\omega) \geq F(\omega),$$

where $V(\omega)$ is the final value of the portfolio with respect to scenario ω and $F(\omega)$ is the pay-off of the option with respect to the same scenario.

Thus each portfolio in this set will provide enough funds to settle the option. We then look for a portfolio in the above set with minimal initial value. *The fact that the above set is non-empty and that such a minimising portfolio exists follows from the so called Duality Theorem from Linear Programming, which we haven't discussed here.* Again, finding such a minimising portfolio is the standard Linear Programming task and there are efficient algorithms.

The example from the previous section carried all the steps of the above strategy for the case of two risky assets each following their own binomial model. In this case the we could provide not so complicated formulae.

Observe that in the above strategy we had to choose the model. This can never be done right and this step is always problematic. Our models are mathematically correct but nobody knows which model one should apply.

6.16 Exercises

6a. Show that $\mathbf{E}_{\mathbf{P}_*}(S_i(1)) = RS_i(0)$ is equivalent to $\mathbf{E}_{\mathbf{P}_*}(K(1)) = r$.

6b. Recall the conditions under which a system of linear equations has

1. no solutions;
2. a unique solution;
3. infinitely many solutions.

Recall methods of solving linear equations and search for a software which can quickly do it.

6c. Read internet resources on polytopes. Convince yourself that the set of martingale measures is a convex bounded polytope. They read about linear programming.

6d. Consider the system of equations (6.5) and:

1. Verify that (6.6) is indeed a solution of the system (6.5).
2. Show that since $0 < q_i < 1$, we get that

$$\max\{p_1 + p_2 - 1, 0\} < t < \min\{p_1, p_2\}.$$

6e. Verify the computation on page 66:

$$\begin{aligned} H(t) &= H(Q_t) = F_1 t + F_2(p_1 - t) + F_3(p_2 - t) + F_4(1 - p_1 - p_2 + t) \\ &= F_2 p_1 + F_3 p_2 + F_4(1 - p_1 - p_2) + t(F_1 - F_2 - F_3 + F_4). \end{aligned}$$

6f. Verify the computations of $\min H(t)$ and $\max H(t)$.

6g. Verify the computations of the efficient portfolio.

6h. Show that charging the maximum value $\max H(t)$ for the option leads to arbitrage. This exercise explains why we need the condition that the risk-neutral probability is equivalent to the initial probability. In other words, martingale measure may lead to arbitrage.

6i. Consider a single period model with two assets following binomial price dynamics with $S_1(0) = 100$, $S_2(0) = 100$, $R = 1.05$ and

$$S_1(1) = \begin{cases} 110 \\ 90 \end{cases} \quad S_2(1) = \begin{cases} 120 \\ 80 \end{cases}.$$

Let $f: \mathbf{R}^2 \rightarrow \mathbf{R}$ be the pay-off function given by

1. $f(x, y) = \max \left\{ 0, \frac{x+y}{2} - 100 \right\};$
2. $f(x, y) = \max \left\{ 0, \frac{x+y}{2} - 105 \right\};$
3. $f(x, y) = \max \left\{ 0, \frac{x+y}{2} - 95 \right\};$
4. $f(x, y) = \max \left\{ 0, \frac{x+y}{2} - 90 \right\}.$

In each case determine the set of martingale measures and compute the minimal and the maximal prices. Check whether these prices lead to arbitrage.

7 Pricing in multi-step binomial model

7.1 Recalling the model

In this section we consider n -step binomial model with 1 risky asset. Recall that the sample space of the model is $\Omega = \{0, 1\}^n$ and the price process is a sequence of random variables $S(t): \Omega \rightarrow \mathbf{R}$ for $t = 0, 1, 2, \dots, n$ defined by

$$S(t)(\omega) = S(0)U^k D^{t-k},$$

where $S(0)$ is known, $\omega = (\omega_1, \omega_2, \dots, \omega_n) \in \Omega$, k is the number of 1's in the sequence $(\omega_1, \omega_2, \dots, \omega_t)$ and $0 < D < R < U$, where $R = 1 + r$ and r is a risk-free rate. The equivalent formula is

$$S(t)(\omega) = S(0)U^{\sum_{i=1}^t \omega_i} D^{t - \sum_{i=1}^t \omega_i},$$

which is straightforward to use for computer programs. The model also has a probability function $\mathbf{P}: 2^\Omega \rightarrow [0, 1]$. As before, however, it will play no role in pricing options.

It is useful to depict elements of the sample space $\Omega = \{0, 1\}^n$ as leaves of a rooted directed binary tree of height n .

Remark 7.2. In the above model we use the letters U and D for the price jumps rather than returns, which in this case are equal to $U - 1$ and $D - 1$ respectively. There are different conventions and which one is used should be always clear from the context.

7.3 Pricing an option

Consider a contingent claim with pay-off $C(n): \Omega \rightarrow \mathbf{R}$. For example, a European call option has the pay-off given by $C(n) = \max\{S(n) - X, 0\}$, where X is the strike price.

Recall from Example 5.12 that in the single-step model an option can be replicated and the replication portfolio is given by

$$x = \frac{C(1)(u) - C(1)(d)}{S(0)(U - D)} \quad y = \frac{C(1)(d)U - C(1)(u)D}{(U - D)A(0)R},$$

provided $U \neq D$. The no arbitrage principle implies that the value of an option is the initial value of the replicating portfolio. In Section 5.15 we

observed that this equal to the risk-neutral expectation of the pay-off:

$$\begin{aligned} C(0) &= xS(0) + yA(0) \\ &= R^{-1}\mathbf{E}_{\mathbf{P}_*}(C(1)) \end{aligned}$$

provided $D < R < U$.

The guiding idea in this section is to apply the single-step argument recursively to obtain a rational value of the contingent claim at each time. Moreover, at each time we shall obtain a suitable replicating portfolio.

Example 7.4. Consider an option with exercise time $n = 2$ and payoff

$$C(2) = \max \left\{ 0, \frac{S(1) + S(2)}{2} - X \right\}.$$

This is a European call with the average price as the underlying security. Let $U = 1.1$, $D = 0.9$, $R = 1.05$, $S(0) = 100$ and $X = 90$. Then

$$C(2) = \begin{cases} C(2)(11) = 25.5, \\ C(2)(10) = 14.5, \\ C(2)(01) = 4.5, \\ C(2)(00) = 0.0. \end{cases}$$

We can consider this 2-step model as three 1-step models combined together which is easy to see on the corresponding tree. Since the returns in each of these single-step models are the same so is the risk neutral probability which is equal to $\mathbf{P}_* = (p_*, 1 - p_*) = (0.25, 0.75)$. Then by applying the single step model three times we get

$$\begin{aligned} C(1)(1) &= \frac{1}{R} (p_* C(2)(11) + (1 - p_*) C(2)(10)) = 21.67, \\ C(1)(0) &= \frac{1}{R} (p_* C(2)(01) + (1 - p_*) C(2)(00)) = 3.21, \\ C(0) &= \frac{1}{R} (p_* C(1)(1) + (1 - p_*) C(1)(0)) = 6.24. \end{aligned}$$

◇

Observe that in the n -step model we need to apply a single-step computation $1 + 2 + 4 + \dots + 2^{n-1}$ times which is unfeasible for large n . Notice that the contingent claim in the above example depends on the price of the risky asset at time $t = 1$ and $t = 2$ which forces the pay-off to attain four different values. Let us make the following definition.

A contingent claim is called ***path-independent*** if its pay-off depends only on the final price of the underlying asset. More precisely, $C(n) = f(S(n))$, for some function $f: \mathbf{R}_+ \rightarrow \mathbf{R}$. Otherwise the contingent claim is ***path-dependent***. So in the above example we considered a path dependent option. Let us consider a more abstract example.

Example 7.5. Let $n = 2$ and let $C(n)$ be the pay-off of a, possibly time dependent, contingent claim. Assume for simplicity that $R = 1$ and let $q_* = (1 - p_*)$.

$$\begin{aligned} C(0) &= \mathbf{E}_{\mathbf{P}_*}(C(1)) \\ &= p_* C(1)(1) + q_* C(1)(0) \\ &= p_*(p_* C(2)(11) + q_* C(2)(10)) \\ &\quad + q_*(C(2)(01) + q_* C(2)(00)) \\ &= p_*^2 C(2)(11) + p_* q_* C(2)(10) + p_* p_* C(2)(01) + q_*^2 C(2)(00) \end{aligned}$$

If the contingent claim is path independent then $C(2)(10) = C(2)(01)$ and we obtain that

$$C(0) = p_*^2 C(2)(11) + 2p_* q_* C(2)(10) + q_*^2 C(2)(00).$$

Similar computation for $n = 3$, in the case of time independent option, yields

$$C(0) = p_*^3 C(3)(111) + 3p_*^2 q_* C(3)(110) + 3p_* q_*^2 C(3)(100) + q_*^3 C(3)(000).$$

◇

Theorem 7.6. *Let $C(n) = f(S(n))$ be the pay-off of a path independent contingent claim. Then its initial value is given by*

$$C(0) = R^{-n} \sum_{k=0}^n \binom{n}{k} p_*^k (1 - p_*)^{n-k} f(S(0)U^k D^{n-k}).$$

Proof. We prove the statement by induction. For $n = 1$ it has been proven in Section 5.15. Assume that the formula is true for $n - 1$. Then for n we obtain the following.

$$\begin{aligned}
C(0) &= R^{-1} (p_* C(1)(1) + (1 - p_*) C(1)(0)) \\
&= R^{-1} p_* \left(R^{n-1} \sum_{k=0}^{n-1} \binom{n-1}{k} p_*^k (1 - p_*)^{n-1-k} f(S(0) U^{k+1} D^{n-1-k}) \right) \\
&\quad + R^{-1} (1 - p_*) \left(R^{n-1} \sum_{k=0}^{n-1} \binom{n-1}{k} p_*^k (1 - p_*)^{n-1-k} f(S(0) U^k D^{n-k}) \right) \\
&= R^{-n} \sum_{k=0}^{n-1} \binom{n-1}{k} p_*^{k+1} (1 - p_*)^{n-1-k} f(S(0) U^{k+1} D^{n-1-k}) \\
&\quad + R^{-n} \sum_{k=0}^{n-1} \binom{n-1}{k} p_*^k (1 - p_*)^{n-k} f(S(0) U^k D^{n-k}) \\
&= R^{-n} \sum_{k=1}^n \binom{n-1}{k-1} p_*^k (1 - p_*)^{n-k} f(S(0) U^k D^{n-k}) \\
&\quad + R^{-n} \sum_{k=0}^{n-1} \binom{n-1}{k} p_*^k (1 - p_*)^{n-k} f(S(0) U^k D^{n-k}) \\
&= R^{-n} p_*^n f(S(0) U^n) \\
&\quad + R^{-n} \sum_{k=1}^{n-1} \left[\binom{n-1}{k-1} + \binom{n-1}{k} \right] p_*^k (1 - p_*)^{n-k} f(S(0) U^k D^{n-k}) \\
&\quad + R^{-n} (1 - p_*)^n f(S(0) D^n) \\
&= R^{-n} \sum_{k=0}^n \binom{n}{k} p_*^k (1 - p_*)^{n-k} f(S(0) U^k D^{n-k})
\end{aligned}$$

It remains to show that $\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k}$ which is left as a straightforward exercise. $\square$

Remark 7.7. Observe that $C(0) = R^{-n} \mathbf{E}_{\mathbf{P}_*^n}(C(n))$, where $\mathbf{P}_*^n$ is the probability measure on $\Omega = \{0, 1\}^n$ which is n -th power of the risk-neutral measure $\mathbf{P}_*$ on $\{0, 1\}$.

7.8 Exercises

7a. Recall the binomial model and the computation of the risk-neutral measure for single-period model with one asset (Section 5.15).

7b. Let $(\Omega_1, \mathbf{P}_1)$ and $(\Omega_2, \mathbf{P}_2)$ be finite probability spaces. Recall what is meant by a product probability measure $\mathbf{P}_1 \otimes \mathbf{P}_2$ on the product sample space $\Omega_1 \times \Omega_2$.

7c. Let $\Omega = \{0, 1\}$ and let $\mathbf{P} = (p, 1 - p)$ and $\mathbf{Q} = (q, 1 - q)$ be two probability measures. Write formulas for the following product measures on Ω^n for suitable n :

- | | | |
|--------------------------------------|--|--|
| 1. $\mathbf{P} \otimes \mathbf{Q}$; | 3. $\mathbf{P}^2$; | 5. $\mathbf{P}^2 \otimes \mathbf{Q}^2$. |
| 2. $\mathbf{P} \otimes \mathbf{P}$; | 4. $\mathbf{P}^2 \otimes \mathbf{Q}$; | |

Evaluate the above examples for $p = 1/4$ and $q = 1/3$.

7d. Let $\Omega = \{0, 1\}$. Give an example of a probability measure on Ω^2 which is not a product measure and justify your answer. For each $n > 2$ give an example of a probability measure on Ω^n which is not a product measure.

Remark 7.9. The purpose of the above exercises is to give you an intuition that a product probability measure is very rare among all measures. Thus the fact that in our n -step binomial model with 1 asset the risk-neutral measure is a product measure is remarkable.

7e. Prove the statement from Remark 7.7. This shows, in particular, that the risk-neutral measure in a multi-step binomial model with one asset is unique. Investigate whether the proof of Theorem 6.12 can be adapted to show that also in the multi-step case uniqueness of risk-neutral measure is equivalent to the existence of a replicating portfolio.

7f. Recall and prove the binomial formula for computing the power of a sum of two numbers $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$. Show that

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k}$$

for $n > k > 0$. Read the Wikipedia article on the binomial coefficient.

https://en.wikipedia.org/wiki/Binomial_coefficient

7g. Let $S(0) = 100$, $U = 1.1$, $D = 0.9$, $R = 1$. Compute the prices of contingent claims with the following pay-offs:

1. $C(2) = \max\{S(2) - 100, 0\};$
2. $C(2) = \max\{S(2) - 102, 0\};$
3. $C(2) = \max\{S(2) - 98, 0\};$
4. $C(3) = \max\{S(3) - 100, 0\};$
5. $C(3) = \max\{S(3) - 102, 0\};$
6. $C(3) = \max\{S(3) - 98, 0\};$
7. $C(n) = \max\{S(n) - 100, 0\};$
8. $C(2) = \max\left\{\frac{S(1)+S(2)}{2} - 100, 0\right\};$

7h. Read about basket options. Consider the following basket option which is a sum of a European call and put

$$C(2) = \max\{S(2) - 100, 0\} + \max\{0, 90 - S(2)\}.$$

Is it path-dependent? Compute its price.

8 Continuous time model: Black-Scholes formula

A general continuous time model with m -risky assets consists of a probability space $(\Omega, \mathcal{F}, \mathbf{P})$ and a family of random variables $S_i(t)$ for $i = 1, 2, \dots, m$ and $t \in [0, T]$. The sample space is not assumed to be finite and in most examples it will be, in fact, infinite with continuous probability distribution.

In the next sections we will present a construction of the Black-Scholes model and we will derive the Black-Scholes equation. Heuristically speaking, the model is obtained as a limit of binomial models. More precisely, we consider a time interval $[0, T]$ which we divide into n -equal subintervals and consider the n -step binomial model on $[0, T]$. Thus each time step has length $h = T/n$. We then take larger and larger values of n (equivalently, smaller time steps h) and pass to the limit as $n \rightarrow \infty$ (or $h \rightarrow 0$).

8.1 Binomial model revisited

Throughout this section we assume that in all binomial models the probability is uniform. That is, it is the product probability of the 1-step probability with $p = 1/2$. Thus in the n -step model the probability of each scenario is equal to $1/2^n$.

Recall that in the n -step binomial model with one asset we have

$$S(k) = S(0)\xi(k),$$

where $\xi(k): \Omega = \{0, 1\}^n \rightarrow \{U^k, U^{k-1}D, \dots, UD^{k-1}, D^k\}$ and $0 < D < U$. Observe (exercise) that $\xi(k) = \xi_1 \xi_2 \dots \xi_k$, where $\xi_j: \Omega \rightarrow \{U, D\}$ is the random variable defined by

$$\xi_j(\omega) = \begin{cases} U & \text{if } \omega_j = 1 \\ D & \text{if } \omega_j = 0. \end{cases}$$

It follows that $\log \xi(k) = \log \xi_1 + \dots + \log \xi_k$ and we have that

$$\mathbf{E}(\log \xi(k)) = k\mathbf{E}(\log \xi_1), \quad (8.1)$$

because $\mathbf{E}(\log \xi_i) = \mathbf{E}(\log \xi_j)$. Moreover, the random variables ξ_j are jointly independent. This implies, that

$$\text{Var}(\log \xi(k)) = \text{Var}(\log \xi_1 + \dots + \log \xi_k) = k \text{Var}(\log \xi_1). \quad (8.2)$$

Take the time interval $[0, T]$ and consider an n -step model by subdividing the interval $[0, T]$ into n equal subintervals $[kh, (k+1)h]$, for $k = 0, 1, \dots, n-1$ and $h = T/n$. Let

$$\mu = \frac{1}{h} \mathbf{E}(\log \xi(1)) \quad \text{and} \quad \sigma^2 = \frac{1}{h} \text{Var}(\log \xi(1)).$$

It follows that $\log \xi(1) = \mu h \pm \sigma \sqrt{h}$ and hence we obtain that

$$U = e^{\mu h + \sigma \sqrt{h}} \quad \text{and} \quad D = e^{\mu h - \sigma \sqrt{h}}.$$

Consequently we get that the price of the asset at time h is given by

$$S(h) = S(0) e^{\mu h + X_1 \sigma \sqrt{h}},$$

where X_1 is a random variable with distribution

$$\mathbf{P}(X_1 = 1) = \frac{1}{2} = \mathbf{P}(X_1 = -1).$$

More generally, for $t = kh$ we have

$$S(t) = S(0) e^{\mu t + \sigma W_n(t)},$$

where $W_n(t) = \sqrt{h}(X_1 + X_2 + \dots + X_k)$ and X_i are independent random variables with equal distribution

$$\mathbf{P}(X_i = 1) = \frac{1}{2} = \mathbf{P}(X_i = -1).$$

Observe that in this approach the sample space of the model became unimportant; all we care about is that the random variables X_i are independent and identically distributed.

Example 8.2. Consider the probability space $([0, 1], \mathcal{B}, \mathbf{P})$, where the sample space is the unit interval $[0, 1] \subset \mathbf{R}$, the σ -algebra of measurable sets is the standard Borel algebra and the probability is defined by $\mathbf{P}([a, b]) = b - a$. For each natural number $k \in \mathbf{N}$ define a random variable $X_k: [0, 1] \rightarrow \{-1, 1\}$ by

$$X_k(\omega) = \begin{cases} 1 & \text{if } \omega \in \bigcup_{\ell=0}^{2^{k-1}-1} \left[\frac{2\ell}{2^k}, \frac{2\ell+1}{2^k} \right) \\ -1 & \text{otherwise.} \end{cases}$$

The following figure shows the graphs of the first three variables. These random variables are (obviously) identically distributed and (less obviously) jointly independent. Thus they can be used in the modified binomial model with any time steps. $\diamond$

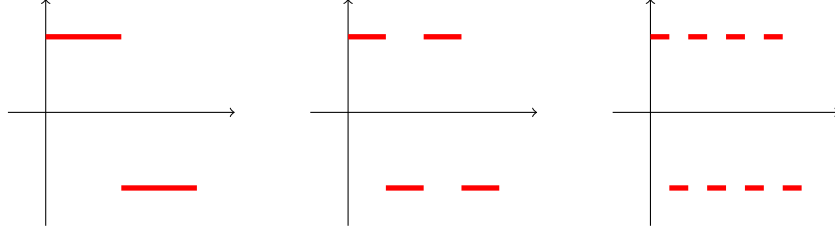


Figure 8.1: Random variables X_1 , X_2 and X_3 .

Example 8.3. Consider the set $\Omega_\infty = \{0, 1\}^\infty$ of infinite binary sequences. Let $X_j: \Omega \rightarrow \{-1, 1\}$ be a function defined by

$$X_j(\omega) = \begin{cases} 1 & \text{if } \omega_j = 1 \\ -1 & \text{if } \omega_j = 0. \end{cases}$$

We will introduce the structure of a probability space on Ω as an exercise at the end of this section. With this structure the variables X_k will be independent and identically distributed. $\diamond$

8.4 Scaled random walks

Let $X_j: \Omega_\infty \rightarrow \{-1, 1\}$ be a sequence of random variables from Example 8.3. Let $M_0 = 0$ and

$$M_k = X_1 + X_2 + \cdots + X_k.$$

The process $(M_k)_{k=0,1,\dots}$ is called a **symmetric random walk**. For $k < \ell$ let $M_{k,\ell} = M_\ell - M_k = \sum_{j=k+1}^\ell X_j$ be the **increment** of the random walk.

Lemma 8.5. *For each choice $0 \leq k_0 < k_1 < \cdots < k_m$ the increments $M_{k_0,k_1}, M_{k_1,k_2}, \dots, M_{k_{m-1},k_m}$ are independent. Moreover,*

$$\begin{aligned} \mathbf{E}(M_{k,\ell}) &= 0 \\ \text{Var}(M_{k,\ell}) &= \ell - k, \end{aligned}$$

for each $k < \ell$.

Heuristically, the increments are independent because they depend on different coin tosses. The proof of the remaining two statements are straightforward and are left as an exercise.

Let $W_n(t) = \sqrt{h}M_k = \sqrt{h}(X_1 + X_2 + \cdots + X_k)$, where $t = hk$, be the family of random variables from the discussion of the binomial model. It is called a **scaled symmetric random walk** with time step h and jump $\pm\sqrt{h}$. Since $T = nh$ we can write

$$\frac{W_n(T)}{\sqrt{T}} = \frac{X_1 + \cdots + X_n}{\sqrt{n}}$$

It follows from the Central Limit Theorem that the distribution $W_n(T)/T$ tends to the normal distribution as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \mathbf{P} \left(a \leq \frac{W_n(T)}{\sqrt{T}} \leq b \right) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx.$$

It follows from Lemma 8.5 and the Central Limit Theorem that the distribution of $W_n(t) - W_n(s)$ converges to the normal distribution with variance equal to $t - s$. This is the motivation for the following definition.

Definition 8.6. A **Wiener process** or a **Brownian motion** is a family of random variables $W(t)$ defined for all non-negative real numbers $t \in [0, \infty)$ such that $W(0) = 0$, the increments $W(t) - W(s)$ have normal distribution with mean 0 and variance $t - s$ for all $s < t$ and the increments $W(t_k) - W(t_{k-1}), W(t_{k-1}) - W(t_{k-2}), \dots, W(t_2) - W(t_1)$ are independent for all sequences $0 \leq t_1 < t_2 < \cdots < t_k$.

Remark 8.7. An intuition for this definition is that a Wiener process *should* be a limit of a scaled random walk if we take smaller and smaller time steps. It is not at all clear that a Wiener process exists and it is nontrivial to prove that it indeed does.

Remark 8.8. Since the increments of a Brownian motion have normal distribution it implies that the sample space on which the process is defined is uncountable. Notice that the sample space in both Example 8.2 and 8.3 are uncountable.

8.9 The Black-Scholes formula

Recall from Section 8.1 that the stock price in the n -step binomial model is given by $S(t) = S(0)e^{\mu t + \sigma W_n(t)}$ for $t = hk$. Assuming the existence of the Wiener process we define

$$S(t) = S(0)e^{\mu t + \sigma W(t)}. \quad (8.3)$$

This is the **Black-Scholes formula**. Observe that taking the logarithm yields

$$\log S(t) = \log S(0) + \mu t + \sigma W(t).$$

Since $W(t)$ has normal distribution with variance t , the random variable $\log S(t)$ has normal distribution $N(\log S(0) + \mu t, \sigma^2 t)$. We thus say that the stock price in the Black-Scholes model has **log normal** distribution.

Remark 8.10. One of the criticism of the Black-Scholes model is that in reality the prices fail to have log normal distribution. For example, it has been argued that stock prices should be governed by, so called, **fat tailed** distributions instead of the normally distributed Wiener process $W(t)$.

8.11 Further comments

Random walks and Brownian motions

Consider a symmetric random walk (M_k) and let $\omega \in \Omega_\infty$ be fixed. Let $w: \mathbf{N} \rightarrow \mathbf{R}$ be a sequence defined by $w(n) = X_1(\omega) + X_2(\omega) + \cdots + X_n(\omega)$. Extend the sequence w to a function $W: [0, \infty) \rightarrow \mathbf{R}$ so that for each $n \in \mathbf{N}$ the restriction of W to the interval $[n, n+1]$ is linear. In other words the graph of W is obtained by joining the dots $(n, w(n))$ and $(n+1, w(n+1))$ with straight lines. Since $|w(n) - w(n+1)| = 1$ these lines are of slope ± 1 .

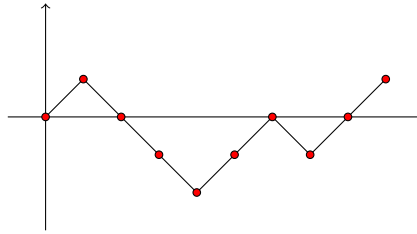


Figure 8.2: Graph of a random walk.

Now consider the scaled random walk $\sqrt{h}M_k$, where $h = 1/2$. Its graph will be similar except that the spikes will be placed more densely and the slopes will be steeper, since $\sqrt{h} > h$ if $h < 1$.

If $W(t)$ is a Wiener process and a scenario $\omega \in \Omega$ is fixed then the function $w: [0, \infty) \rightarrow \mathbf{R}$ given by $w(t) = W(t)(\omega)$ has remarkable properties. It is continuous and it is nowhere differentiable. Intuitively speaking it has spikes everywhere.

The drift

The Black-Scholes formula $S(t) = S(0)e^{\mu t + \sigma W(t)}$ contains the summand μt in the exponent. Since the Brownian motion $W(t)$ has mean equal to zero, the summand μt is responsible for the drift of prices. That is, if $\mu > 0$ then the prices will tend to grow, which usually happens in real markets.

Observe that the drift can be created in two different ways. For example, by choosing the price jumps U, D so that $\mathbf{E}(\xi(1)) > 0$ if $p = 1/2$. Or by keeping the price jumps nicely balanced and choose $p \neq 1/2$. However, choosing the probability measure for the model is, essentially, arbitrary and it hardly corresponds to reality, one can take a risk-neutral probability and a piece of data determined by the price jumps and the interest rate. This will be briefly discussed in the next section.

8.12 Exercises

8a. Write an explicit formula for $\xi(1)$, $\xi(2)$, $\xi(3)$ in an 3-step binomial model. Show that $\xi(k) = \xi_1 \xi_2 \dots \xi_k$ in a general n -step model.

8b. Compute $\mathbf{E}(\xi_i)$, $\text{Var}(\xi_i)$ and show directly (without using the independence of ξ_j 's) that $\text{Var}(\log \xi(k)) = k \text{Var}(\log \xi_1)$.

8c. Let $\mu = h^{-1}\mathbf{E}(\log \xi(1))$ and $\sigma = h^{-1} \text{Var}(\log \xi(1))$. Show that $\log \xi(1) = \mu h \pm \sigma \sqrt{h}$ and deduce that $U = e^{\mu h + \sigma \sqrt{h}}$ and $D = e^{\mu h - \sigma \sqrt{h}}$.

8d. Show that $S(t) = S(0)e^{\mu t + \sigma W_n(t)}$ in a general n -step binomial model, where $W_n(t) = \sqrt{h}(X_1 + \dots + X_k)$ and $t = hk$.

8e. Show that the variables $X_j: [0, 1] \rightarrow \{-1, 1\}$ from Example 8.2 are identically distributed. Then show that the following variables are independent:

- | | |
|----------------------|---|
| 1. X_1 and X_2 ; | 3. X_1 and X_k for any $k > 3$. |
| 2. X_1 and X_3 ; | 4. X_i and X_j for any $i \neq j$; |

Show that the variables X_j are jointly independent.

8f. Recall the definition of a probability space. In particular, of the σ -algebra of measurable sets.

8g. This exercise is based on Example 1.1.4 from the book [?] Let $\Omega = \{0, 1\}^\infty$ be the set of all infinite sequences of 0's and 1's as in Example 8.3.

1. Let $A_0 = \{\omega \in \Omega \mid \omega_1 = 0\}$ be the set of sequences with first entry equal to 0, and, similarly, A_1 be the set of sequences starting with 1. Show that the collection

$$\mathcal{F}_1 = \{\emptyset, A_0, A_1, \Omega\}$$

is a σ -algebra.

2. Let $\mathbf{P}_1: \mathcal{F}_1 \rightarrow [0, 1]$ be defined by $\mathbf{P}_1(\emptyset) = 0$, $\mathbf{P}_1(A_i) = 1/2$ and $\mathbf{P}_1(\Omega) = 1$. Show that $(\Omega, \mathcal{F}_1, \mathbf{P}_1)$ is a probability space.
3. Let $A_{00} = \{\omega \in \Omega \mid \omega_1 = \omega_2 = 0\}$ be the set of all sequences starting with 00. Define analogously the sets A_{01} , A_{10} and A_{11} . Let $\mathcal{F}_2$ be the smallest σ -algebra of subsets of Ω containing the sets A_{ij} , where $i, j \in \{0, 1\}$. Write down all the elements of $\mathcal{F}_2$.
4. Observe that a function $\mathbf{P}_2$ defined on the sets A_{ij} by $\mathbf{P}_2(A_{ij}) = 1/4$ extends uniquely to a probability measure on $\mathcal{F}_2$ and hence $(\Omega, \mathcal{F}_2, \mathbf{P}_2)$ is a probability space.
5. Observe that $\mathcal{F}_1 \subset \mathcal{F}_2$ and $\mathbf{P}_2(A_i) = \mathbf{P}_1(A_i)$, for $i = 0, 1$.
6. Let $A_{i_1 i_2 \dots i_n} \subset \Omega$ be the set of all sequences whose first n -entries are equal to $(i_1, i_2, \dots, i_n)$. Observe that there are 2^n such sets. Let $\mathcal{F}_n$ be the σ -algebra they generate. Show that $(\Omega, \mathcal{F}_n, \mathbf{P}_n)$ is a probability space, where the probability measure is defined by $\mathbf{P}_n(A_{i_1 \dots i_n}) = 1/2^n$.
7. Let $\mathcal{F} = \bigcup_{n=1}^{\infty} \mathcal{F}_n$. Show that $\mathcal{F}$ is a σ -algebra. Define $\mathbf{P}: \mathcal{F} \rightarrow [0, 1]$ by $\mathbf{P}(A) = \mathbf{P}_n(A)$ if $A \in \mathcal{F}_n$ and show that $(\Omega, \mathcal{F}, \mathbf{P})$ is a probability space.

8h. Read the wikipedia article (or other resources) about fat tailed distributions and their relevance to financial models.

9 Black-Scholes as the limit of binomial models

The Black-Scholes formula in the previous section has been *guessed* in the followings sense. We first observed that the price in the binomial model with increasing number of steps had the form $S(t) = S(0)e^{ut+\sigma W_n(t)}$ where W_n is a random walk. It follows from the Central Limit Theorem that the distributions of $W_n(T)/\sqrt{T}$ converge to the normal distribution which motivated the definition of the Brownian motion. Finally the Brownian motion replaced random walks in the new formula for the price. In this section we state and prove a theorem which says that the distribution of the prices in the sequence of finer and finer binomial models converge to the distribution of the price given by the Black-Scholes formula.

Let Ω_∞ be the space of all binary sequences. Let $t > 0$ be a fixed time which, for simplicity, we assume to be an integer. Consider the binomial model which takes n -steps per unit of time, so that at time t there will be nt number of steps. Assume, further, that the price jumps are given by

$$U_n = 1 + \frac{\sigma}{\sqrt{n}} \quad \text{and} \quad D_n = 1 - \frac{\sigma}{\sqrt{n}}, \quad (9.1)$$

where $\sigma > 0$.

Remark 9.1. Notice that this is a very special choice of price jumps which in the proof of Theorem 9.2 below will lead to a very convenient cancellation in the main computation.

Let $u_k(\omega)$ denote the number of 1's among the first k entries of the sequence ω and let $d_k(\omega)$ be the number of 0's. Thus $u_k(\omega) + d_k(\omega) = k$ and

$$u_k(\omega) - d_k(\omega) = X_1(\omega) + \cdots + X_k(\omega) = M_k(\omega),$$

where we use the notation from the previous section. Consequently, we get that

$$u_k = \frac{1}{2}(k + M_k) \quad \text{and} \quad d_k = \frac{1}{2}(k - M_k) \quad (9.2)$$

and, finally, the price at time T is

$$S_n(t) = S(0)U_n^{u_{nt}}D_n^{d_{nt}} = S(0)\left(1 + \frac{\sigma}{\sqrt{n}}\right)^{\frac{1}{2}(nt+M_{nt})}\left(1 - \frac{\sigma}{\sqrt{n}}\right)^{\frac{1}{2}(nt-M_{nt})}. \quad (9.3)$$

Theorem 9.2. *As $n \rightarrow \infty$, the distribution of $S_n(t)$ converges to the distribution of*

$$S(t) = S(0)e^{\sigma W(t) - \frac{1}{2}\sigma^2 t},$$

where $W(t)$ is a random variable with normal distribution with zero mean and variance t .

Proof. Notice that it is enough to show that the suitable convergence of distribution of logarithms. That is, to show that the distribution of

$$\log S_n(t) = \log S(0) + \frac{1}{2}(nt + M_{nt}) \log \left(1 + \frac{\sigma}{\sqrt{n}} \right) - \frac{1}{2}(nt - M_{nt}) \log \left(1 - \frac{\sigma}{\sqrt{n}} \right)$$

converges, as $n \rightarrow \infty$ to the distribution of

$$\log S(t) = \log S(0) + \sigma W(t) - \frac{1}{2}\sigma^2 t.$$

Recall that the Taylor expansion of the function $\log(1+x)$ is given by

$$\log(1+x) = x - \frac{1}{2}x^2 + O(x^3).$$

By evaluating this on $x = \pm \frac{\sigma}{\sqrt{n}}$ and substituting to the formula for $S_n(T)$ we obtain

$$\begin{aligned} \log S_n(t) &= \log S(0) + \frac{1}{2}(nt + M_{nt}) \left(\frac{\sigma}{\sqrt{n}} - \frac{\sigma^2}{2n} + O\left(n^{-\frac{3}{2}}\right) \right) \\ &\quad + \frac{1}{2}(nt - M_{nt}) \left(-\frac{\sigma}{\sqrt{n}} - \frac{\sigma^2}{2n} + O\left(n^{-\frac{3}{2}}\right) \right) \\ &= \log S(0) + nt \left(-\frac{\sigma^2}{2n} + O\left(n^{-\frac{3}{2}}\right) \right) + M_{nt} \left(\frac{\sigma}{\sqrt{n}} + O\left(n^{-\frac{3}{2}}\right) \right) \\ &= \log S(0) - \frac{1}{2}\sigma^2 t + O\left(n^{-\frac{1}{2}}\right) + \sigma \frac{M_{nt}}{\sqrt{n}} + O\left(n^{-1}\right) \frac{M_{nt}}{\sqrt{n}} \end{aligned}$$

Since $\frac{M_{nt}}{\sqrt{n}} = \frac{\sqrt{t}(X_1 + \dots + X_{nt})}{\sqrt{nt}}$ is a scaled symmetric random walk, its distribution converges to the normal distribution with zero mean and variance t , due to the Central Limit Theorem. Denoting this limit random variable by $W(t)$ we obtain that

$$\lim_{n \rightarrow \infty} \log S_n(t) = \log S(0) - \frac{1}{2}\sigma^2 t + \sigma W(t)$$

as claimed. □

9.3 A general case

Theorem 9.2 dealt with a special case of very symmetric price jumps and uniform distribution. The following, more general version, has a similar proof but a more cumbersome computation. It can be found in [?, Exercise 3.8].

Consider the binomial model with n -steps per unit of time and price jumps $U_n = e^{\frac{\sigma}{\sqrt{n}}}$ and $D_n = e^{-\frac{\sigma}{\sqrt{n}}}$. Assume moreover that the interest rate per r per unit of time, so r/n per time step in the model. This yields the risk-neutral probability:

$$p_n = \frac{\frac{r}{n} + 1 - e^{-\frac{\sigma}{\sqrt{n}}}}{e^{\frac{\sigma}{\sqrt{n}}} - e^{-\frac{\sigma}{\sqrt{n}}}} \quad \text{and} \quad 1 - p_n = \frac{e^{\frac{\sigma}{\sqrt{n}}} - \frac{r}{n} - 1}{e^{\frac{\sigma}{\sqrt{n}}} - e^{-\frac{\sigma}{\sqrt{n}}}} \quad (9.4)$$

Let $M_{nt,n} = \sum_{k=1}^{nt} X_{k,n}$, where the variables $X_{1,n}, \dots, X_{n,n}$ are iid with distribution

$$\mathbf{P}(X_{k,n} = 1) = p_n, \quad \text{and} \quad \mathbf{P}(X_{k,n} = -1) = 1 - p_n,$$

for $k = 1, 2, \dots, n$. Consequently, the stock price at time t is given by

$$\begin{aligned} S_n(t) &= S(0) U_n^{\frac{1}{2}(nt + M_{nt,n})} D_n^{\frac{1}{2}(nt - M_{nt,n})} \\ &= S(0) e^{\frac{\sigma}{\sqrt{n}} M_{nt,n}}. \end{aligned} \quad (9.5)$$

Theorem 9.4. *As $n \rightarrow \infty$, the distribution of $S_n(t)$ converges to the distribution of*

$$S(t) = S(0) e^{\sigma W(t) + (r - \frac{1}{2}\sigma^2)t},$$

where $W(t)$ is a random variable with normal distribution with zero mean and variance t .

9.5 Exercises

9a. Recall the big-O notation and apply it to Taylor expansions of the following functions

- | | | |
|---------------|----------------|-----------------------------|
| 1. e^x ; | 3. $\cos x$; | 5. $\cosh x$; |
| 2. $\sin x$; | 4. $\sinh x$; | 6. $\sum_{k=0}^n a_k x^k$. |

9b. Consider the family of binomial models with $U_n = 1 + \frac{\sigma}{\sqrt{n}}$ and $D_n = 1 - \frac{\sigma}{\sqrt{n}}$ all equipped with uniform probability as in the beginning of this section. Compare these models. More precisely, try to answer the following questions.

1. What is the maximal price of the asset at time t in each model? Are they equal?
2. What is the expected price of the asset at time t in each model? Are they equal?
3. Compare the distributions of $S_1(2)$ and $S_2(2)$.
4. Ask your own question and investigate until you feel that you know what is going on and that you can explain that to others.

9c. Draw the graphs of several log-normal distributions as accurately as possible. Develop intuitive meaning of the future price of a risky asset in the Black-Scholes model.

9d. Recall everything you know about the normal distribution.

9e. Let $\varphi: \mathbf{R} \rightarrow \mathbf{R}$ be the density function of the standard normal distribution (i.e. the one with mean 0 and variation 1):

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

Prove that the density function φ_{μ, σ^2} of the normal distribution $N(\mu, \sigma^2)$ with mean μ and variance σ^2 is equal to $\frac{1}{\sigma} \varphi\left(\frac{x-\mu}{\sigma}\right)$.

9f. The normal distribution $N(\mu, \sigma^2)$ defines a probability measure $\mathbf{P}$ on the set $\mathbf{R}$ of real numbers by

$$\mathbf{P}[a, b] = \frac{1}{\sigma\sqrt{2\pi}} \int_a^b e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx.$$

What is a random variable $X: \mathbf{R} \rightarrow \mathbf{R}$ with mean $\mathbf{E}(X) = 0$ and variance $\text{Var}(X) = \sigma^2$?

9g. The book by Shreve [?], which goes beyond the scope of this course, is recommended as a further reading. At this stage it may be useful to read some parts of it. In particular, read the historical notes about the Black-Scholes model and why it is sometimes called Black-Scholes-Merton model.

10 The Black-Scholes-Merton equation

The purpose of this section is to derive the following Black-Scholes-Merton equation

$$c_t(x, t) + rx c_x(t, x) + \frac{1}{2} \sigma^2 x^2 c_{xx}(t, x) = rc(t, x), \quad (10.1)$$

where $c: [0, T) \times [0, \infty) \rightarrow \mathbf{R}$ is a function which is at least twice differentiable. A solution $c(t, x)$ of this partial differential equation with *terminal* condition $c(T, x) = \max\{x - K, 0\}$ give the value of the European call option with strike price K and expiry date T and any time $0 \leq t \leq T$. The solution is given by

$$c(t, x) = xN(d_+(T - t, x)) - Ke^{-r(T-t)}N(d_-(T - t, x)), \quad (10.2)$$

where

$$d_{\pm}(\tau, x) = \frac{1}{\sigma\sqrt{\tau}} \left(\log \frac{x}{K} + \left(r \pm \frac{\sigma^2}{2} \right) \tau \right) \quad (10.3)$$

and N is the cumulative standard normal distribution

$$N(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-\frac{x^2}{2}} dx = \frac{1}{\sqrt{2\pi}} \int_{-y}^{\infty} e^{-\frac{x^2}{2}} dx. \quad (10.4)$$

The main difficulty in applying methods of mathematical analysis in the Black-Scholes model is that the price of an asset $S(t)$ is not differentiable as a function of t and a new essential intellectual input is needed. This is provided by the Ito-Döblin formula which we also discuss in this section.

10.1 The Ito-Döblin formula

The argument presented in this section is only a sketch and is missing details on passing to various limits. More details are presented in [?, Section 4].

Recall the Black-Scholes formula

$$S(t) = S(0)e^{\mu t + \sigma W(t)}.$$

It is a random variable and hence evaluating it on a fixed scenario $\omega \in \Omega$ we obtain a function of $t \in [0, \infty)$. Since $t \mapsto W(t)(\omega)$ is nowhere differentiable we cannot apply the standard methods of mathematical analysis. However, we proceed as follows. Let $f: \mathbf{R}^2 \rightarrow \mathbf{R}$ be a function defined by

$$f(t, x) = S(0)e^{\mu t + \sigma x}.$$

It is a smooth (i.e. infinitely differentiable) function and we can consider its Taylor expansion:

$$f(t+h, x+\Delta(x)) = f(t, x) + hf_t(t, x) + \Delta(x)f_x(t, x) + \frac{1}{2}f_{xx}(t, x)\Delta(x)^2 + \dots, \quad (10.5)$$

where the subscripts denote suitable partial derivatives. The idea is to try to apply this Taylor expansion to $f(t, W(t))$. This will be done by applying it first to $f(t, W_n(t))$ and passing to the limit as $n \rightarrow \infty$. Recall that the price of an asset in the n -step binomial model is given by

$$S_n(t) = S(0)e^{\mu t + \sigma W_n(t)},$$

where $t = kh$. In other words, $S(t) = f(t, W_n(t))$. Let

$$\begin{aligned} \Delta f(t, W_n(t)) &= f(t+h, W_n(t+h)) - f(t, W_n(t)) \\ \Delta W_N(t) &= W_n(t+h) - W_n(t) = \pm\sqrt{h}. \end{aligned}$$

It follows from the Taylor expansion (10.6) that

$$\begin{aligned} \Delta f(t, W_n(t)) &= f_t(t, W_n(t))h + f_x(t, W_n(t))\Delta W_n(t) \\ &\quad + \frac{1}{2}f_{xx}(t, W_n(t))(\Delta W_n(t))^2 + O\left(h^{\frac{3}{2}}\right), \end{aligned}$$

as $h \rightarrow 0$, where $t = kh$. Here we use the fact that $\Delta W_N(t) = \pm\sqrt{h}$ and hence the square of this increment is equal to h and the higher order terms are $O\left(h^{\frac{3}{2}}\right)$ as $h \rightarrow 0$. Using the above formula we obtain the following:

$$\begin{aligned} f(T, W_n(T)) &= f(0, W_n(0)) + \sum_{k=1}^{n-1} \Delta f(kh, W_n(kh)) \\ &= f(0, W_n(0)) + \sum_{k=1}^{n-1} f_t(kh, W_n(kh))h \\ &\quad + \sum_{k=1}^{n-1} f_x(kh, W_n(kh))\Delta W_n(kh) \quad (10.6) \\ &\quad + \frac{1}{2} \sum_{k=1}^{n-1} f_{xx}(kh, W_n(kh))h + O\left(h^{\frac{1}{2}}\right). \end{aligned}$$

Passing to the limit as $n \rightarrow \infty$ we obtain the **Ito-Döbblin formula** (cf. [?, Theorem 4.4.1]):

$$\begin{aligned} f(T, W(T)) &= f(0, W(0)) + \int_0^T f_t(t, W(t))dt \\ &\quad + \int_0^T f_x(t, W(t))dW(t) + \frac{1}{2} \int_0^T f_{xx}(t, W(t))dt, \end{aligned} \quad (10.7)$$

where the term

$$\int_0^T f_x(t, W(t))dW(t) = \lim_{n \rightarrow \infty} \sum_{k=1}^{n-1} f_x(kh, W_n(kh))\Delta W_n(kh) \quad (10.8)$$

by definition. It is called the **Ito integral**. Notice that this is an equality between random variables which becomes an equality between numbers only after evaluating it on an element ω of the sample space.

Remark 10.2. The right hand side of the above equation defining the Ito integral is a limit of a sequence of random variables which has not been defined. Definition of such a limit and proving its existence can be done with the use of a bit more advanced mathematical machinery. From now on assume that such a limit exists and that the Ito integral is well defined. A more detailed argument for the above formula is presented in [?, Section 4.4]. In particular, the following useful identities are proven there

$$dW(t)dW(t) = dt, \quad dtdW(t) = dW(t)dt = 0, \quad dtdt = 0 \quad (10.9)$$

The Ito-Döbblin formula is applicable in a more general situation in which the Brownian motion $W(t)$ is replaced by a more general process called the **Ito process**. It is a stochastic process $X(t)$ defined for all $t \geq 0$ by

$$X(t) = X(0) + \int_0^t \Delta(u)dW(u) + \int_0^t \Theta(u)du$$

where $X(0) \in \mathbf{R}$, $W(t)$ is a Brownian motion, $\Delta(u)$ and $\Theta(u)$ are processes adapted to a filtration $\mathcal{F}(t)$ associated with the process $W(t)$.

Theorem 10.3 (Ito-Döbblin formula). *Let $X(t)$ be an Ito process and let $f: \mathbf{R}^2 \rightarrow \mathbf{R}$ be a twice differentiable function. Then*

$$\begin{aligned} f(T, X(T)) &= f(0, X(0)) + \int_0^T f_t(t, X(t))dt + \int_0^T f_x(t, X(t))\Delta(t)dW(t) \\ &\quad + \int_0^T f_x(t, X(t))\Theta(t)dt + \frac{1}{2} \int_0^T f_{xx}(t, X(t))\Delta^2(t)dt. \end{aligned} \quad (10.10)$$

for every $T \geq 0$

The following is a simplified differential notation for the Ito-Döblin formula:

$$df(t, X(t)) = f_t(t, X(t))dt + f_x(t, X(t))dX(t) + \frac{1}{2}f_{xx}(t, X(t))dX(t)dX(t)dt. \quad (10.11)$$

It is obtained by *pretending* we can differentiate the original formula (we cannot!). This differential version is simpler and more convenient to use in applications. However, one has to remember that it has no mathematical meaning beyond being a convenient notation of the original formula.

Example 10.4. Let $W(t)$ be a Brownian motion and let

$$X(t) = \int_0^t \sigma(s)dW(s) + \int_0^t \left(\alpha(s) - \frac{1}{2}\sigma^2(s) \right) ds$$

be an Ito process. We have that

$$dX(t) = \sigma(t)dW(t) + \left(\alpha(t) - \frac{1}{2}\sigma^2(t) \right) dt \quad (10.12)$$

$$dX(t)dX(t) = \sigma^2(t)dW(t)dW(t) = \sigma^2(t)dt, \quad (10.13)$$

where in the computation we used the identities (10.9). Let the price process be given by

$$S(t) = S(0)e^{X(t)} = S(0)e^{\int_0^t \sigma(s)dW(s) + \int_0^t (\alpha(s) - \frac{1}{2}\sigma^2(s))ds}.$$

It can be written as $S(t) = f(X(t))$ for $f(x) = S(0)e^x$. Consequently, we have that $f'(x) = f''(x) = S(0)e^x$. By applying the Ito-Döblin formula and using the above identities we get that

$$\begin{aligned} dS(t) &= df(X(t)) \\ &= f'(X(t))dX(t) + \frac{1}{2}f''(X(t))dX(t)dX(t) \\ &= S(0)e^{X(t)}dX(t) + \frac{1}{2}S(0)e^{X(t)}dX(t)dX(t) \\ &= S(t)dX(t) + \frac{1}{2}S(t)dX(t)dX(t) \\ &= \alpha(t)S(t)dt + \sigma(t)S(t)dW(t). \end{aligned} \quad (10.14)$$

◇

10.5 Deriving the Black-Scholes-Merton equation

The strategy to derive the Black-Scholes-Merton equation is to assume that there exists a replicating portfolio and derive the relevant equation. It will then follow that solutions of the obtained equation will yield a replicating portfolio and its initial value will be the price of the option.

10.5.1 Initial computations

Consider a portfolio of value $V(t)$ at time $t \in [0, T]$ with position in stock at time t denoted by $\Delta(t)$; thus the position in money at time t is given by $V(t) - \Delta(t)S(t)$. Let $f(t, x) = e^{-rt}x$. Applying the Ito-Döblin formula and the identity (10.14) we compute first the differential of the discounted price:

$$\begin{aligned} d(e^{-rt}S(t)) &= df(t, S(t)) \\ &= f_t(t, S(t))dt + f_x(t, S(t))dS(t) + \frac{1}{2}f_{xx}(t, S(t))dS(t)dS(t) \\ &= -re^{-rt}S(t)dt + e^{-rt}dS(t) \\ &= (\alpha - r)e^{-rt}S(t)dt + \sigma e^{-rt}S(t)dW(t). \end{aligned} \quad (10.15)$$

And then we use it to compute the differential of the discounted value of the portfolio:

$$\begin{aligned} d(e^{-rt}V(t)) &= df(t, V(t)) \\ &= f_t(t, V(t))dt + f_x(t, V(t))dV(t) + \frac{1}{2}f_{xx}(t, V(t))dV(t)dV(t) \\ &= -re^{-rt}V(t)dt + e^{-rt}dV(t) \\ &= \Delta(t)(\alpha - r)e^{-rt}S(t)dt + \Delta(t)\sigma e^{-rt}S(t)dW(t) \\ &= \Delta(t)d(e^{-rt}S(t)) \end{aligned} \quad (10.16)$$

Consider a European call option with pay-off $\max\{S(T) - K, 0\}$ and time T . According to Black-Scholes-Merton the value of such an option at time t should be a function of t and the price $S(t)$ of the asset at time t . Let $c(t, x)$ be such a hypothetical function which for $x = S(t)$ gives the value of the option. We shall investigate the properties of such a function. First we apply Ito-Döblin to compute differentials of $c(t, S(t))$ and the discounted

versions.

$$\begin{aligned}
dc(t, S(t)) &= c_t(t, S(t))dt + c_x(t, S(t))dS(t) + \frac{1}{2}c_{xx}(t, S(t))dS(t)dS(t) \\
&= c_t(t, S(t))dt + c_x(t, S(t))(\alpha S(t)dt + \sigma S(t)dW(t)) \\
&\quad + \frac{1}{2}c_{xx}(t, S(t))\sigma^2 S^2(t)dt \\
&= \left(c_t(t, S(t)) + \alpha S(t)c_x(t, S(t)) + \frac{1}{2}\sigma^2 S^2(t)c_{xx}(t, S(t)) \right) dt \\
&\quad + \sigma S(t)c_x(t, S(t))dW(t) \tag{10.17}
\end{aligned}$$

For the discounted version let $f(t, x) = e^{-rt}x$. Then

$$\begin{aligned}
d(e^{-rt}c(t, S(t))) &= df(t, c(t, S(t))) \\
&= f_t(f, c(t, S(t)))dt + f_x(t, c(t, S(t)))dc(t, S(t)) \\
&\quad + \frac{1}{2}f_{xx}(t, c(t, S(t)))dc(t, S(t))dc(t, S(t)) \\
&= -re^{-rt}c(t, S(t))dt + e^{-rt}dc(t, S(t)) \\
&= e^{-rt} \left(-rc(t, S(t)) + c_t(t, S(t)) + \alpha S(t)c_x(t, S(t)) \right. \\
&\quad \left. + \frac{1}{2}\sigma^2 S^2(t)c_{xx}(t, S(t)) \right) dt + e^{-rt}\sigma S(t)c_x(t, S(t))dW(t) \tag{10.18}
\end{aligned}$$

10.5.2 Comparing portfolio with the option

The assumption that the call option has a replicating portfolio means that the value of such a portfolio is equal to the value of the option at each time: $V(t) = c(t, S(t))$. In particular, there is an equality between the discounted values which implies that

$$d(e^{-rt}V(t)) = d(e^{-rt}c(t, S(t)))$$

and $X(0) = c(0, S(0))$. Using the computations from the previous section and substituting (10.16) and (10.18) to the above equality we obtain

$$\begin{aligned}
\Delta(t)(\alpha - r)S(t)dt + \Delta(t)\sigma S(t)dW(t) &= \\
&= e^{-rt} \left(-rc(t, S(t)) + c_t(t, S(t)) + \alpha S(t)c_x(t, S(t)) \right. \\
&\quad \left. + \frac{1}{2}\sigma^2 S^2(t)c_{xx}(t, S(t)) \right) dt + e^{-rt}\sigma S(t)c_x(t, S(t))dW(t) \tag{10.19}
\end{aligned}$$

By comparing the factors of $dW(t)$ we get that

$$\Delta(t) = c_x(t, S(t)) \quad (10.20)$$

and by comparing the factors of dt and cancelling relevant terms we obtain that

$$rc(t, S(t)) = c_t(t, S(t)) + rS(t)c_x(t, S(t)) + \frac{1}{2}\sigma^2 S^2(t)c_{xx}(t, S(t)). \quad (10.21)$$

This is how the following **Black-Scholes-Merton** (or simply BSM) equation is derived:

$$c_t(t, x) + rxc_x(t, x) + \frac{1}{2}\sigma^2 x^2 c_{xx}(t, x) = rc(t, x) \quad (10.22)$$

for all $t \in [0, T]$ and $x \geq 0$. Suppose that $c(t, x)$ is a function which satisfies this equation and the terminal condition

$$c(T, x) = \max\{x - K, 0\}.$$

Then it follows from the above considerations that $\Delta(t) = c_x(t, S(t))$ is the position in the risky asset in the replicating portfolio. In particular, a replicating portfolio exists and its value $V(t)$ is equal to the value of the option $c(t, S(t))$ at each time $t \in [0, T]$.

10.5.3 The solution and its properties

The Black-Scholes-Merton equation is of a special type called *parabolic* and there are mathematical theories (beyond the scope of this course) providing solutions of such equations. In particular the solution of our equation with the above boundary conditions is given by

$$c(t, x) = xN(d_+(T - t, x)) - Ke^{-r(T-t)}N(d_-(T - t, x)),$$

where

$$d_{\pm}(\tau, x) = \frac{1}{\sigma\sqrt{\tau}} \left(\log \frac{x}{K} + \left(r \pm \frac{\sigma^2}{2} \right) \tau \right)$$

and N is the cumulative standard normal distribution

$$N(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-\frac{x^2}{2}} dx = \frac{1}{\sqrt{2\pi}} \int_{-y}^{\infty} e^{-\frac{x^2}{2}} dx.$$

The partial derivatives of the function $c(t, x)$ are called the **Greeks**, so that, c_x is called *delta*, c_t - *theta* and c_{xx} - *gamma*; this is an especially bad naming which makes no sense. The following lemma, whose proof is left as an exercise, computes these derivatives.

Lemma 10.6. *Let $c(t, x)$ be the solution of the BSM-equation. Then*

$$c_x(t, x) = N(d_+(T - t, x)) \quad (10.23)$$

$$c_t(t, x) = -rKe^{-r(T-t)}N(d_-(T - t, x)) - \frac{\sigma x}{2\sqrt{T-t}}N'(d_+(T - t, x)) \quad (10.24)$$

$$c_{xx}(t, x) = \frac{1}{\sigma x\sqrt{T-t}}N'(d_+(T - t, x)). \quad (10.25)$$

□

The formula for the function $c(t, x)$ is quite complicated and it is not clear how the function really looks like. However, by looking at the derivatives we can find out some of its features.

Since the cumulative distribution N is always positive, the delta, $c_x(t, x)$, is always positive which implies that for each fixed t , the function $x \mapsto c(t, x)$ is strictly increasing. Moreover, since the derivative N' of the cumulative normal distribution is positive, the gamma, $c_{xx}(t, x)$ is positive. This implies that the above function is not only increasing but also convex.

Finally, the theta, $c_t(t, x)$ is always negative. Consequently, for each fixed x the function $t \mapsto c(t, x)$ is strictly decreasing.

10.7 Put-Call parity

Recall that a **forward** contract with delivery price K and expiration time T obligates the holder to buy one share of stock for the price K at time T . Thus at the expiration time the contract is worth $S(T) - K$, which we may think of as of the pay-off of the contract. Let $f(t, x)$ denote the value of the contract at time t is the stock price at that time is $x = S(t)$. The usual no-arbitrage argument (assuming the constant interest rate) implies that $f(t, x) = x - e^{-r(T-t)}K$. Recall that the forward price $K = e^{rT}S(0)$ which implies that

$$f(t, S(t)) = S(t) - e^{rt}S(0). \quad (10.26)$$

Consider both European call and put option with expiry date T and strike price K . Observe that the pay-off of the call option minus the pay-off of the put option is equal to the pay-off of the forward contract:

$$f(T, S(T)) = c(T, S(T)) - p(T, S(T)).$$

It implies that

$$f(t, x) = c(t, x) - p(t, x),$$

and this equation is called **put-call parity**. Consequently, we can compute the value of the put option which is given by

$$p(t, x) = Ke^{-r(T-t)}N(-d_-(T-t, x)) - xN(-d_+(T-t, x)). \quad (10.27)$$

10.8 Exercises

10a. Check the computations in Sections 10.5.1 and 10.5.2. Try to do them without looking up.

10b. Notice that the function $c(t, x)$ is not defined for $t = T$. Compute the limit $\lim_{t \rightarrow T} c(t, x)$.

10c. Sketch the graph of the cumulative normal distribution N . What is the value $N(0)$?

10d. Let $x = K$, $T = 1$ and $r = 0$. Derive the formula for $c(0, x)$ and investigate the relation between the option price $c(0, x)$ and the volatility.

10e. Let $S(0) = 100$ be the current price of a risky asset. Consider a European call option on this asset with the strike price $K = 100$ and expiry time $T = 1$ (year). Suppose that $r = 0$ and $\sigma = 20\%$.

1. What is the current price of this option?
2. What is the probability that the option will be exercised?
3. Suppose you are the seller of this option. What is your hedging portfolio at each time $t \in [0, T]$?
4. The same questions for $K = 105$.

10f. Let $S(0) = 50$ be the current price of a risky asset. Compute the price of a three-months European call option with strike price $K = 50$, where the interest rate is $r = 10\%$ per annum and the volatility is $\sigma = 30\%$ per annum. Compute the probability that the option will be exercised.

10g. The purpose of this exercise is to verify that the function $c(t, x)$ defined by formula (10.2) is a solution of the BSM-equation. We use the notation of this section.

1. Compute $N'(x)$.
2. Show that $xN'(d_+) = Ke^{-r(T-t)}N'(d_-)$.
3. Compute $\frac{\partial d_+}{\partial x}$ and $\frac{\partial d_-}{\partial x}$.
4. Assume that $c = xN(d_+) - Ke^{-r(T-t)}N(d_-)$. Show that

$$\frac{\partial c}{\partial t} = -rKe^{-r(T-t)}N(d_-) - xN'(d_+)\frac{\sigma}{2\sqrt{T-t}}.$$

5. Show that $\frac{\partial c}{\partial x} = N(d_+)$.
6. Show that $c(t, x)$ satisfies the Black-Scholes-Merton equation.
7. Show that $c(t, x)$ satisfies the boundary condition of the European call option. That is, $\lim_{t \rightarrow T} c(t, x) = \max\{x - K, 0\}$.

10h. Verify equations (10.26) and (10.27).

11 Volatility surface and hedging

11.1 The volatility surface

There is a very simple way to verify if a model works in practice. Just apply the model to some past data and check whether it could predict the actual behaviour. A less harsh question is to ask to what extent the model is correct. In this section we show that the Black-Scholes model is wrong.

Recall that the Black-Scholes formula provides the price $c(t)$ at time $t \leq T$ of a European option with exercise time T , the strike price K . The price $c(t)$ depends on T, K , the current price $S = S(t)$ of the asset, the interest rates $r > 0$ and the variance σ^2 .

Suppose that $t = 0$ is *now* and let $c(S(0), K, T)$ denote the *actual* price now ($t = 0$) of such a European option with strike K and maturity $T > 0$. Let $\sigma: \mathbf{R}_+^2 \rightarrow \mathbf{R}_+$ be the function such that

$$C(S, K, T) = \text{BS}(S, T, r, K, \sigma(K, T))$$

where the right hand side denote the price of the option predicted by the Black-Scholes under the assumption that the risk is given by $\sigma(K, T)$. In other words we substitute to the Black-Scholes formula the value for the risk such that the formula gives the *correct* prediction. The graph of the function σ is called the ***volatility surface***.

If the Black-Scholes model is correct then, since it assumes constant σ , the volatility surface should be flat. However, in concrete examples with the use of the actual data it often looks as in the following figure.

Figure 11.1: Typical volatility surface

Of course, the shape of the volatility surface depends on the experimental data. However, many of these experiments have the following common features:

- Options with lower strike prices tend to have bigger volatility.
- For a fixed strike price the volatility tends to converge to some constant as $T \rightarrow \infty$ (for small T there behaviour various type of behaviour can occur).

The Black-Scholes model is a very interesting and beautiful concept but it *does not agree with experiments*. Many physical theories do not agree with

experiments but among those who don't are theories for which errors are negligible in certain applications. For example, Newton's theory of gravitation is good enough to send humans to the Moon. It is not good enough to make satellite navigation practical (for this we need Einstein's General Relativity). The Black-Scholes model, however, is still being used despite its limitations.

11.2 Hedging

Consider a portfolio consisting of stock and contingent claims (options etc) based on this stock. The value of this portfolio depends on S, t, r and σ (once the option is issued the strike K and expiry T are fixed). Let $V(S, t, r, \sigma)$ denote this value as a function of the above variables. We consider the following partial derivatives called ***Greek parameters***:

$$\begin{aligned} \text{delta}_V &= \frac{\partial V}{\partial S} & \text{gamma}_V &= \frac{\partial^2 V}{\partial^2 S} \\ \text{theta}_V &= \frac{\partial V}{\partial t} & \text{vega}_V &= \frac{\partial V}{\partial \sigma} \\ \text{rho}_V &= \frac{\partial V}{\partial r} \end{aligned}$$

so that for small changes in the arguments (using Taylor's formula) we have that

$$\Delta V \approx \text{delta}_V \Delta S + \text{theta}_V \Delta t + \text{vega}_V \Delta \sigma + \text{rho}_V \Delta r + \frac{1}{2} \text{gamma}_V (\Delta S)^2. \quad (11.1)$$

The main idea is to construct the portfolio so that some of its Greek parameters are zero which makes the portfolio resistant to the changes of the corresponding variable. We compute the Greek parameters for a single

European option at time $t = 0$:

$$\begin{aligned}
\text{delta}(c(0, x)) &= \frac{\partial c(t, x)}{\partial x} \Big|_{t=0} = N(d_+), \\
\text{gamma}(c(0, x)) &= \frac{\partial^2 c(t, x)}{\partial x^2} \Big|_{t=0} = \frac{1}{S\sigma\sqrt{2\pi T}} e^{-\frac{d_+^2}{2}}, \\
\text{theta}(c(0, x)) &= \frac{\partial c(t, x)}{\partial t} \Big|_{t=0} = -\frac{S\sigma}{2\sqrt{2\pi T}} e^{-\frac{d_+^2}{2}} - rKe^{-rT}N(d_-), \quad (11.2) \\
\text{vega}(c(0, x)) &= \frac{\partial c(t, x)}{\partial \sigma} \Big|_{t=0} = \frac{S\sqrt{T}}{\sqrt{2\pi}} e^{-\frac{d_+^2}{2}}, \\
\text{rho}(c(0, x)) &= \frac{\partial c(t, x)}{\partial r} \Big|_{t=0} = TK e^{-rT} N(d_-).
\end{aligned}$$

In what follows we discuss and compare various hedging strategies. Bear in mind that the main idea for the hedging is a bit naive because even if a function of one variable has derivative zero at some point it can change very quickly. For example, consider the family of functions $f(x) = ax^2$, where $a > 1$.

11.3 Delta hedging

We are concerned with dependence of the value of a portfolio on the asset price S . For simplicity we denote the value of the portfolio by $V(S)$. We know from the above considerations that $\Delta V(S) \approx \frac{d}{dS} V(S) \Delta S$. Consider a portfolio made from the stock, bonds (of current value 1, for simplicity) and some derivative security of our stock:

$$V(S) = xS + y + zc(t, S).$$

Assume that $z = -1$ (there is just one option) and then at $t = 0$ we have

$$\frac{d}{dS} V(S) = x - \frac{d}{dS} c(t, S) = x - N(d_+).$$

It follows that the portfolio $(N(d_+), 1, -1)$ has its *delta* equal to zero and hence it is immune on the changes in the price of the asset. More precisely,

$$V(S) = N(d_+)S + y - c(0, S)$$

does not vary much if the change in the stock price is small. It is also convenient to choose $y = -Ke^{-rT}N(d_-)$ such that the value of the portfolio is equal to zero.

Example of delta hedging

Let $r = 8\%$, $T = 90$, $X = 60$, $S = 60$ and $\sigma = 0.3$. That is, we consider a call option with strike price $X = 60$ and expiry $T = 90$ on a share of stock currently at the price $S = 60$. The Black-Scholes gives the current price of the option $C_E = 4.14452$ and the delta of the option is equal to 0.581957.

Suppose we write and sell 1000 options, so that $z = -1000$ in the portfolio we want to construct. We compute the values of x and y so that the portfolio (x, y, z) has value zero. They are $x = 581.96$ and $y = -30,772.88$. What we do is:

- sell 1000 options
- borrow 30,772.88
- buy 581.96 shares

We now discuss what happens after one day depending on the change of the price of the stock and compare it with the unhedged portfolio ($x = 0, y = 4,118.33, z = -1000$).

The price of stock does not change. The price of the option has decreased as well as our debt.

stock	34,917.39
money	-30,779.62
options	-4,118.33
total	19.45

For the unhedged portfolio we have 27.10, so no hedging is better in this case.

The price goes up to 61. For the hedged portfolio we have:

stock	35,499.35
money	-30,779.62
options	-4,721.50
total	-1.77

The unhedged portfolio has -576.07 .

The price goes down to 59. The hedges portfolio brings a small loss.

stock	34,335.44
money	-30,779.62
options	-3,559.08
total	-3.26

The unhedged portfolio has 586.35.

The following table shows more data.

S	Hedged	Unhedged
50.00	-2,233.19	3,594.03
55.00	-554.65	2,362.79
58.00	-71.35	1,100.22
58.50	-31.56	849.03
59.00	-3.26	586.35
59.50	13.69	312.32
60.00	19.45	27.10
60.50	14.22	-269.11
61.00	-1.77	-576.07
61.50	-28.24	-893.53
62.00	-64.93	-1,221.19
65.00	-481.60	-3,383.73
70.00	-1,765.15	-7,577.06

Conclusion: The delta hedging prevented the huge loss in the case of an increase of the price but also did not allow for a nice gain in the case of the price decrease. With respect to other variable changes the delta-hedged portfolio is not satisfactory at all.

11.4 Exercises

11a. What could be the reasons for the above behaviour of the volatility? Speculate and search in available resources for suggestions. See, for example, https://en.wikipedia.org/wiki/Volatility_smile

11b. Make formula (11.1) super-precise with the use of the O -notation.

11c. Verify the formulas (11.2) for the Greek parameters of the price of the option.

11d. Similarly to the delta-hedging analyse other hedging strategies. This is done, for example, in [?, Section 8.5.2]

12 Risk-neutral measure in the Black-Scholes-Merton model

In Section 10 we derived the Black-Scholes-Merton equation and stated the formula for its solution. This also yielded the replication portfolio. In this section we shall construct a risk-neutral measure and find the solution of the BSM-equation as the risk-neutral expectation of the pay-off on the option.

The initial set up is a probability space $(\Omega, \mathcal{F}, \mathbf{P})$, the price process $S(t): \Omega \rightarrow \mathbf{R}_+$, for $0 \leq t \leq T$ and the pay-off $c(T, S(T))$ of a European call option. The price process is given by

$$S(t) = S(0)e^{(r - \frac{1}{2}\sigma^2)t + \sigma W(t)},$$

where $W(t)$ is a random variable with normal distribution $N(0, t)$.

12.1 The Radon-Nikodym process

Let $\mathcal{F}(t) \subseteq \mathcal{F}$ be a filtration of Ω , where $0 \leq t \leq T$. Let $Z: \Omega \rightarrow \mathbf{R}$ be a positive random variable with $\mathbf{E}Z = 1$. It defines a probability measure $\mathbf{P}_Z$ by

$$\mathbf{P}_Z(A) = \int_A Z \, d\mathbf{P},$$

for all $A \in \mathcal{F}$. So in the language of Theorem B.4 the function $Z = \frac{d\mathbf{P}_Z}{d\mathbf{P}}$ is the Radon-Nikodym derivative of $\mathbf{P}_Z$ with respect to $\mathbf{P}$. We define the **Radon-Nikodym derivative process** $Z(t)$ by

$$Z(t) = \mathbf{E}(Z | \mathcal{F}(t)).$$

Lemma 12.2. *The Radon-Nikodym process is a martingale.*

Proof. Let $s \leq t$. The first and the third equality below is the definition of the process and the second is a property of the conditional expectation (see Proposition B.6).

$$\begin{aligned}\mathbf{E}(Z(t)|\mathcal{F}(s)) &= \mathbf{E}[\mathbf{E}(Z|\mathcal{F}(t))|\mathcal{F}(s)] \\ &= \mathbf{E}(Z|\mathcal{F}(s)) = Z(s)\end{aligned}$$

□

Lemma 12.3. *Let Y be an $\mathcal{F}(t)$ -measurable random variable. Then*

$$\mathbf{E}_Z(Y) = \mathbf{E}(YZ(t)), \quad (12.1)$$

where $\mathbf{E}_Z$ is the expectation with respect to measure $\mathbf{P}_Z$.

Proof. This is again definitions and properties of the conditional expectation.

$$\mathbf{E}_Z(Y) = \mathbf{E}(YZ) = \mathbf{E}(\mathbf{E}(YZ|\mathcal{F}(t))) = \mathbf{E}(Y\mathbf{E}(Z|\mathcal{F}(t))) = \mathbf{E}(YZ(t)).$$

□

Lemma 12.4. *Let Y be an $\mathcal{F}(t)$ -measurable random variable and let $s \leq t$. Then*

$$\mathbf{E}_Z(Y|\mathcal{F}(s)) = \frac{1}{Z(s)}\mathbf{E}(YZ(t)|\mathcal{F}(s)). \quad (12.2)$$

Proof. According to the uniqueness of the conditional expectation we need to check whether the function on the right hand side is $\mathcal{F}(s)$ -measurable (it clearly is) and satisfies the defining equation for the conditional expectation. That is, we need to verify the equation

$$\int_A \frac{1}{Z(s)}\mathbf{E}(YZ(t)|\mathcal{F}(s))d\mathbf{P}_Z = \int_A Yd\mathbf{P}_Z$$

for every $A \in \mathcal{F}(s)$. It is a consequence of the following computation in which various properties of the conditional expectation are applied.

$$\begin{aligned}\int_A \frac{1}{Z(s)}\mathbf{E}(YZ(t)|\mathcal{F}(s))d\mathbf{P}_Z &= \mathbf{E}_Z\left(I_A \frac{1}{Z(s)}\mathbf{E}(YZ(t)|\mathcal{F}(s))\right) \\ &= \mathbf{E}(I_A \mathbf{E}(YZ(t)|\mathcal{F}(s))) \\ &= \mathbf{E}(\mathbf{E}(I_A YZ(t)|\mathcal{F}(s))) \\ &= \mathbf{E}(I_A YZ(t)) \\ &= \mathbf{E}_Z(I_A Y) = \int_A Yd\mathbf{P}_Z.\end{aligned}$$

□

12.5 Girsanov's theorem

The proof of the following result can be found in [?, Section 5.2.1].

Theorem 12.6. *Let $0 \leq t \leq T$. Let $W(t)$ be a Brownian motion with associated filtration $\mathcal{F}(t)$. Let $\Theta(t)$ be an adapted process. Define*

$$Z(t) = \exp \left(- \int_0^t \Theta(u) dW(u) - \frac{1}{2} \int_0^t \Theta^2(u) du \right) \quad (12.3)$$

$$W_Z(t) = W(t) + \int_0^t \Theta(u) du \quad (12.4)$$

$$(12.5)$$

and assume that

$$\mathbf{E} \int_0^T \Theta^2(u) Z^2(u) du < \infty.$$

Let $Z = Z(T)$. Then $\mathbf{E}Z = 1$ and with respect to the probability $\mathbf{P}_Z$ the process $(W_Z(t))$ is a Brownian motion.

12.7 Risk-neutral pricing

Let $(S(t))$ for $0 \leq t \leq T$ be a **price process** given by

$$S(t) = S(0) \exp \left[\int_0^t \sigma(s) dW(s) + \int_0^t \left(\alpha(s) - \frac{\sigma^2(s)}{2} \right) ds \right],$$

where $\alpha(t)$ is the mean rate of return and $\sigma(t)$ is the volatility and both are adapted processes allowed to vary in time. Let $(D(t))$ be the **discount process** defined by

$$D(t) = \exp \left[- \int_0^t R(s) ds \right],$$

where $R(t)$ is the adapted interest rate process (also allowed to vary in time). They both combine to the **discounted price process**:

$$D(t)S(t) = S(0) \exp \left[\int_0^t \sigma(s) dW(s) + \int_0^t \left(\alpha(s) - R(s) - \frac{\sigma^2(s)}{2} \right) ds \right]$$

The differentials of these processes are computed with the use of Ito-Döblin formula and are given by:

$$dS(t) = \alpha(t)S(t)dt + \sigma(t)S(t)dW(t) \quad (12.6)$$

$$dD(t) = -R(t)D(t)dt \quad (12.7)$$

$$\begin{aligned} d(D(t)S(t)) &= (\alpha(t) - R(t))D(t)S(t)dt + \sigma(t)D(t)S(t)dW(t) \\ &= \sigma(t)D(t)S(t) (\Theta(t)dt + dW(t)), \end{aligned} \quad (12.8)$$

where the process

$$\Theta(t) = \frac{\alpha(t) - R(t)}{\sigma(t)}$$

is called the **market price of risk**.

We use the process $\Theta(t)$ in Girsanov's theorem to define the process $Z(t)$, the probability measure $\mathbf{P}_Z$ and the Brownian motion W_Z so that

$$dW_Z(t) = \Theta(t)dt + dW(t)$$

and the equality (12.8) has the form

$$d(D(t)S(t)) = \sigma(t)D(t)S(t)dW_Z(t). \quad (12.9)$$

It follows that

$$D(t)S(t) = S(0) + \int_0^t \sigma(u)D(u)S(u)dW_Z(u). \quad (12.10)$$

The second summand is an Ito integral and hence it is a martingale with respect to the measure $\mathbf{P}_Z$ which implies that the discounted process is a martingale with respect to the measure $\mathbf{P}_Z$. Since the measure $\mathbf{P}_Z$ is equivalent to the measure $\mathbf{P}$ (they have the same sets of measure zero), it is **risk-neutral**.

Remark 12.8. A proof of the theorem that the Ito integral is a martingale can be found in [?, Section 4.2 and 4.3].

The price process can be expressed with the use of the new Brownian motion as follows:

$$S(t) = S(0) \exp \left[\int_0^t \sigma(s)dW_Z(s) + \int_0^t \left(R(s) - \frac{\sigma^2(s)}{2} \right) ds \right].$$

Let $V(T): \Omega \rightarrow \mathbf{R}$ be the pay-off of a contingent claim with underlying risky asset described by the process $(S(t))$. Having a risk-neutral measure we can find a no-arbitrage value $V(t)$ of the contingent claim at time $t \in [0, T]$ by computing the conditional expectation:

$$D(t)V(t) = \mathbf{E}_Z(D(T)V(T)|\mathcal{F}(t)),$$

which yields

$$V(t) = \mathbf{E}_Z \left(\exp \left[- \int_t^T R(u) du \right] V(T) | \mathcal{F}(t) \right). \quad (12.11)$$

This is a very general formula for the value of a contingent claim under no restriction on the mean return, interest rate and volatility which are allowed to vary. Also the contingent claim may be time dependent.

12.9 Deriving the Black-Scholes-Merton formula

In the case of constant interest rates, mean return and volatility. The value of the European call option can be computed explicitly as follows.

Let $V(T) = (S(T) - K)^+ = \max \{0, S(T) - K\}$ be the pay-off of the European call option. The price process is now given by

$$S(t) = S(0) \exp \left[\sigma W_Z(t) + \left(r - \frac{\sigma^2}{2} \right) t \right]$$

which implies that

$$\begin{aligned} S(T) &= S(t) \exp \left[\sigma (W_Z(T) - W_Z(t)) + \left(r - \frac{\sigma^2}{2} (T - t) \right) \right] \\ &= S(t) \exp \left[-\sigma \sqrt{\tau} Y + \left(r - \frac{\sigma^2}{2} \right) \tau \right], \end{aligned}$$

where $\tau = T - t$ is the time to expiration and

$$Y = -\frac{W_Z(T) - W_Z(t)}{\sqrt{\tau}},$$

is the standard normal variable. Thus $S(T)$ is a product of a random variable $S(t)$ which is $\mathcal{F}(t)$ measurable and a factor which is independent of $\mathcal{F}(t)$.

It then follows from the properties of the conditional expectation that the formula (12.11) has the form

$$V(t) = c(t, S(t)) = \mathbf{E}_Z \left(e^{-r\tau} \left(S(t) \exp \left[-\sigma\sqrt{\tau}Y + \left(r + \frac{\sigma^2}{2} \right) \tau \right] - K \right)^+ \right).$$

We use this formula to define the suitable function $c(t, x)$ and we make the following computation

$$\begin{aligned} c(t, x) &= \mathbf{E}_Z \left(e^{-r\tau} \left(x \exp \left[-\sigma\sqrt{\tau}Y + \left(r + \frac{\sigma^2}{2} \right) \tau \right] - K \right)^+ \right) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-r\tau} \left(x \exp \left[-\sigma\sqrt{\tau}y + \left(r + \frac{\sigma^2}{2} \right) \tau \right] - K \right)^+ e^{-\frac{1}{2}y^2} dy \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{d_-(\tau, x)} e^{-r\tau} \left(x \exp \left[-\sigma\sqrt{\tau}y + \left(r + \frac{\sigma^2}{2} \right) \tau \right] - K \right)^+ e^{-\frac{1}{2}y^2} dy \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{d_-(\tau, x)} x \exp \left[-\frac{y^2}{2} - \sigma\sqrt{\tau}y - \frac{\sigma^2\tau}{2} \right] dy \\ &\quad - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{d_-(\tau, x)} e^{-r\tau} K e^{-\frac{1}{2}y^2} dy \\ &= \frac{x}{\sqrt{2\pi}} \int_{-\infty}^{d_-(\tau, x) + \sigma\sqrt{\tau}} e^{-\frac{z^2}{2}} dz - e^{-r\tau} K N(d_-(\tau, x)) \\ &= xN(d_+(\tau, x) - e^{-r\tau} K N(d_-(\tau, x)), \end{aligned}$$

where

$$d_{\pm}(\tau, x) = \frac{1}{\sigma\sqrt{\tau}} \left(\log xK + \left(r \pm \frac{1}{2}\sigma^2 \right) \tau \right).$$

The third equality in the above computation follows from the fact that

$$x \exp \left[-\sigma\sqrt{\tau}y + \left(r + \frac{\sigma^2}{2} \right) \tau \right] - K > 0$$

if and only if $y < d_-(\tau, x)$.

This computation shows how to obtain the solutions of the Black-Scholes-Merton equation and the closed formula for the value of an European call option.

12.10 Exercises

12a. Verify the all the formulas presented in Section 12.7.

12b. Show that $d_-(\tau, x) = d_+(\tau, x) - \sigma\sqrt{\tau}$.

A Basics of Probability Theory

A.1 Probability space

A **probability space** is a triple $(\Omega, \mathcal{F}, \mathbf{P})$ where:

- Ω is a set, called the **sample space**.
- $\mathcal{F}$ is a σ -algebra of subsets of Ω . This means that $\mathcal{F}$ is a collection of subsets of Ω such that:
 - $\emptyset$ and Ω belong to $\mathcal{F}$,
 - if $A \in \mathcal{F}$ then $\Omega \setminus A \in \mathcal{F}$,
 - if $A_1, A_2, A_3, \dots \in \mathcal{F}$ then $A_1 \cup A_2 \cup A_3 \cup \dots \in \mathcal{F}$.
- $\mathbf{P}: \mathcal{F} \rightarrow [0, 1]$ is a function called **probability measure** such that:
 - $\mathbf{P}(\Omega) = 1$.
 - If $A_1, A_2, \dots \in \mathcal{F}$ are pairwise disjoint subsets then

$$\mathbf{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbf{P}(A_i).$$

An element of $A \in \mathcal{F}$ is called an **event** and the value $\mathbf{P}(A)$ is called the **probability of the event** A . A probability space $(\Omega, \mathcal{F}, \mathbf{P})$ is called finite if the sample space is finite. Otherwise it is called infinite.

Example A.2. Consider a finite probability space $(\Omega = \{\omega_1, \dots, \omega_n\}, 2^\Omega, \mathbf{P})$ in which the σ -algebra of events consists of all subsets of Ω . The probability measure is uniquely defined by the numbers $\mathbf{P}(\{\omega_i\}) = p_i$. It follows that

$$\mathbf{P}(A) = \sum_{\omega_i \in A} p_i$$

for every event $A \subset \Omega$. The function $p: \Omega \rightarrow [0, 1]$ given by $p(\omega_i) = p_i$ is called the probability *mass function* of *density*.

Since $\mathbf{P}(\Omega) = 1$ we get that $\sum_{i=1}^n p_i = 1$. Hence the probability measure can be considered as a vector $(p_1, p_2, \dots, p_n) \in \mathbf{R}^n$ such that the sum of its coordinates is equal to 1. So the set

$$\Delta = \{(p_1, \dots, p_n) \in \mathbf{R}^n \mid p_1 + \dots + p_n = 1 \text{ and } 0 \leq p_i \leq 1\}$$

consists of all probability measures on a sample space with n elements. $\diamond$

Let $\mathcal{B}$ be the smallest σ -algebra of subsets of the set $\mathbf{R}$ of real numbers containing all intervals (a, b) . It is called the *Borel* σ -algebra. Analogously, we define the Borel σ -algebra for the interval $[0, 1]$ and any other interval.

Example A.3. Let $([0, 1], \mathcal{B}, \mathbf{P})$, where the probability measure is uniquely defined by the condition $\mathbf{P}(a, b) = b - a$, for every interval $(a, b) \subset [0, 1]$. $\diamond$

Example A.4. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be a function such that $\int_{\mathbf{R}} f \, dx = 1$. Let $(\mathbf{R}, \mathcal{B}, \mathbf{P})$ be the probability space where the probability measure is defined by

$$\mathbf{P}(A) = \int_A f \, dx.$$

One of the most important example is the normal distribution defined by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}},$$

where $\mu \in \mathbf{R}$ is called the *mean* and $\sigma^2 > 0$ is the *variance*. $\diamond$

Example A.5. Let $x_1, \dots, x_n \in \mathbf{R}$ and let $p_1, \dots, p_n \in [0, 1]$ be numbers such that $\sum_i p_i = 1$. Define $\mathbf{P}: \mathcal{B} \rightarrow [0, 1]$ by

$$\mathbf{P}(A) = \sum_{x_i \in A} p_i.$$

This probability measure is called *atomic* and the points $x_i \in \mathbf{R}$ are called atoms. $\diamond$

Example A.6. Let Ω be any set and let $p: \Omega \rightarrow [0, 1]$ be a function which attains positive values on at most countable subset $\Sigma \subset \Omega$ and such that

$\sum_{\omega \in \Omega} p(\omega) = 1$. It is then a probability mass function and it defines a probability measure by

$$\mathbf{P}(A) = \sum_{\omega \in \Sigma \cap A} p(\omega).$$

As a concrete example consider $\Omega = \mathbf{R}$ and $\Sigma = \mathbf{N}_+$, the set of all positive integers. Define $p: \mathbf{R} \rightarrow [0, 1]$ by

$$p(x) = \begin{cases} \frac{1}{2^x} & \text{if } x \in \mathbf{N}_+ \\ 0 & \text{otherwise.} \end{cases}$$

◇

A.7 Random variables

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. A function $X: \Omega \rightarrow \mathbf{R}$ is called a **random variable** if for every interval $(a, b) \subset \mathbf{R}$ the preimage $X^{-1}(a, b) \in \mathcal{F}$. Examples of random variables:

- If the σ -algebra $\mathcal{F} = 2^\Omega$ consists of all subsets of Ω then any function $X: \Omega \rightarrow \mathbf{R}$ is a random variable.
- Let $(\mathbf{R}, \mathcal{B}, \mathbf{P})$ be as in Example A.4. Then any continuous function is a random variable. More generally, every piecewise continuous function with discrete discontinuity points is a random variable. In general, not every function is a random variable, in fact, most of them are not, but they require some mathematical acrobatics to define. The intuition is that every *sensibly* defined function is a random variable.

The **expectation** of a random variable $X: \Omega \rightarrow \mathbf{R}$ is defined by

$$\mathbf{E}(X) = \int_{\Omega} X(\omega) d\mathbf{P}.$$

If the sample space is finite $\Omega = \{\omega_1, \dots, \omega_n\}$ and the probability is defined by the mass function $p(\omega_i) = p_i$ then the expectation is given by

$$\mathbf{E}(X) = \sum_{\omega \in \Omega} X(\omega) p_i = \sum_{i=1}^n X(\omega_i) p_i.$$

The expectation is **linear** in the sense that if $X_1, \dots, X_m: \Omega \rightarrow \mathbf{R}$ are random variables and $a_1, \dots, a_m \in \mathbf{R}$ then

$$\mathbf{E} \left(\sum_{i=1}^m a_i X_i \right) = \sum_{i=1}^m a_i \mathbf{E}(X_i). \quad (\text{A.1})$$

The **covariance** of two random variables $X, Y: \Omega \rightarrow \mathbf{R}$ is defined by

$$\text{cov}(X, Y) = \mathbf{E}(XY) - \mathbf{E}(X)\mathbf{E}(Y).$$

The **variance** of the random variable X is defined by

$$\text{var}(X) = \text{cov}(X, X) = \mathbf{E}(X^2) - \mathbf{E}(X)^2.$$

The **standard deviation** of a random variable X is defined by

$$\sigma_X = \sqrt{\text{var}(X)}.$$

The **correlation** of two random variables:

$$\text{cor}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Proposition A.8. *Let $X, Y: \Omega \rightarrow \mathbf{R}$ be random variables and let $a, b \in \mathbf{R}$ be real numbers. The following identities are true:*

$$\mathbf{E}(aX + bY) = a\mathbf{E}(X) + b\mathbf{E}(Y) \quad (\text{A.2})$$

$$\text{cov}(X, Y) = \mathbf{E}((X - \mathbf{E}(X))(Y - \mathbf{E}(Y))) \quad (\text{A.3})$$

$$\text{var}(X) = \mathbf{E}((X - \mathbf{E}(X))^2) \quad (\text{A.4})$$

$$|\text{cov}(X, Y)|^2 \leq \text{var}(X) \text{var}(Y) \quad (\text{A.5})$$

$$-1 \leq \text{cor}(X, Y) \leq 1 \quad (\text{A.6})$$

$$\text{var}(aX + bY) = a^2 \text{var}(X) + b^2 \text{var}(Y) + 2ab \text{cov}(X, Y). \quad (\text{A.7})$$

□

The **distribution** of a random variable $X: \Omega \rightarrow \mathbf{R}$ is a probability space $(\mathbf{R}, \mathcal{B}, X_*\mathbf{P})$, where the probability measure $X_*\mathbf{P}$ is defined by

$$X_*\mathbf{P}[a, b] = \mathbf{P}(X^{-1}[a, b]) = \mathbf{P}(a \leq X \leq b).$$

We say that a function $F_X: \mathbf{R} \rightarrow \mathbf{R}$ is a **cumulative distribution** or a **distribution function** of a random variable X if

$$F_X(x) = \mathbf{P}(X \leq x) = X_*\mathbf{P}(-\infty, x].$$

In many scientific problems one works with distributions rather than with concrete random variables. This is because the probability theory require a sample space to define a random variable which is impractical. In practice, various measurements are made and from the pattern of their values and frequency of occurrences one *guesses* the distribution which often has a *nice* distribution function or even a density as in Example A.4.

Let X be a random variable. The distribution function $F: \mathbf{R} \rightarrow \mathbf{R}$ of X is called:

1. **discrete** iff there exist a countable set $\{x_i\} \subset \mathbf{R}$ and $p_i \in [0, 1]$ such that

$$F(x) = \sum_{x_i \leq x} p_i.$$

2. **continuous** if F is continuous.
3. **absolutely continuous** if there exists a non-negative, Lebesgue integrable function f , such that

$$F(b) - F(a) = \int_a^b f(x)dx.$$

4. **singular** if $F \neq 0$ and F' is defined almost everywhere and $F' = 0$ almost everywhere.

Digression on absolute continuity. In general, a function $F: I \rightarrow \mathbf{R}$ defined on an interval $I \subset \mathbf{R}$ is called **absolutely continuous** if for every $\epsilon > 0$ there exists $\delta > 0$ such that for every collection of pairwise disjoint intervals $(a_1, b_1), \dots, (a_n, b_n) \subset I$ with $\sum(b_i - a_i) < \delta$ we have

$$\sum_i |f(b_i) - f(a_i)| < \epsilon.$$

Proposition A.9. Let $F: [a, b] \rightarrow \mathbf{R}$ be a function defined on a closed interval. The following conditions are equivalent:

(1) F is absolutely continuous.

(2) F has a derivative almost everywhere, the derivative is integrable and

$$F(x) = F(a) + \int_a^x F'(t)dt.$$

(3) there exists a Lebesgue integrable function $f: [a, b] \rightarrow \mathbf{R}$ such that

$$F(x) = F(a) + \int_a^x f(t)dt$$

almost everywhere. $\square$

The following theorem says that the above distributions are building block for every distribution function.

Theorem A.10 (Theorem 2.4 in [?]). *Every distribution function F is a unique convex combination of a discrete, absolutely continuous and continuous singular distribution functions:*

$$F = aF_{ac} + bF_d + cF_{cs},$$

where $a, b, c \geq 0$, $a + b + c = 1$ and:

- $F_{ac}(x) = \int_{-\infty}^x f(t)dt$, where $F'(t) = f(t)$ almost everywhere;
- F_d is a pure jump function with at most countable number of jumps;
- F_{cs} is continuous and $F'_{cs} = 0$ almost everywhere. $\square$

Example A.11. The following two tables show examples of frequently used discrete and continuous distributions. In the first table we set $q = 1 - p$.

Discrete distributions

Distribution	Notation	Probability function	Domain	Mean	Variance
One point	$\delta(a)$	$\mathbf{P}(a) = 1$	any	a	0
Sym. Bernoulli		$\mathbf{P}(-1) = \mathbf{P}(1) = \frac{1}{2}$	any	0	1
Bernoulli	$\text{Be}(p)$	$\mathbf{P}(0) = q, \mathbf{P}(1) = p$	any	p	pq
Binomial	$\text{Bin}(n, p)$	$\mathbf{P}(k) = \binom{n}{k} p^k q^{n-k}$	$k = 0, 1, \dots, n$	np	npq
Geometric	$\text{Ge}(p)$	$\mathbf{P}(k) = pq^k$	$k = 0, 1, \dots$	$\frac{q}{p}$	$\frac{q}{p^2}$
First success	$\text{Fs}(p)$	$\mathbf{P}(k) = pq^{k-1}$	$k = 1, 2, \dots$	$\frac{1}{p}$	$\frac{q}{p^2}$
Poisson	$\text{Po}(m)$	$\mathbf{P}(k) = e^{-m} \frac{m^k}{k!}$	$k = 0, 1, \dots$	m	m

The distribution $\text{Be}(p)$ describes the outcome of the coin tossing experiment or the binomial pricing model; $\text{Bin}(n, p)$ - the number of successes in n trials (or how many times prices grew in the n -step binomial model); $\text{Ge}(p)$ - the number of failures prior to the first success; $\text{Fs}(p)$ - the number of trials required to succeed once.

Absolutely continuous distributions

Distribution	Notation	Density function	Domain	Mean	Variance
Uniform	$U(a, b)$	$f(x) = \frac{1}{b-a}$	$a < x < b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
	$U(0, 1)$	$f(x) = 1$	$0 < x < 1$	$\frac{1}{2}$	$\frac{1}{12}$
	$U(-1, 1)$	$f(x) = \frac{1}{2}$	$ x < 1$	0	$\frac{1}{3}$
Triangular	$\text{Tri}(-1, 1)$	$f(x) = 1 - x $	$ x < 1$	0	$\frac{1}{6}$
Exponential	$\text{Exp}(\theta)$	$f(x) = \frac{e^{-\frac{x}{\theta}}}{\theta}$	$x > 0$	θ	θ^2
Gamma	$\Gamma(p, \theta)$	$f(x) = \frac{x^{p-1} e^{-\frac{x}{\theta}}}{\Gamma(p) \theta^p}$	$x > 0$	$p\theta$	$p\theta^2$
Beta	$\beta(r, s)$	$f(x) = \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)} x^{r-1} (1-x)^{s-1}$	$0 < x < 1$	$\frac{r}{r+s}$	$\frac{rs}{(r+s)^2(r+s+1)}$
Normal	$N(\mu, \sigma^2)$	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$x \in \mathbf{R}$	μ	σ^2
	$N(0, 1)$	$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$	$x \in \mathbf{R}$	0	1
	$LN(\mu, \sigma^2)$	$f(x) = \frac{1}{\sigma x \sqrt{2\pi}} e^{-\frac{(\log \frac{x-\mu}{2\sigma^2})^2}{2}}$	$x > 0$	$e^{\mu+\frac{1}{2}\sigma^2}$	$s^{2\mu} (e^{2\sigma^2} - e^{\sigma^2})$
Cauchy	$C(0, 1)$	$f(x) = \frac{1}{\pi} \frac{1}{1+x^2}$	$x \in \mathbf{R}$	undefined	undefined
Levy	$L(\mu, c)$	$f(x) = \sqrt{\frac{c}{2\pi}} \frac{e^{\frac{c^2(x-\mu)}{2}}}{(x-\mu)^{\frac{3}{2}}}$	$x \in [\mu, \infty)$	∞	∞

A.12 Independence

Definition A.13.

- Events $A_1, \dots, A_n$ are *independent* if

$$P\left(\bigcap_{k \in S} A_k\right) = \prod_{k \in S} P(A_k)$$

for any $S \subset \{1, 2, \dots, n\}$.

- Events $\{A_n, n \geq 1\}$ are independent if for any $n \geq 1$ the events $A_1, \dots, A_n$ are independent.
- Random variables $X_1, \dots, X_n$ are *independent* if

$$P\left(\bigcap_{k=1}^n \{X_k \in A_k\}\right) = \prod_{k=1}^n P(X_k \in A_k)$$

for any Borel sets $A_1, \dots, A_n$.

In what follows we state several criteria for independence of random variables.

Theorem A.14. *Two random variables X and Y are independent if and only if*

$$E(g(X)h(Y)) = E(g(X))E(h(Y))$$

for all continuous bounded functions $g, h: \mathbf{R} \rightarrow \mathbf{R}$.

Warning A.15. It follows from the above theorem that if the random variables X, Y are independent then they are *uncorrelated*: $E(XY) = E(X)E(Y)$. The converse is not true. For example, let $\Omega = [-1, 1]$ and let $X, Y: [-1, 1] \rightarrow \mathbf{R}$ be given by $X(t) = t$ and $Y(t) = t^2$ respectively. Then $E(X) = 0$ and $E(Y) = \frac{2}{3}$ and $E(XY) = 0$ which shows that X and Y are uncorrelated. On the other hand, let $A = (-\frac{1}{2}, \frac{1}{2})$ and $B = [\frac{1}{4}, 1]$ then

$$\begin{aligned} P(\{X \in A\} \cap \{Y \in B\}) &= P\left(\left(-\frac{1}{2}, \frac{1}{2}\right) \cap \left(\left[-1, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, 1\right]\right)\right) \\ &= P(\emptyset) = 0. \end{aligned}$$

On the other hand

$$P(X \in A)P(Y \in B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

which shows that X and Y are not independent.

There are, however, situations when being uncorrelated implies independence:

Proposition A.16. *Let $X = (X_1, \dots, X_n)$ be a normal random vector. This means that the sum $a_1X_1 + \dots + a_nX_n$ is normally distributed for every $(a_1, \dots, a_n) \in \mathbf{R}^n$. If the variables $X_1, \dots, X_n$ are pairwise uncorrelated then they are independent.*

Theorem A.17. *Random variables $X_1, \dots, X_n$ are independent if and only if*

$$F_X(x_1, \dots, x_n) = \prod_{k=1}^n F_{X_k}(x_k),$$

where $X = (X_1, \dots, X_n)$ and $x_i \in \mathbf{R}$.

Theorem A.18. *Let $X_1, \dots, X_n$ be absolutely continuous random variables. They are independent if and only if*

$$f_X(x_1, \dots, x_n) = \prod_{k=1}^n f_{X_k}(x_k),$$

for $x_i \in \mathbf{R}$.

Proof. We present a sketch only for two variables X, Y . Assume that X and Y are independent. Consider the collection

$$\mathcal{E} = \left\{ A \times B \mid \iint_{A \times B} f_{X,Y}(u, v) du dv = \iint_{A \times B} f_X(u) f_Y(v) du dv \right\}$$

of sets $A \times B$ over which the integrals of the densities has the required behaviour. Observe that for $A = (-\infty, a]$ and $B = (-\infty, b]$ the set $A \times B$ belongs to $\mathcal{E}$. Indeed, we have

$$\begin{aligned} \iint_{A \times B} f_{X,Y}(u, v) du dv &= \int_{-\infty}^a \int_{-\infty}^b f_{X,Y}(u, v) du dv \\ &= F_{X,Y}(a, b) = F_X(a) F_Y(b) && \text{independence} \\ &= \int_A f_X(u) du \int_B f_Y(v) dv \\ &= \iint_{A \times B} f_X(u) f_Y(v) du dv. && \text{Fubini} \end{aligned}$$

The next step is to show that arbitrary rectangles $[a_1, a_2] \times [b_1, b_2]$ belong to $\mathcal{E}$ and then countable unions and intersections of rectangles and deduce that $\mathcal{E}$ is the whole σ -algebra of Borel sets. Finally, if the integrals of two functions agree over any Borel set then these functions are equal up to a set of measure zero.

The converse implication follows directly from the Fubini theorem:

$$\begin{aligned} F_{X,Y}(x, y) &= \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) \, dudv \\ &= \int_{-\infty}^x \int_{-\infty}^y f_X(u) f_Y(v) \, dudv \\ &= \int_{-\infty}^x f_X(u) \, du \int_{-\infty}^y f_Y(v) \, dv \\ &= F_X(x) F_Y(y) \end{aligned}$$

□

B Advanced topics in Probability Theory

Throughout this section $(\Omega, \mathcal{F}, \mathbf{P})$ is a probability space.

B.1 Conditional expectation

Let $X: \Omega \rightarrow \mathbf{R}$ be a random variable and let $\mathcal{G} \subseteq \mathcal{F}$ be a σ -subalgebra. The **conditional expectation** of X with respect to $\mathcal{G}$ is a $\mathcal{G}$ -measurable random variable $\mathbf{E}(X|\mathcal{G}): \Omega \rightarrow \mathbf{R}$ which satisfies

$$\int_A \mathbf{E}(X|\mathcal{G}) d\mathbf{P} = \int_A X \, d\mathbf{P}, \quad (\text{B.1})$$

for every $A \in \mathcal{G}$. Before proving that the conditional expectation exists let us look at simple examples.

Example B.2. Let $\mathcal{F}_0 = \{\emptyset, \Omega\}$ be the trivial σ -algebra. Then every $\mathcal{F}_0$ -measurable function is constant and

$$\mathbf{E}(X|\mathcal{F}_0) = \mathbf{E}(X).$$

That is, the conditional expectation is the ordinary expectation. ◇

Example B.3. Let $\Omega = \{0, 1\}^2 = \{00, 01, 10, 11\}$ and let

$$\begin{aligned}\mathcal{F}_0 &= \{\emptyset, \Omega\} \\ \mathcal{F}_1 &= \{\emptyset, \Omega, \{00, 01\}, \{10, 11\}\} \\ \mathcal{F}_2 &= 2^\Omega\end{aligned}$$

and let $\mathbf{P}: \mathcal{F}_2 \rightarrow [0, 1]$ be the uniform probability. Let $X: \Omega \rightarrow \mathbf{R}$ be an $\mathcal{F}_2$ -measurable random variable defined by

$$X(00) = 0, \quad X(01) = 1, \quad X(10) = 2, \quad X(11) = 3.$$

It is easy to verify that

$$\mathbf{E}(X|\mathcal{F}_0) = \mathbf{E}(X) = \frac{3}{2} \quad \text{and} \quad \mathbf{E}(X|\mathcal{F}_1)(0*) = \frac{1}{2}, \quad \mathbf{E}(X|\mathcal{F}_1)(1*) = \frac{5}{2},$$

◇

It may be useful to think that the conditional expectation of an $\mathcal{F}$ -measurable random variable is a $\mathcal{G}$ -measurable random variable $\mathbf{E}(X|\mathcal{G})$ with the same expectation as X . The existence of the conditional expectation is proven with the use of the following fundamental result of measure theory.

Theorem B.4 (Radon-Nikodym). *Let $(\Omega, \mathcal{F})$ be a measurable space and let $\mu, \nu: \mathcal{F} \rightarrow \mathbf{R}$ be two σ -finite measures such that ν is absolutely continuous with respect to μ . Then there is a measurable function $Z: \Omega \rightarrow \mathbf{R}$ such that*

$$\nu(A) = \int_A Z \, d\mu$$

for every measurable set $A \in \mathcal{F}$. □

The function Z in the above theorem is called the **Radon-Nikodym** derivative of ν with respect to μ and it is denoted by

$$Z = \frac{d\nu}{d\mu}.$$

Corollary B.5. *Let $(\Omega, \mathbf{F}, \mathbf{P})$ be a probability space, $X: \Omega \rightarrow \mathbf{R}$ a random variable with $\mathbf{E}(X) < \infty$ and $\mathcal{G} \subseteq \mathcal{F}$ a subalgebra. Then there exist the conditional probability $\mathbf{E}(X|\mathcal{G})$ and it is unique up to sets of measure zero.*

Proof. Let $Q_{\pm}: \mathcal{G} \rightarrow \mathbf{R}$ be a measure defined by

$$Q_{\pm}(A) = \int_A X_{\pm} d\mathbf{P},$$

where $X_+(\omega) = \max\{0, X(\omega)\}$ and $X_-(\omega) = -\min\{0, X(\omega)\}$. Both $X_{\pm}$ are non-negative functions and $X = X_+ - X_-$. It follows from definition that $Q_{\pm}$ are absolutely continuous measures with respect to $\mathbf{P}$.

Let $\mathbf{E}(X|\mathcal{G}) = \frac{dQ_+}{d\mathbf{P}|_{\mathcal{G}}} - \frac{dQ_-}{d\mathbf{P}|_{\mathcal{G}}}$. We check that it satisfies (B.1):

$$\begin{aligned} \int_A \mathbf{E}(X|\mathcal{G}) d\mathbf{P} &= \int_A \frac{dQ_+}{d\mathbf{P}|_{\mathcal{G}}} d\mathbf{P} - \int_A \frac{dQ_-}{d\mathbf{P}|_{\mathcal{G}}} d\mathbf{P} \\ &= \int_A dQ_+ - \int_A dQ_- = \int_A (X_+ - X_-) d\mathbf{P} = \int_A X d\mathbf{P}. \end{aligned}$$

To prove uniqueness, assume that $Y, Z: \Omega \rightarrow \mathbf{R}$ are $\mathcal{G}$ -measurable random variables satisfying the condition (B.1). Let $A = \{\omega \in \Omega \mid Y(\omega) > Z(\omega)\} \in \mathcal{G}$. Then

$$\int_A (Y - Z) d\mathbf{P} = 0$$

which implies that $\mathbf{P}(A) = 0$. Similarly for $A = \{\omega \in \Omega \mid Y(\omega) < Z(\omega)\}$ and we conclude that the conditional expectation is uniquely defined up to sets of measure zero. $\square$

Proposition B.6 (Properties of conditional expectation). *Let X, Y be integrable random variables, $\mathcal{H} \subset \mathcal{G} \subset \mathcal{F}$, and $a, b, c \in \mathbf{R}$. Then*

1. $\mathbf{E}(\mathbf{E}(X|\mathcal{G})) = \mathbf{E}(X)$;
2. $\mathbf{E}(aX + bY|\mathcal{G}) = \mathbf{E}(aX|\mathcal{G}) + \mathbf{E}(bY|\mathcal{G})$;
3. If X is $\mathcal{G}$ -measurable then $\mathbf{E}(X|\mathcal{G}) = X$;
4. $\mathbf{E}(c|\mathcal{G}) = c$;
5. $\mathbf{E}(X|\{\emptyset, \Omega\}) = \mathbf{E}X$;
6. If $X \geq 0$ then $\mathbf{E}(X|\mathcal{G}) \geq 0$;
7. If $X \leq Y$ then $\mathbf{E}(X|\mathcal{G}) \leq \mathbf{E}(Y|\mathcal{G})$;
8. $|\mathbf{E}(X|\mathcal{G})| \leq \mathbf{E}(|X| |\mathcal{G})$;
9. If X is independent of $\mathcal{G}$ then $\mathbf{E}(X|\mathcal{G}) = EX$;

10. $(\mathbf{E}(X|\mathcal{G}) | \mathcal{H}) = \mathbf{E}(X|\mathcal{H}) = \mathbf{E}(\mathbf{E}(X|\mathcal{G}) | \mathcal{H})$.
11. If X and XY are integrable and Y is $\mathcal{G}$ -measurable then $\mathbf{E}(XY|\mathcal{G}) = Y\mathbf{E}(X|\mathcal{G})$;
12. $\mathbf{E}(\mathbf{E}(X|\mathcal{G})|\mathcal{H}) = \mathbf{E}(X|\mathcal{H}) = \mathbf{E}(\mathbf{E}(X|\mathcal{H})|\mathcal{G})$.

B.7 Filtrations

A family of σ -algebras $\mathcal{F}(t)$ for $0 \leq t \leq T$ is called a **filtration** of Ω if

$$s \leq t \implies \mathcal{F}(s) \subseteq \mathcal{F}(t).$$

Sometimes it is easier (and more useful) to parametrise partitions with integers or any other well ordered set.

Example B.8. It is useful to construct filtrations from partitions. Recall that a partition of a set Ω is a collection of its subsets $S_i \subset \Omega$ such that S_i 's are pairwise disjoint and their union is $\Omega = \bigsqcup_i S_i$. A partition $\mathcal{P}$ is finer than partition $\mathcal{P}'$ if every set of $\mathcal{P}$ is contained in some set of $\mathcal{P}'$. We then write $\mathcal{P} \leq \mathcal{P}'$.

Let $\mathcal{P}_0 = \{\Omega\} \geq \mathcal{P}_1 \geq \mathcal{P}_2 \dots$ be a sequence of finer and finer partitions. Let $\mathcal{F}(t)$ be the σ -algebra generated by the partition $\mathcal{P}_{\lfloor t \rfloor}$. Recall that a σ -algebra $\mathcal{G}$ is generated by a collection of subsets of Ω if it is the smallest σ -algebra containing this collection. For example, $\mathcal{F}(0) = \{\emptyset, \Omega\}$ is the trivial σ -algebra. Furthermore, since for $s \leq t$ the partition $\mathcal{P}_t$ is finer than $\mathcal{P}_s$, we get that $\mathcal{F}(s) \subseteq \mathcal{F}(t)$ and hence the family $\mathcal{F}(t)$ is a filtration. In this example it is useful to parametrise the algebras of the partition with integers rather than real numbers. $\diamond$

Example B.9. Let $\Omega = \{0, 1\}^n$ and let $\mathcal{P}_k$ be the partition consisting of sequences with fixed prefix of length k . For example, $\mathcal{P}_0 = \{\Omega\}$, $\mathcal{P}_1$ has n sets and, in general, $\mathcal{P}_k$ comprises of 2^k sets. It is a sequence of finer and finer partitions and hence they generate a filtration as in the previous example. $\diamond$

Let $(\Omega, \mathbf{F}, \mathbf{P})$ be a probability space. Suppose that it is equipped with a filtration $\mathcal{F}(t)$ for $0 \leq t \leq T$ with $\mathcal{F}(T) = \mathcal{F}$. We can restrict the probability measure to each of the $\mathcal{F}(t)$ and for $s \leq t$ we see that the probability space $(\Omega, \mathcal{F}(s), \mathbf{P})$ has less events than $(\Omega, \mathcal{F}(t), \mathbf{P})$. Also the first one has less random variables or, more precisely, the vector space of random variables on $(\Omega, \mathcal{F}(s), \mathbf{P})$ is a subspace of the vector space of random variables on

$(\Omega, \mathcal{F}(t), \mathbf{P})$. In the extreme case of $\mathcal{F}(0) = \{\emptyset, \Omega\}$ the only random variables are constant functions.

Since we are modelling random phenomena, it is useful to think of a σ -algebra as being related to the information available at a given time.

Example B.10. Let $\Omega = \{0, 1\}^n$ be equipped with a filtration $\mathcal{F}(k)$ as in Example B.9. Recall that this sample space is used in the binomial model. Observe that the price at time k is a random variable $S(k): \Omega \rightarrow \mathbf{R}$ which is $\mathcal{F}(k)$ -measurable. In general, we consider any random variable which is $\mathcal{F}(k)$ -measurable as a part of information available at time k . $\diamond$

Example B.11. If $X: \Omega \rightarrow \mathbf{R}$ is a random variable it defines a σ -algebra $\sigma(X)$ on Ω which is the smallest σ -algebra containing all the sets $X^{-1}(a, b) = \{\omega \in \Omega \mid X(\omega) \in (a, b)\}$.

Suppose we have a stochastic process $(W(t))$, for example, a Brownian motion. It defines a filtration by $\mathcal{F}(t) = \sigma(\bigcup_{s \leq t} \sigma(W(s)))$. That is, $\mathcal{F}(t)$ is the smallest σ -algebra containing all algebras $\sigma(W(s))$ for $0 \leq s \leq t$. It is clearly a filtration. Moreover, $W(t)$ is $\mathcal{F}(t)$ -measurable, which is also immediate from definition. What is less clear is that for $0 \leq s < t$ the increment $W(t) - W(s)$ is independent of $\mathcal{F}(s)$. A filtration with these three properties is called a *filtration for the Brownian motion* $(W(t))$. $\diamond$

B.12 Martingales

Let $\mathcal{F}(t)$ be a filtration of a probability space $(\Omega, \mathcal{F}, \mathbf{P})$. A stochastic process X_t adapted to $\mathcal{F}(t)$, where all X_t are integrable is called a *martingale* if

$$\mathbf{E}(X(t) | \mathcal{F}(s)) = X(s) \quad (\text{B.2})$$

for all $0 \leq s \leq t$.

Example B.13. Let $W(t)$ be a Brownian motion and $\mathcal{F}(t)$ its filtration. It is a martingale. Indeed, this follows from the properties of the conditional expectation and definition of the Brownian motion.

$$\begin{aligned} \mathbf{E}(W(t) | \mathcal{F}(s)) &= \mathbf{E}(W(t) - W(s) + W(s) | \mathcal{F}(s)) \\ &= \mathbf{E}(W(t) - W(s) | \mathcal{F}(s)) + \mathbf{E}(W(s) | \mathcal{F}(s)) \\ &= \mathbf{E}(W(t) - W(s)) + W(s) = W(s) \end{aligned}$$

$\diamond$

B.14 Zero-one laws

Version for events

Let $\{A_n, n \geq 1\}$ be a collection of events. We consider the following σ -subalgebras:

- $\mathcal{A}_n = \sigma\{A_1, \dots, A_n\}$ - the σ -algebra generated by $A_1, \dots, A_n$;
- $\mathcal{A}'_n = \sigma\{A_{n+1}, A_{n+2}, \dots\}$;
- $\mathcal{T} = \bigcap_{n=0}^{\infty} \mathcal{A}'_n$.

If we think of the parametered n as time variable then $\mathcal{A}_n$ is the algebra containing the information up to time n , the algebra $\mathcal{A}'_n$ contains an information after time n . The algebra $\mathcal{T}$ contains information about events beyond any time and it is called the algebra of *remote events* or the *tail-algebra*.

Theorem B.15. *If the events A_n for $n \geq 1$ are independent then for any $A \in \mathcal{T}$ we have that either $P(A) = 0$ or $P(A) = 1$.*

Sketch of proof: If $A \in \mathcal{T}$ then $A \in \mathcal{A}'_n$ for every $n = 0, 1, 2, \dots$ and hence A is independent of $\mathcal{A}_n$ for all n . This implies, on the one hand, that $A \in \sigma\{A_1, A_2, \dots\} = \mathcal{A}'_0$ and that A is independent of $\sigma\{A_1, A_2, \dots\}$ because it is independent of $\mathcal{A}_n = \sigma\{A_1, \dots, A_n\}$ for every $n = 1, 2, \dots$ (this step requires a more careful proof). In particular, A is independent of itself and

$$P(A) = P(A \cap A) = P(A) \cdot P(A)$$

which implies that $P(A)$ can be equal to either 0 or 1 only. □

Version for random variables

Let $\{X_k, k \geq 1\}$ be a collection of random variables. We consider the following sub-algebras:

- $\mathcal{F}_n = \sigma\{X_1, \dots, X_n\}$, that is the σ -algebra generated by the sets of the form $X_k^{-1}(B)$, where $B \subset \mathbf{R}$ is a Borel set.
- $\mathcal{F}'_n = \sigma\{X_{n+1}, X_{n+2}, \dots\}$.
- $\mathcal{T} = \bigcap_{n=0}^{\infty} \mathcal{F}'_n$.

Theorem B.16. *If $X_1, X_2, \dots$ are independent then for any $A \in \mathcal{T}$ we have that either $P(A) = 0$ or $P(A) = 1$.*

Example B.17. Let $S(n)$ be the price of a risky asset at time $n = 1, 2, 3, \dots$. Suppose that U is the event that the price will increase infinitely many times. Intuitively, U is a remote event because whether it occurs or not does not depend on any finite time behaviour of prices. The Zero-one Law implies that the probability of U is either 0 or 1. Which possibility is it? $\diamond$

Ba. Prove that U in the above example is a remote event.

B.18 Borel-Cantelli Lemmas

Let $\{A_n, n \geq 1\}$ be a collection of events. Recall the definition of the limit inferior and the limit superior for sequences of sets:

$$A_* = \liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} A_m \qquad A^* = \limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m$$

If $\omega \in \limsup A_n$ then it belongs to the union $\bigcup_{m=n}^{\infty} A_m$ for every $n \geq 1$. Equivalently, $\omega \in A_m$ for infinitely many m or, equivalently, $\omega \in A_m$ for arbitrarily large m . A convenient notation is

$$\omega \in \limsup_{n \rightarrow \infty} A_n \implies \omega \in \{A_n \text{ i.o.}\} = \{A_n \text{ infinitely often}\}.$$

The first Borel-Cantelli Lemma

Lemma B.19. *Let $\{A_n, n \geq 1\}$ be arbitrary events. Then*

$$\sum_{n=1}^{\infty} P(A_n) < \infty \implies P(A_n \text{ i.o.}) = 0.$$

Proof. Observe that the condition $\sum_{n=1}^{\infty} P(A_n) < \infty$ implies that

$$\lim_{n \rightarrow \infty} \left(\sum_{m=n}^{\infty} P(A_m) \right) = 0$$

because $P(A_m) \geq 0$ for all m . The statement follows from the following computation

$$\begin{aligned} P(A_n \text{ i.o.}) &= P(\limsup A_n) = P\left(\bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m\right) \\ &\leq P\left(\bigcup_{m=n}^{\infty} A_m\right) \leq \sum_{m=n}^{\infty} P(A_m) \rightarrow 0 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

□

The second Borel-Cantelli Lemma

Lemma B.20. *Let $\{A_n, n \geq 1\}$ be a collection of independent events. Then*

$$\sum_{n=1}^{\infty} P(A_n) = \infty \quad \implies \quad P(A_n \text{ i.o.}) = 1.$$

Proof.

$$\begin{aligned} P(A_n \text{ i.o.}) &= P\left(\limsup_{n \rightarrow \infty} A_n\right) = P\left(\bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m\right) \\ &= 1 - P\left(\left(\bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m\right)^c\right) \\ &= 1 - P\left(\bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} A_m^c\right) \\ &= 1 - \lim_{n \rightarrow \infty} P\left(\bigcap_{m=n}^{\infty} A_m^c\right) \\ &= 1 - \lim_{n \rightarrow \infty} \prod_{m=n}^{\infty} P(A_m^c) \quad \text{independence} \\ &= 1 - \lim_{n \rightarrow \infty} \prod_{m=n}^{\infty} (1 - P(A_m)). \end{aligned}$$

It remains to show that $\lim_{n \rightarrow \infty} \prod_{m=n}^{\infty} (1 - P(A_m)) = 0$. Recall that $1 - x \leq e^{-x}$ for $x \geq 0$. Using this fact we get (even stronger conclusion than vanishing

of the limit)

$$\prod_{m=n}^{\infty} (1 - P(A_m)) \leq \prod_{m=n}^{\infty} e^{-P(A_m)} = e^{-\sum_{m=n}^{\infty} P(A_m)} = 0.$$

□

Example B.21 (Optimistic). Let $S(n)$ denote the price of a risky asset at time $n = 1, 2, 3, \dots$. Consider the binomial model in which the probability that the price increases at time n is equal to $\frac{1}{n}$. In other words $P(U_n) = \frac{1}{n}$, where U_n is the event that the price increases at time n . The events U_n are independent and, moreover, $\sum_{n=1}^{\infty} P(U_n) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$, so the second Borel-Cantelli applies. We conclude that $P(U_n \text{ i.o.}) = 1$, that is, the price *surely* will increase infinitely many times.

Notice that in the above situation the prices will decrease surely infinitely many times. ◇

Example B.22 (Pesimistic). The setup is the same as in the previous example with the difference that $P(U_n) = \frac{1}{n^\alpha}$, where $\alpha > 1$. Then the series $\sum_{n=1}^{\infty} \frac{1}{n^\alpha}$ converges and, by the first Borel-Cantelli Lemma, the price increase can surely occur at most finitely many times. ◇

Corollary B.23 (Zero-one Law again). *If the events $\{A_n, n \geq 1\}$ are independent then*

$$P(A_n \text{ i.o.}) = \begin{cases} 0 & \text{if } \sum_{n=1}^{\infty} P(A_n) < \infty \\ 1 & \text{if } \sum_{n=1}^{\infty} P(A_n) = \infty. \end{cases}$$

In particular, if $P(A_n \text{ i.o.}) > 0$ then $P(A_n \text{ i.o.}) = 1$.

Corollary B.24. *Suppose that $X_1, X_2, \dots$ are independent random variables. Then*

$$X_n \xrightarrow{a.s.} 0 \quad \Longleftrightarrow \quad \sum_{n=1}^{\infty} P(|X_n| < \epsilon) < \infty \quad \text{for all } \epsilon > 0.$$

Remark B.25. We used the above fact in the proof of the existence of the Wiener process.

Proof. Let $A_n = \{|X_n| < \epsilon\}$ for $n \geq 1$ and $\epsilon > 0$. Then

- $\omega \in \liminf A_n^c$ means that ω is such that $|X_n(\omega)| \leq \epsilon$ for all sufficiently large n .
- $\omega \in \limsup A_n$ means that ω is such that there are arbitrarily large n such that $|X_n(\omega)| > \epsilon$.

It follows that if ω is such that $\lim_{n \rightarrow \infty} X_n(\omega) = 0$ then for every $\epsilon > 0$ the inequality $|X_n(\omega)| > \epsilon$ holds for at most finitely many n . Hence

$$\begin{aligned}
 P\left(\{\omega \mid \lim_{n \rightarrow \infty} X_n(\omega) = 0\}\right) = 1 & \iff P(|X_n| > \epsilon \text{ i.o.}) = 0 \quad \text{for all } \epsilon > 0 \\
 & \iff \sum_{n=1}^{\infty} P(|X_n| < \epsilon) < \infty \quad \text{for all } \epsilon > 0.
 \end{aligned}$$

□

Bb. Let $S(n)$ be the price of a share of stock at time $n \in \mathbf{N}$. Suppose that the probability of prices going up at time n is independent of n and it is equal to p and that the probability of the price going down is $1 - p$. Prove that surely:

1. prices will go up infinitely many times;
2. prices will go down infinitely many times;
3. prices will go up 1000 times in a row infinitely many times;
4. any finite sequence of price changes will occur infinitely many times.

Bc. The last part in the previous exercise has the following funny counterpart. Suppose that an immortal monkey is given an indestructible typewriter and an infinite supply of paper. The monkey plays with the typewriter by pressing keys at random. Show that the monkey will surely write your favourite book infinitely many times.

Bd. Let $\{X_n, n \geq 1\}$ be a collection of independent identically distributed continuous random variables. Let $T(1) = 1$ and

$$T(n) = \min\{n \in \mathbf{N} \mid X_n > X_{L(n-1)}\}.$$

Think of $T(n)$ as the time of the *record*. For example, if X_n is the price of stock at time n we record the time instances when the price is the highest in history so far.

Consider the first n variables $X_1, \dots, X_n$ and show that $P(X_k \text{ is a record}) = \frac{1}{k}$. Deduce that there will be infinitely many records.

Be. The setup is as above and we are asking the question of frequency of double records, that is, the event where two consecutive variables X_n and X_{n+1} are records. Show that there will be most only finitely many double records.

B.26 Existence of the Wiener process

Product measures

Definition B.27. Let $(\Omega_i, \mathcal{F}_i, \mathbf{P}_i)_{i \in A}$ be a collection of measurable spaces. The **product** σ -algebra $\mathcal{F}_A$ on the product $\Omega_A = \prod_{i \in A} \Omega_i$ is defined to be the smallest σ -algebra containing all the sets of the form $\pi_i^{-1}(E_i)$ for $i \in A$ and $E_i \in \mathcal{F}_i$, where $\pi_i: \Omega_A \rightarrow \Omega_i$ is the projection. The **product measure** $\mathbf{P}_A$ is defined by $\mathbf{P}_A(\pi_i^{-1}(E_i)) = \mathbf{P}_i(E_i)$.

The product measure, if it exists, is unique. The existence, however, is a nontrivial matter and there are examples of families of probability spaces for which the product measure does not exist.

Theorem B.28. Let $([0, 1], \mathcal{B}, \mathbf{P})$ be the standard Borel probability space on the interval. Then there exists a unique probability measure on an arbitrary product $\prod_{i \in A} ([0, 1], \mathcal{B}, \mathbf{P})$.

Density of dyadic rationals in $\mathbf{R}$

Definition B.29. A subset $S \subset \mathbf{R}$ is called **dense** in $\mathbf{R}$ if for every $t \in \mathbf{R}$ and every $\epsilon > 0$ there exist $s \in S \cap (t - \epsilon, t + \epsilon)$. Density of $S \subset [a, b]$ is defined analogously.

Bf. Show that the set $\mathbf{D} = \{\frac{j}{2^k} \in \mathbf{Q} \mid j \in \mathbf{Z}, k \in \mathbf{N}\}$ of dyadic rationals is dense in $\mathbf{R}$.

The Borel-Cantelli Lemma

Lemma B.30. Let X_n, X be random variables such that $\sum_n \mathbf{P}(|X_n - X|) \geq \epsilon) < \infty$ for every $\epsilon > 0$. Then X_n converges almost surely to X .

More precisely, the lemma is saying that the hypothesis implies that for ω in the set of probability 1 the sequence $X_n(\omega)$ converges to $X(\omega)$:

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega).$$

In particular, for each of such ω and every $\epsilon > 0$ there exist $M \in \mathbf{N}$ such that for all $n > M$ we have

$$|X_n(\omega) - X(\omega)| \leq \epsilon.$$

Hölder continuity

Definition B.31. Let $\Sigma \subset \mathbf{R}$ be a subset. A function $f: \Sigma \rightarrow \mathbf{R}$ is called **Hölder continuous** if there exist constants $\theta > 0$ and $C > 0$ such that

$$|f(s) - f(t)| \leq C|s - t|^\theta$$

for all $s, t \in \Sigma$.

Bg. Show that a Hölder continuous function with exponent $\theta > 1$ is constant. Give several examples of Hölder continuous functions.

Theorem B.32. Let $\Sigma \subset \mathbf{R}$ be a dense subset and let $f: \Sigma \rightarrow \mathbf{R}$ be a Hölder continuous function. Then f is uniformly continuous. In particular, it can be uniquely extended to a uniformly continuous function $\hat{f}: \mathbf{R} \rightarrow \mathbf{R}$.

The above theorem means that the function $\hat{f}$ restricted to the set Σ is equal to the initial function f .

Wiener process

Definition B.33. Let $\mu \in \mathbf{R}$, $\Sigma \subseteq [0, \infty)$ and $0 \in \Sigma$. A one-dimensional **Wiener process** on Σ with initial position μ is a collection $(X_t)_{t \in \Sigma}$ of real random variables X_t such that:

1. $X_0 = \mu$.
2. Almost surely the map $t \mapsto X_t$ is a continuous function on Σ .
3. For every $0 \leq t_- < t_+$ in Σ , the increment $X_{t_+} - X_{t_-}$ has the distribution of $N(0, t_+ - t_-)$. In particular, $X_t \equiv N(\mu, t)$ for every $t > 0$.
4. For every $0 \leq t_1 \leq t_2 \leq \dots \leq t_n$ in Σ , the increments $X_{t_i} - X_{t_{i-1}}$ for $i = 1, \dots, n$ are jointly independent.

If Σ is discrete then $(X_t)_{t \in \Sigma}$ is called a **discrete Wiener process**. A Wiener process is also called a **Brownian motion**.

Theorem B.34. Let $\Sigma \subseteq [0, \infty)$ be either a discrete subset or all of $[0, \infty)$ and let $\mu \in \mathbf{R}$. Then there exists a Wiener process $(X_t)_{t \in \Sigma}$ with initial position μ . Any other Wiener process $(X'_t)_{t \in \Sigma}$ has the same same distribution as $(X_t)_{t \in \Sigma}$.

Proof. Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. In the process of the proof we will be extending it whenever necessary.

Existence: the case of discrete Σ . Let $\Sigma = \{t_0, t_1, t_2, \dots\}$, where

$$t_0 = 0 < t_1 < t_2 < \dots$$

is an increasing sequence of real numbers. Let $(dX_i)_{i=1}^\infty$ be a collection of jointly independent random variables such that $dX_i \equiv N(0, t_i - t_{i-1})$. These random variables will serve as increments of the Wiener process we are looking for.

Such random variables can be constructed as follows. Let $\mathbf{m}_i$ be a measure on $\mathbf{R}$ defining the normal distribution $N(0, t_i - t_{i-1})$. Define random variables $dX_i: \Omega \times \prod_{i=1}^\infty (\mathbf{R}, \mathbf{m}_i) \rightarrow \mathbf{R}$ by

$$dX_i(\omega, x) = dX_i(\omega, x_1, x_2, \dots) = x_i,$$

where $x = (x_1, x_2, \dots)$. Here the infinite product $\prod_{i=1}^\infty (\mathbf{R}, \mathbf{m}_i)$ is equipped with the *product* measure $\mathbf{m}$. It is immediate from the definition that

$$(dX_i)_*(\mathbf{P} \times \mathbf{m}) = \mathbf{m}_i,$$

which means that $dX_i \equiv N(0, t_i - t_{i-1})$. In order to check that the variables dX_i are jointly independent we let $dX: \Omega \times \prod_{i=1}^\infty (\mathbf{R}, \mathbf{m}_i) \rightarrow \prod_{i=1}^\infty \mathbf{R}$ to be defined by

$$dX(\omega, x) = (dX_1(\omega, x), dX_2(\omega, x), \dots).$$

Again, it is immediate from the definition of dX_i that the pushforward measure $dX_*(\mathbf{P} \times \mathbf{m}) = \mathbf{m}$ which shows that dX_i are jointly independent.

Remark B.35. Observe that the above construction gives us more than we need because the definition of the Wiener process requires only that any *finite* collection of increments consists of jointly independent variables. Thus we could avoid the difficult construction of the *infinite* product space and instead define dX_i on finite products and extend if necessary.

Let $X_{t_i} = \mu + dX_1 + \cdots + dX_i$ for all $i = 0, 1, 2, \dots$. This is a Wiener process.

Bh. Verify that (X_{t_i}) is a Wiener process. The solution will be added here after we discuss this in the tutorial.

Existence: the case of $\Sigma = [0, \infty)$. We shall construct a sequence of discrete Wiener processes on finer and finer sets Σ and then pass to the limit. Let $(X_t)_{t \in \mathbf{N}}$ be a discrete Wiener process constructed as in the previous part. We extend it to the set $\frac{1}{2}\mathbf{N}$ by setting

$$X_{t+\frac{1}{2}} = \frac{X_t + X_{t+1}}{2} + Y_{t,0},$$

where $t = 0, 1, 2, \dots$ and $Y_{t,0}$ are iid copies of $N(0, \frac{1}{4})$ which are jointly independent of $(X_t)_{t \in \mathbf{N}}$.

Bi. Provide a construction of the random variables $Y_{t,0}$ with the required properties.

Bj. Verify that $(X_t)_{t \in \frac{1}{2}\mathbf{N}}$ is a Wiener process.

We extend the process further to the set $\frac{1}{4}\mathbf{N}$ by setting

$$X_{t+\frac{1}{4}} = \frac{X_t + X_{t+\frac{1}{2}}}{2} + Y_{t,1},$$

where $t \in \frac{1}{2}\mathbf{N}$ and $Y_{t,1}$ are iid random variables with distribution $N(0, \frac{1}{8})$ which are jointly independent of $(X_t)_{t \in \frac{1}{2}\mathbf{N}}$.

Bk. Provide the construction of random variables $Y_{t,1}$ satisfying the above conditions and prove that $(X_t)_{t \in \frac{1}{4}\mathbf{N}}$ is a Wiener process.

Iterating this procedure infinitely many times, we obtain a sequence of Wiener processes $(X_t)_{t \in \frac{1}{2^k}\mathbf{N}}$ for $k = 0, 1, 2, \dots$ which are consistent in the sense that the previous processes are restrictions of the later ones. This yields a process $(X_t)_{t \in \mathbf{D}}$, where $\mathbf{D} = \{\frac{j}{2^k} \in \mathbf{Q} \mid j, k \in \mathbf{N}\}$ is the set of nonnegative *dyadic* rational numbers. Notice that this set is neither discrete nor equal to $[0, \infty)$ so the process is not Wiener.

In the next step we extend the process further to the whole set $[0, \infty)$. The main idea is that the functions $t \mapsto X_t(\omega)$ for $t \in \mathbf{D}$ and $\omega \in E$, where

the set E has probability 1, satisfy certain inequality which allows to extend them to continuous functions on the whole of $[0, \infty)$.

Let $0 < \theta < \frac{1}{2}$ and let $T > 0$. Since for any $j, k \in \mathbf{N}$ we have

$$X_{\frac{j+1}{2^k}} - X_{\frac{j}{2^k}} \equiv N\left(0, \frac{1}{2^k}\right),$$

it follows from the fact that the normal distribution is *subgaussian* that there exist constants $c, C > 0$ such that

$$\mathbf{P}\left(|X_{\frac{j+1}{2^k}} - X_{\frac{j}{2^k}}| \geq 2^{-k\theta}\right) \leq C \exp(-c 2^k(1 - 2\theta)).$$

B1. Prove that

$$\sum_{\{j,k \mid \frac{j}{2^k} \leq T\}} C \exp(-c 2^k(1 - 2\theta)) < \infty.$$

It then follows from the *Borel-Cantelli Lemma* that, for a fixed T , the following inequality

$$\left|X_{\frac{j+1}{2^k}} - X_{\frac{j}{2^k}}\right| \leq 2^{-k\theta}.$$

holds almost surely for all but finitely many $j, k \in \mathbf{N}$ with $\frac{j}{2^k} \leq T$.

Bm. Verify this claim. Observe that the way we stated the Borel-Cantelli Lemma does not directly apply here, so we need some explanation.

This implies that for each T the function $[0, T] \cap \mathbf{D} \rightarrow \mathbf{R}$ given by $t \mapsto X_t(\omega)$ is *Hölder continuous* for all ω from a set of probability 1. It follows that the function $t \mapsto X_t(\omega)$ is almost surely *locally* Hölder continuous with exponent θ on the set $\mathbf{D}$ of all nonnegative dyadic rationals. In particular, these functions are almost surely locally uniformly continuous on $\mathbf{D}$.

Since the dyadic rationals $\mathbf{D}$ is a dense subset of $[0, \infty)$ we extend the function $t \mapsto X_t(\omega)$ to a unique continuous function on $[0, \infty)$.

Bn. Show that $(X_t)_{t \in [0, \infty)}$ is a Wiener process.

□